

Political Preferences and Migration Decisions of College-Educated Workers

Mitch Downey

Jinci Liu*

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Abstract

We study the consequence of political polarization along educational lines in the United States. Descriptively, we show that college graduates are now well to the left of non-college voters on economic and social issues and much more so than 15 years ago. We then estimate the causal effect of a Republican governor on college graduates' interstate migration rates, finding that conservative governance reduces the inflow of college-educated workers by about 8%. Finally, we analyze a structural model of migration that quantifies the implications of plausible changes in political control for cross-state spillovers and college/non-college earnings inequality.

Keywords: Political polarization, migration, spatial sorting, inequality

JEL Classification Numbers: D31, D72, H31, H73, J61, P16

*Downey: Institute for International Economic Studies, Stockholm University, mitch.downey@iies.su.se.
Liu: Institute for International Economic Studies, Stockholm University, jinci.liu@iies.su.se. We are grateful to Jay Lee for invaluable guidance and Adrien Bilal, Simon Görlach, Seth Hill, Benny Kleinman, Martin Koenen, Kieran Larkin, Kurt Mitman, Enrico Moretti, Alessandra Peter, Giacomo Ponzetto, David Strömberg, Horng Chern Wong, Noam Yuchtman, and seminar participants at SU, IESEG, Illinois, Arkansas, the UEA, the Barcelona Summer Forum, and the MYPEERS Workshop for helpful feedback.

1 Introduction

How do people choose where to live, and what are the consequences for the pattern of economic activity? Over the last 40 years, US college graduates have become increasingly concentrated in specific places (Moretti, 2012; Diamond and Gaubert, 2022). Because high-skilled workers and sectors generate local economic spillovers (Moretti, 2004; Giannone, 2022), this spatial sorting has slowed regional convergence (Kleinman, Liu, and Redding, 2023) and exacerbated welfare inequality between college and non-college workers (Diamond, 2016).

At the same time, Democrats and Republicans increasingly live in different places (Brown et al., 2022; Kaplan et al., 2022). Because neighbors and neighborhoods affect individual attitudes (Perez-Truglia and Cruces, 2017; Perez-Truglia, 2018; Martin and Webster, 2020; Cantoni and Pons, 2022), this spatial sorting can exacerbate political polarization (Bishop, 2009; Brown and Enos, 2021). It also reduces democratic competition, which exacerbates the election of political extremists (Hopkins, 2017) and undermines economic growth (Besley, Persson, and Sturm, 2010).

In this paper, we show that these two patterns of geographic sorting—by education and by political attitudes—are deeply intertwined, and we quantify the consequences for the pattern of economic activity. Divisions in political attitudes increasingly fall along educational lines. In 2020, the college/non-college gap in Biden/Trump voting was as large as the gap between New York and Mississippi. This is part of a long-running worldwide trend in the realignment of political coalitions (Gethin, Martínez-Toledano, and Piketty, 2022). Our main goal in this paper is to document that college graduates are reluctant to move into Republican-governed states, which reduces the human capital of those states, and to quantify the consequences of this effect for the pattern of economic activity.

We begin by using survey data and state-of-the-art tools borrowed from political science to create comparable indices of policy views on economic and social/cultural issues over the last 15 years.¹ We find that the gap in policy views between college and non-college voters has grown dramatically since 2010. On economic issues, as recently as 2010, there was *no* college/non-college gap in policy views, while today, the average college voter is .4 standard deviations to the left of the average non-college voter. On social and cultural issues, while the gap shrank during the later years of the Bush Administration and was roughly stable

¹Our approach is similar to Draca and Schwarz (2024) and Danieli, Gidron, Kikuchi, and Levy (2024), and contrasts with the approach based on inferring multidimensional preferences from voters’ choices and the positions of candidates (Longuet-Marx, 2024) or parties (Gethin et al., 2022). See Fiorina and Abrams (2008) for some discussion about the difficulty of inferring polarization of continuous preferences from binary vote choices.

from 2010-2015, it more than doubled from 2015-2020 (to .5 standard deviations). We show that this implies that higher earners are, on average, far to the left of lower-earning voters on both social and economic issues.

We next estimate the migration effects of transitioning from a Democratic governor to a Republican one. As we discuss below, we focus on governors because they are very salient and have a large influence on policy outcomes. Using a difference-in-difference strategy, we find that a Republican governor, relative to a Democratic one, reduces the in-migration of college graduates employed in the private sector by about 8% per year.² This implies a 0.25 percentage point annual decrease in the share of college graduates who moved into the state within the last year. Over four years, this amounts to a 1% decline in the stock of college-educated workers, roughly equal to one year of growth in the average state’s college-educated workforce.

In interpreting this magnitude, it is important to emphasize that neither our empirical results nor our model implies that individuals are uprooting their families to flee Republican governors. Indeed, like most of the literature (Monras, 2020), we find no effects on out-migration. Our results instead entirely reflect decreased in-migration under Republican governors. This suggests that once graduates have decided to move, politics is one of the factors they take into account in choosing *where* to live.

We find larger and significant effects from a regression discontinuity design, though we prefer the difference-in-difference estimates. We also rule out the possibility that migrants are responding to economic conditions or beliefs about voters instead of the governor. At the same time, however, we do not find that larger policy shifts lead to larger migration responses.³ Thus, we interpret our results as reflecting worsened attitudes towards the Republican Party as a whole, rather than responding to the specific policies that any particular governor actually implements.

We are not the first to suggest that the Republican Party’s core policy priorities might push away college-educated professionals.⁴ Conservative states spend millions advertising low

²Focusing on states that change gubernatorial partisanship is a limitation in the sense that we cannot speak to the full range of political variation in the United States since much of the US policy variation is a persistent difference between liberal and conservative states (Caughey and Warshaw, 2016). At the same time, however, drawing identification from “switchers” speaks directly to the effects of changes in partisanship, which we consider policy relevant.

³If anything, we present suggestive evidence that more pro-market policy reforms offset some of the declining in-migration a Republican governor brings. This would suggest a *revealed* preference among college graduates for less redistributive policies, despite the descriptive evidence we present about their *stated* preferences. This is important because these preferences against redistribution have been used to interpret some existing findings in this literature (e.g., Gethin et al. (2022)), despite their conflict with directly observable survey measures.

⁴In a 2017 survey, Smith, Hibbing, and Hibbing (2019) find that one in four respondents report that politics had led them to consider moving. The challenge of attracting college graduates to conservative states

taxes in liberal states (Moretti and Wilson, 2017). As an example, Republican-controlled Indiana ran a billboard in neighboring Democratic-controlled Illinois asking whether residents were “Illinoisised by higher taxes” and encouraging them to “Come to Indiana: A state that works.” At the same time, however, Indiana Governor Mike Pence was signing into law one of the most controversial anti-LGBTQ laws in the nation, and one of Indianapolis’ Democratic City Councilmembers noted, “The real harm, to all [of us], is that it makes us all look like backwater hicks. Indiana is losing jobs and young professionals like crazy. How much more can our state government make Indiana uninviting (Eason, 2015)?” The rhetoric of making Indiana “uninviting” is notably similar to the rhetoric conservative policymakers use to discuss high taxes. In a sense, then, our paper shows that this councilmember was correct and that conservative policies are an implicit preference tax on the average college graduate.

To understand the consequences of this, we analyze a structural model of migration that builds on Bryan and Morten (2019). College and non-college workers choose where to live based on wages, migration costs, and general amenities, and are combined in state-level production. Our key innovation is adding a “political amenity” that depends on the partisanship of the governor. We estimate most parameters following the approach developed by Bryan and Morten (2019) and calibrate the political amenity to match our reduced form results. We use the calibrated model to simulate various counterfactuals reflecting plausible shifts in partisan control, and find that four years under a Republican governor can reduce a state’s GDP by up to 2%.

We also use our model to study inequality, both because inequality is an important trend in the US economy (Acemoglu and Autor, 2011) and because it plays a key role in political polarization.⁵ Increasing in-migration of college graduates has two primary effects on college/non-college earnings inequality. On the one hand, increasing the relative supply of college graduates lowers their relative wages (a price effect). On the other hand, migrants are positively selected relative to non-migrants, since they must receive employment opportunities sufficient to justify the move. Thus, increasing the share of migrants among college graduates increases the average productivity of college graduates and thereby increases earnings inequality (a composition effect).⁶

has garnered considerable attention since the Supreme Court’s *Dobbs* decision gave states more authority over abortion laws (Cain et al., 2022; Hagelgans and Basi, 2022; Keshner, 2021; Leonhardt, 2021). Our results show that this challenge is real but broader than abortion laws alone.

⁵The political consequences of income inequality have been a major research area in economics (Gurieff and Papaioannou, 2022), political science (McCarty, Poole, and Rosenthal, 2008; Iversen and Soskice, 2015), and psychology (Power, Madsen, and Morton, 2020). Indeed, the possibility that inequality fuels political polarization dates back at least to James Madison, who wrote that “the most common and durable source of factions has been the various and unequal distribution of property” (Gunderson, 2022).

⁶There is also a third channel that is quantitatively small. The marginal migrant is less positively

Theoretically, it is ambiguous which force will be most important, but in our counterfactuals the composition effect dominates the price effect. As a result, Democratic control of a state attracts more college graduates but also raises their average earnings relative to non-college workers. This occurs even without agglomeration forces or knowledge spillovers, but solely because cross-state migrant college graduates are more positively selected than in-state college graduates. This is an interesting observation because many of the most progressive areas of the country (e.g., coastal cities and states) have seen growing college workforces (especially due to in-migration from other states) along with rising college/non-college inequality. Since wages and employment of college graduates have both increased, this has often been attributed to demand-side forces (e.g., competitive productivity advantages). Our model can explain these trends through supply-side forces alone, once one accounts for the selectivity of the in-migrating workers.

Finally, our model lets us study the cross-state spillovers inherent to migration. Deterred would-be migrants choose to live somewhere else, so changes in the incentives to move to one state ripple throughout other states. These spillovers can be considerable and illustrate how the regionalization of party success (the tendency for Democrats or Republicans to win across all states within a given region) influences the migration responses we estimate.

At the broadest level, our results speak to the real-world consequences of political polarization. Despite the widely held view that polarization is an important social phenomenon, there is little evidence of its effects on tangible economic outcomes.⁷ As US politics grows more hostile and divisive (Boxell, Gentzkow, and Shapiro, 2024), it is important to understand its consequences beyond purely political outcomes.

Our results also contribute to recent work exploring the role of education in divides on political attitudes (Gethin, Martínez-Toledano, and Piketty, 2022; Hall, Evans, and Kim, 2023; Kuziemko, Longuet-Marx, and Naidu, 2026; Longuet-Marx, 2024; Firoozi, 2026) and social, cultural, and lifestyle choices (Easterbrook, Kuppens, and Manstead, 2016; Bertrand and Kamenica, 2023; Desmet, Ortuño-Ortín, and Wacziarg, 2025). Recently, Gethin et al. (2022) combined data from 70 years and 21 countries to show that increased college graduates' support for left-of-center parties is part of a universal and long-run trend. This suggests that the forces we focus on will likely continue to be relevant, regardless of short-term changes in the influence of specific candidates or issues. It also illustrates that these

selected than inframarginal migrants, and so increasing in-migration lowers the average human capital of college graduate in-migrants.

⁷Some exceptions include experiment evidence that workers require significant compensating differentials to work for employers whose political views oppose their own (McConnell et al., 2018; Chinoy and Koenen, 2024), and a growing range of evidence about the health consequences of polarization on issues like COVID-19 and the Affordable Care Act (Wallace et al., 2022; Bursztyn et al., 2022).

forces are relevant beyond the United States. For instance, Brox and Krieger (2021) show that anti-immigration protests in Germany led to declines in in-migration into affected cities which was driven by college graduates, and Pickard, Bove, and Efthymoulou (2022) show that voters opposed to Brexit began migrating away from pro-Brexit districts after the vote.

Finally, our results contribute to work on the role of politics in spatial sorting. A growing literature using stated preference experiments studies the role of politics in neighborhood choice (Mummolo and Nall, 2017; Koşar, Ransom, and Van der Klaauw, 2022) and the role of specific policies—like abortion laws (Baumle, Miller, and Gregory, 2023) or political reforms (Nelson and Witko, 2022)—in state choice. Our reduced form empirical results imply that the insights from these studies aggregate in two important ways. First, aversion to policies that are disproportionately enacted by Republican politicians implies aversion to the Republican Party. Second, a stronger aversion to these things among Democrats (who are disproportionately college-educated) implies a stronger aversion, on average, among college graduates. Our structural analysis then quantifies the consequences of these findings.

The rest of the paper is organized as follows. Section 2 documents our descriptive results. Section 3 estimates the causal effects of governors on migration. Section 4 presents our model, estimation, and counterfactual analysis. Section 5 concludes.

2 Descriptive facts about education polarization

We begin by documenting trends in “education polarization,” by which we mean the difference in policy preferences between college graduates and non-college graduates. Though these trends are not the main contribution of our paper, we believe they help to interpret the migration responses we document below.

2.1 Data and methods

Our main data is from the Cooperative Election Study (CES). The CES began in 2006 with roughly 35,000 respondents and today includes roughly 60,000 respondents per even-numbered year (much larger than any similar survey). During each wave, the CES asks 15-50 different policy questions, far more than other political surveys. Using these policy questions, we create indices of policy preferences on social and economic policies. We classify questions using the topic lists developed by Caughey and Warshaw (2018); Appendix Tables A1 and A2 contain details.⁸ The specific set of policy questions changes from year to year so

⁸Examples of social issues include gun control, abortion, immigration, and policing; examples of economic issues include financial regulation, environmental policy, taxation, and government spending. The average survey wave includes 13.5 social policy questions and 13.9 economic policy questions.

as to capture the central issues in political debate. This is an advantage in the sense that the questions important for political divides naturally change over time, and our index reflects disagreement on the issues relevant at any given moment.⁹

At the same time, this is a disadvantage in the sense that we cannot distinguish between changes in specific views and changes in which social and economic issues are debated. This implies that we cannot determine whether the trends are supply-driven (i.e., led by politicians) or demand-driven (i.e., voters changing their views for other fixed issues), despite the centrality of this question to the literature (e.g., Longuet-Marx 2024). In Appendix Section B.2.3, we present results for each question individually and find trends that are similar to our indices, though the changes in questions over time does play some role.

To estimate indices, we follow the best practices in political science and estimate ideal points using Item Response Theory (IRT). IRT was developed for use in standardized testing, where not all test takers receive the identical test, and not all questions are equally informative about underlying ability. It is widely used in political opinion research because (1) it does not require the analyst to specify, ex-ante, which questions are more or less informative about ideology, and (2) unlike other methods (such as principal component analysis), it generates comparable scores across respondents even when the set of questions changes over time, so long as there are “bridge” questions that are continuous from one survey wave to the next (see Caughey and Warshaw (2015) for some discussion). For this reason, the CES intentionally includes bridge questions in each survey wave.

We estimate our indices using an Expectation Maximization algorithm (Imai, Lo, and Olmsted, 2016), and normalize them to be mean zero with a standard deviation of one (across all respondents from all years) and so that more negative values indicate more left-leaning views and more positive numbers denote more right-leaning views. To ensure that we capture completed education, we restrict our trends to those age 26 and older.¹⁰ To ensure that our trends actual reflect the sample typically making migration decisions, we focus on those aged 45 and younger.¹¹

⁹For example, in 2006, there were major political divides over whether school science classes should teach intelligent design and whether social security should be privatized. By 2020, these issues were irrelevant; voters instead debated whether schools should require face masks and whether Medicare should be universal. Changing the questions over time allows the index to reflect actual political disagreement. Some of the more permanent divides in US politics (such as questions about tax rates and government spending) appear continuously in each CES wave.

¹⁰Throughout the paper, we focus on those with a Bachelor’s degree or more, but Appendix Figure B5 presents trends for the Bachelor’s only sample separate from those with a post-graduate degree.

¹¹Appendix Figure B1 presents inter-state migration rates by age and education. Appendix Figure B4 shows that the trends are similar among the full population of college graduates aged 26+.

2.2 Differences in policy preferences by education

Figure 1 presents our main results about “education polarization” (i.e., the gap in policy views between college and non-college respondents). Panels (a) and (b) show the evolution of college and non-college respondents’ social and economic policy views over time.¹²

On social issues, both college graduates and non-college graduates have drifted considerably left over time, consistent with widely recognized changes in views on issues like divorce, gay marriage, and marijuana legalization. However, while college graduates have always had more liberal views, the gap has not been constant. Panel (c) shows that the gap fell during the end of the Bush Administration (2006-2008) and was roughly stable from 2008-2015, though it has been growing steadily since the candidacy/election of Donald Trump.¹³ From 2015 to 2020, it more than doubled from $.2\sigma$ to roughly $.5\sigma$.¹⁴

[Figure 1 about here.]

Panel (b) shows that the education gap in economic policy views emerges only after 2010 and then grows quickly. Before 2010, average views were similar across education groups. After 2010, when the Tea Party rose to power with a right-wing economic policy agenda and a large non-college base, college-educated respondents moved left by $.25\sigma$ while non-college voters tended to have moved modestly to the right. Panel (c) shows that the gap in views has grown to roughly $.4\sigma$. Thus, college graduates today report views that are considerably left of non-college respondents on both social and economic issues.¹⁵

These gaps in policy views translate into meaningful differences in political approval. Panel (d) shows results from regressions of gubernatorial approval on state and year fixed effects and interactions between governor partisanship and respondent education.¹⁶ The coefficients and 95% confidence intervals are plotted in panel (d). Relative to a non-college voter with a Republican governor (the omitted category), there is virtually no difference in approval of Democratic governors among non-college voters. In contrast, college graduates report .1 points lower approval of Republican governors and .2 points higher approval of

¹²It is not our goal to determine the *reason* for these changes, but in Appendix Section B.2.4 we summarize several potential theories and present descriptive evidence consistent with some and inconsistent with others.

¹³The coincidence with the timing of Donald Trump’s presidential campaign is noteworthy, although various commentators have suggested the trends pre-dated his campaign (Haidt and Lukianoff, 2021; Mounk, 2023; Yglesias, 2019). Moreover, similar trends are seen in many other countries (Gethin et al., 2022).

¹⁴We see similar trends when we look at the gap in *median* (rather than mean) views in Appendix Figure B4. This is important because the ability to distinguish extreme views has changed over time as more policy questions have been added to the CES.

¹⁵Differences in *average* views are necessarily a simplifying summary. Appendix Figure B6 compares the full distribution of social policy preferences in 2010 and 2020. In 2010, college graduates were 50% more likely than non-college to fall left of the median Democrat; by 2020, they were 120% more likely.

¹⁶ $Y_{i,s,t} = \alpha_s + \delta_t + \beta_{ND}(Noncoll_i \times DemGov_{s,t}) + \beta_{CR}(Coll_i \times RepGov_{s,t}) + \beta_{CD}(Coll_i \times DemGov_{s,t}) + \varepsilon_{i,s,t}$.

Democratic governors. Thus, while non-college respondents' gubernatorial approval does not tend to change with partisanship, college graduates report .3 points higher approval of Democratic governors than Republican ones. This .3 point difference large: It is two-thirds as large as the .44 difference in pre-election approval ratings for governors just before a partisan flip compared to when the party retains power.

Educational gaps in political views are not simply driven by a minority of far-left, college-educated low-earners. In Appendix Section B.2.2, we show that higher earners are to the left of lower earners on both social and economic issues. This is true across virtually all industries, and it is entirely explained by education polarization; holding fixed education, higher earners are more right-leaning than lower earners. This is inconsistent with some interpretations of recent evidence that observes party choice but not policy preferences (Gethin et al., 2022), and implies driving away college graduates likely entails some economic cost to states.

3 Reduced form estimates of migration responses

Our main interest is how college-educated workers respond to changes in the partisan control of state government. For this reason, we focus on the partisanship of the governor. It has long been recognized that citizens know more about their governor than other elected officials and that governors garner more media attention (Hinckley, Hofstetter, and Kessel, 1974; Squire and Fastnow, 1994). Even today, although politics has nationalized, 78% of 2020 CES respondents could correctly identify the partisanship of their governor, compared to only 45-60% for their senators or representatives in the House.

Moreover, governors have significant effects on state policy outcomes, and the size of these effects has grown over time (Caughey, Xu, and Warshaw, 2017).¹⁷ In part, this is because governors face few checks and balances or constraints on authority (Seifter, 2017); in part, it is because they use their veto power much more actively than presidents do; in part, it is because they are typically the *de facto* leaders of their political party within the state, letting them set agendas and drive legislation. Below, we show that gubernatorial transitions have large policy effects. At the same time, however, we recognize that governors also get attention for controversial statements and political performances that are not actual policy changes and these might also drive migration behavior. In Section 3.4, we attempt to

¹⁷In an influential paper, Ferreira and Gyourko (2009) use a regression discontinuity and find that *mayors'* partisanship does not matter for policy outcomes. A subsequent literature uses additional outcomes and larger datasets and finds that it does (Gerber and Hopkins, 2011; de Benedictis-Kessner and Warshaw, 2016; Dippel, 2022; de Benedictis-Kessner, Jones, and Warshaw, 2025). Regardless of whether *mayoral* partisanship matters, the same regression discontinuity design shows that *governors'* partisanship is consequential (Caughey et al., 2017).

distinguish between these explanations.

3.1 Data

Our primary migration data is from the American Community Survey (ACS), a nationally representative 1% sample of the US population conducted by the Census Bureau, accessed via IPUMS (Ruggles et al., 2022). Microdata includes respondents’ state-of-residence, age, education, employment details, and last year’s state-of-residence. This allows us to calculate annual interstate migration rates by education. Relative to studies using state of birth and residence to measure migration (e.g., Bryan and Morten (2019)), a key advantage is that we observe the specific timing of the move.¹⁸ We use the ACS back to 2000 (when it was introduced), although from 2000-2004, the sample size was considerably smaller than the 1% sample design used since 2005.

Our primary focus is on migration flows, calculated separately by whether the respondent has at least a Bachelor’s degree. We define in-migration rates in state s in year t as the fraction of respondents living in state s in year t who report living in a different state in year $t - 1$. Likewise, to define out-migration from s in year t , we aggregate over all respondents interviewed in year t who report living in state s during $t - 1$, and calculate the share who *does not* live in s at the time of the year t interview. We always use ACS sample weights when calculating migration flows.

Throughout the paper, we never analyze migration from 2020 onwards because we are concerned that during the main pandemic years, respondents’ state-of-residence is either not well defined, not well measured, or is only temporary. Thus, we always end our migration analysis window in 2019. Finally, our primary analyses focus on respondents who are at least age 26 (so that educational attainment has been completed) and employed in the private sector (and thus most relevant for economic activity), although we also show results for more restrictive and less restrictive samples.

We observe migration from 2000-2019. Since all elections occur in November and governors take office in January, we treat migration during the election year as pre-election and the year after the election as the first post-election year. For our regression discontinuity, we use elections 2000-2016 (governors take office 2001-2017) so that we have at least three post-election years to measure migration outcomes. For our difference-in-difference estimators, we restrict further and only use elections occurring 2002-2016 (governors take office

¹⁸Some migration studies use the IRS migration files. This data does not contain education, and the college/non-college earnings distributions are not sufficiently distinct to impute it. Other recent work studies migration using voter registration data or change of address data (Brown et al., 2023). This data also does not contain education.

2003-2017) so that we also have at least three pre-election years to test pre-trends.

3.2 Methods

The regression discontinuity design (RDD) is usually considered the gold standard for estimating the effects of election outcomes but uses very few elections. Therefore, it is imprecise and not well suited for studying how effects vary over time or across different types of flips that further divide our sample. Moreover, as we discuss below, we believe that close elections are not representative of the average gubernatorial flip.

For this reason, although we present RDD results as a robustness check, our main specification uses a difference-in-difference (DiD) specification to estimate the effects of a gubernatorial flip relative to keeping the same gubernatorial party. Because different states elect different governors at different times, this is a staggered rollout problem. Thus, we use a stacked DiD estimator (as proposed by Cengiz et al. (2019)) and adjust the weights (as proposed by Callaway and Sant’Anna (2021)) to yield a consistent estimator of the Average Treatment Effect on the Treated (ATT).

As a concrete example, in 2012 North Carolina elected a Republican governor to replace the outgoing Democrat. This governor took office in January 2013. At the time, there were six other states that had Democratic governors in office during the entire ten years surrounding this election (i.e., 2008-2017). We estimate the effects of this flip by comparing the change in migration inflows into North Carolina from 2013 to 2016 with the simultaneous changes experienced in these six other states.

Dube and Zipperer (2014), Cengiz et al. (2019), and Callaway and Sant’Anna (2021) propose this as an estimator for the treatment effect of the gubernatorial flip in North Carolina, which avoids the problems with using a two-way fixed effects (TWFE) estimator (De Chaisemartin and d’Haultfoeuille, 2020; Goodman-Bacon, 2021). Callaway and Sant’Anna (2021) show that averaging many such estimates yields a consistent estimator for the ATT. Following Cengiz et al. (2019), we estimate this ATT by stacking all treatment events together (in comparable treatment time) within one regression. This means that many state-years will appear multiple times (serving as controls for different treatments in different states at different times), so we cluster our standard errors at the state level. Also, as shown by Callaway and Sant’Anna (2021) and Wing, Freedman, and Hollingsworth (2024), this only yields a consistent estimator for the ATT when using adjusted weights such that each state-year’s weight is proportional to the number of states that were treated during that particular treatment cohort.

Thus, letting $\tau \in \{2003, \dots, 2017\}$ denote a treatment cohort (i.e., the year in which the

flip occurred), we estimate the following stacked DiD regression:

$$Y_{s,t,\tau} = \alpha_{s,\tau} + \delta_{t,\tau} + \sum_{k \neq -1} \beta_k (Flip_{s,\tau} \times 1\{t = k\}) + \varepsilon_{s,t,\tau} \quad (1)$$

where s denotes state, $t \equiv y - \tau$ denotes treatment time (where y is the calendar year), state-years are weighted by the number of ACS respondents but with the adjustment to be proportional to the number of treated states in the treatment cohort,¹⁹ $1\{t = k\}$ is an indicator variable that $t = k$, and $Flip_{s,\tau}$ is an indicator that state s 's governor is a different party in year τ ($t = 0$) than they were in year $\tau - 1$ ($t = -1$). Our sample includes any treatment cohorts that include both treatment and control states (in practice, there are typically many control states per treatment state), all treated states from that cohort, all control states that had stable governor partisanship from $t = -5$ to $t = 4$ (i.e., five years pre-treatment and five years post-treatment, at a minimum), and all state-years four years before or after treatment (i.e., $-4 \leq t \leq 3$). The set of states flipping gubernatorial partisanship over time (and by closeness of election) can be seen in Appendix Figure C1.

We estimate this separately for samples where treatment and control states both had Democratic governors at $t = -1$ and therefore $Flip_{s,\tau}$ reflects a flip to a Republican governor, as well as when both had Republicans at $t = -1$ and $Flip_{s,\tau}$ denotes a switch to a Democrat. We refer to these as the effects of D-to-R and R-to-D transitions, respectively. We sometimes present dynamic coefficient plots and sometimes pool all post-treatment coefficients into a single estimate using the following simplified version of Equation (1):

$$Y_{s,t,\tau} = \alpha_{s,\tau} + \delta_{t,\tau} + \sum_{k < -1} \beta_k (Flip_{s,\tau} \times 1\{t = k\}) + \beta_{post} (Flip_{s,\tau} \times 1\{t \geq 0\}) + \varepsilon_{s,t,\tau} \quad (2)$$

which pools post-treatment periods and estimates a single constant post-treatment effect.

The time effects $\delta_{t,\tau}$ are important because migration patterns vary over time, including in response to the large recession in the middle of our sample (Jia et al., 2023). We assume that these temporal fluctuations are proportional,²⁰ and therefore use log migration rates in

¹⁹Formally, let $p_{s,t}$ be the number of ACS respondents in the state-year. Let $N_{T=0,\tau}$ be the number of control states for treatment cohort τ and $N_{T=1,\tau}$ be the number of treated states. Then the weights for any observation denoted by s, t, τ are given by $p_{s,t}$ if s is treated in cohort τ and $\frac{N_{T=1,\tau}}{N_{T=0,\tau}} p_{s,t}$ if state s is a control in cohort τ .

²⁰For instance, we assume that a recession might reduce all states' in-migration rates by half, and find this more plausible than assuming that a recession might reduce in-migration rates by one percentage point equally for states with baseline migration rates of 2% per year and 5% per year (in which case we might estimate specifications with migration rates in levels). In practice, when we estimate our specifications with migration rates in levels, estimated treatment effects are similar and significant, but pre-trends are sometimes worse.

our DiD specifications (Roth and Sant’Anna, 2023).

We use the regressions above – equations (1) and (2) – to estimate the treatment effect of gubernatorial flips and to test whether pre-trends are parallel between treated and control states. However, it is also useful to visualize these changes over time to understand whether migration (or some other outcome) was increasing or decreasing leading up to the flip. To do so, we present the raw plots of levels over time for treatment and control states by normalizing levels by the pre-treatment mean: $\tilde{Y}_{s,t,\tau} = Y_{s,t,\tau} / \bar{Y}_{s,pre,\tau}$ where $\bar{Y}_{s,pre,\tau} = \frac{1}{4}(Y_{s,-4,\tau} + Y_{s,-3,\tau} + Y_{s,-2,\tau} + Y_{s,-1,\tau})$.²¹

We also estimate a modified version of these regressions that pools both types of treatment effects, imposing the assumption that they have symmetric effects:

$$Y_{s,t,\tau,F} = \alpha_{s,\tau,F} + \delta_{t,\tau,F} + \sum_{k < -1} \beta_{F,k} (Flip_{s,\tau,F} \times 1\{t = k\}) + \beta_R Republican_{s,t,\tau} + \varepsilon_{s,t,\tau,F} \quad (3)$$

where $F \in \{DR, RD\}$ indicates whether the treatment event is a D-to-R transition or an R-to-D transition, and both types of treatment events are pooled into the same sample. Comparing (3) to (2), we need no additional identification assumptions (because we allow the set of fixed effects to differ by type of treatment) but impose symmetry for estimated election effects because $Republican_{s,t,\tau} = Flip_{s,\tau} \times 1\{t \geq 0\}$ for D-to-R flips, while $Republican_{s,t,\tau} = 1 - Flip_{s,\tau} \times 1\{t \geq 0\}$ for R-to-D flips.²²

3.3 Main results

We begin by estimating the effects of all types of partisan turnover events on total migration outcomes. To do so, we start with specification (3), which pools Democrat-to-Republican and Republican-to-Democrat transitions. These results are presented in Panel A of Table 1 (pretrends are shown below). Column (1) estimates the effects of a Republican governor on net log migration of college graduates, defined as the log of graduates’ in-migration rate minus the log of their out-migration rate. We find that a Republican governor reduces college graduates’ net migration by about 9 log points. In columns (2) and (3), we decompose this into effects on out-migration and in-migration. While we estimate a small and non-significant increase in out-migration, the results are clearly driven by a large decline in in-migration. This is consistent with evidence from other contexts that migration adjusts to shocks primarily through in-migration rather than out-migration (Howard, 2020; Monras,

²¹Sometimes we analyze outcomes that have no natural units, such as policy indices, which are mean zero by construction. In this case, we normalize by subtracting, rather than dividing by, the pre-treatment mean.

²²In Appendix Table C2, we also present a specification which is simpler but not a direct extension of Equation (2): $Y_{s,t,\tau,F} = \alpha_{s,\tau,F} + \delta_{t,\tau,F} + \beta_R Republican_{s,t,\tau} + \varepsilon_{s,t,\tau,F}$. Results are very similar.

2020; Ransom, 2026).

[Table 1 about here.]

In column (4), we restrict to the sample of college graduates who are US citizens aged 26-45. We do this because politics might be more salient to citizens than non-citizens, and because our descriptive evidence focused on the political opinions of those aged 26-45 since they capture typical interstate migration.²³ In this sample, we find a 9.5 log point decline in the in-migration rate.

Finally, in column (5), we ask whether the migration responses of college graduates differ significantly from those of non-college workers. To do so, we calculate the differential in-migration rate of college graduates as the log of graduates' in-migration rate minus the log of non-graduates' in-migration rate. We find that the in-migration responses of college graduates are significantly stronger than those of non-graduates ($p < .01$). In fact, our magnitudes for *differential* in-migration responses are nearly identical to the effects for in-migration because non-college workers' in-migration is not responsive to changes in governor partisanship (see Appendix Figure C2 and Table C1). We should acknowledge, however, that the *differential* response of college graduates becomes marginally significant or non-significant when we expand the sample to also include non-citizens and respondents over age 45 ($\hat{\beta} = -.05$, $p = .08$) or to use the simpler specification presented in Appendix Table C2 ($\hat{\beta} = -.05$, $p = .11$).

How should we interpret the magnitude of this 8% annual decline in in-migration under Republican governors? On average, the baseline share of a state's college educated workers who moved within the last year is about 3.2%, so an 8% decline is a .25 percentage point decline in annual in-migration. One way to interpret this magnitude is to compare it with estimates from Moretti and Wilson (2017), who study high-skilled inventors and use tax cuts to identify the elasticity of migration with respect to net-of-tax income. They find an elasticity of 1.8, so an 8% decline in inflows is equivalent to a 4.4% reduction in after-tax income. As an alternative, Monras (2020) uses the cross-city heterogeneity in the severity of the Great Recession to estimate the elasticity of migration with respect to average wages and unemployment. Using his estimates, the 0.25 percentage point decline in in-migration that we document would require a 1.25% decline in average wages or a 0.8 percentage point increase in the unemployment rate, shifts which are one-fourth to one-third as large as what the average city experienced during the Great Recession. As a final interpretation, one can observe that the average state sees its college educated population grow by about

²³In Appendix Table C3, we consider all college graduates regardless of employment status, as well as other dimensions of heterogeneity.

1.2% per year. Thus, four years under a Republican governor sets a state’s human capital accumulation back by almost one year.

Panels B and C of Table 1 replicate Panel A, but separately for Democrat-to-Republican (D-to-R) transitions in Panel B and Republican-to-Democrat (R-to-D) transitions in Panel C. The signs of coefficients are systematically symmetric results: switching to a Republican reduces in-migration while switching to a Democrat increases it. However, it is notable that the D-to-R magnitudes tend to be larger than the R-to-D estimates (though they are not significantly different), and that the D-to-R coefficients are more robustly significant. Thus, our results seem to be particularly driven by switches from Democratic to Republican governors, although we cannot reject the null of symmetric effects.

In Figures 2 and 3 we plot the normalized levels of migration in treatment and control states across event-time, as well as conventional event study coefficient plots corresponding to specification (1). Figures 2 and 3 also speak to a very natural question: How have these effects changed over time?

3.3.1 Effects over time

To answer understand changes over time, we split our 15 year sample (elections 2002-2016) into three five-year periods (2002-2006, 2007-2011, 2012-2016), although the pre- and post-treatment windows for these periods overlap one another.²⁴ Table 2 shows corresponding coefficient estimates.

[Figure 2 about here.]

[Figure 3 about here.]

For D-to-R transitions, we find large and marginally significant in-migration declines during both the 2002-2006 election period and the 2012-2016 election period, with smaller and non-significant declines during the intermediate period. Looking at the difference-in-difference figures, during these two periods, we see fairly flat pre-trends followed by noticeable and near-instantaneous declines in migration. For the intermediate period, the pre-trends are noisier, and it is difficult to say whether migration responses are truly different or not.

For R-to-D transitions, on the other hand, it is *only* during this intermediate period where we find that in-migration significantly increased in response to a Democratic governor, although for the earliest period (elections 2002-2006) we find a similar magnitude that is

²⁴For instance, migration in 2009 is post-treatment for elections that occurred in 2006 (which falls in the first period) and those that occurred in 2007 (which falls in the second).

less precisely estimated.²⁵ In the latest period, however, we find no migration increase in response to a Democratic governor. In fact, when using the full sample, we actually find a significant *decrease* in in-migration in response to a Democratic governor. As we show in Appendix Figure C3, this is entirely due to Louisiana, a state that flipped from a Republican to a Democratic governor after the 2015 election. Louisiana had a rapid increasing in in-migration before that election, and so its inclusion leads to considerable pre-trends that make it difficult to draw any causal conclusions. Excluding Louisiana leads to fairly flat pre-trends (which are presented in Panels g and h of Figure 3), and a precisely estimated zero migration response.

[Table 2 about here.]

In summary, we find consistent evidence that switching to a Republican governor reduces the in-migration of college-educated workers (without affecting their out-migration or the in-migration of non-college workers), while switching to a Democratic governor increases it. But these effects have not been constant over time; different periods have seen different migration responses. A natural question is why? We return to this below, but first address the concern that our results reflect time-varying unobservables instead of the causal effect of the governor.

3.3.2 Robustness: Regression discontinuity estimates

In studying the effects of politicians, Regression Discontinuity Design (RDD) estimates have long been considered the gold standard. These specifications yield valid inference as long as unobservable confounds (those which influence both election results and the outcomes of interest) are continuous in a narrow bandwidth around the 50% threshold that determines winner-take-all election outcomes. This is generally seen as the most conservative identification assumption in studying election outcomes. However, the clear tradeoff is that these specifications are typically underpowered because focusing only on very close elections uses only a small share of the total variation (Appendix Figure C1 shows exactly which state-years drive identification in each specification).

Figure 4 presents a binned scatterplot showing college graduates' in-migration rates as a function of the Republican vote share in the most recent gubernatorial election. We see a large and notable decline in the in-migration rate of roughly 1 percentage point. That is, during the four years after a Democratic governor's very close win, roughly 4% of college

²⁵As noted above, the ACS was a considerably smaller sample size from 2000-2004, so it is unsurprising that this early period consistently yields larger standard errors.

graduates are recent migrants into the state. During the four years after a Republican’s narrow win, on the other hand, only about 3% are. This aggregates to a 4% decline in the college population after four years, which is larger than our difference-in-difference estimates. One possible explanation is that many voters have moderate political preferences (Fowler et al., 2023), so moderate governors often win by large margins (which contribute to identification in the DiD), while relatively extreme governors are often only able to win by narrow margins (which drive identification in the RDD). In this case, the RDD results are disproportionately identified by the most extreme governors, and it is plausible that graduates are particularly responsive to these.²⁶

[Figure 4 about here.]

In Appendix Figure C9 and Table C5, we show that these results are statistically significant and robust to a range of choices about bandwidths, kernels, functional forms, and controls. Nonetheless, as mentioned above, RDD estimates use only a small share of the available data, and so they will always be subject to concerns about the influence of a small number of elections. Thus, we see the RDD as a robustness check and treat the DiD estimates as our primary specification.

In considering threats to identification in the DiD design, it is important to mention that lots of variation in election outcomes is due to idiosyncratic shocks likely to be orthogonal to migration incentives. To illustrate this, consider three elections when the partisanship of the governor changed for seemingly exogenous reasons, even though the elections were not necessarily close.

In 2016, Missouri replaced its term-limited Democratic governor, Jay Nixon, with a Republican (who won by 6 points). Many argued at the time that poor Democratic Party performance was due to the 2014 murder of Michael Brown at the hands of police in Ferguson and mistakes made by party leaders (including Nixon and the Democrats’ 2016 gubernatorial candidate, who had been the Attorney General at the time) in responding to the protests (Ortiz, 2016).

In 2012, North Carolina elected a Republican governor for the first time since 1988 (who won by 11 points). The Democratic candidate was the incumbent Lieutenant Governor, and commentary at the time focused on the challenges he faced campaigning at the same time that several of the Governor’s staff were facing obstruction of justice charges (Catanese, 2012) and while the Democratic Party’s senior staff was embroiled in sexual harassment proceedings (Bass, 2012).

²⁶See George (2024) for evidence that the lowest quality leaders are disproportionately selected into the RDD sample.

In 2007, Kentucky (one of the most conservative states in the country) replaced its incumbent Republican governor with a Democrat (who won by 18 points) largely because 13 members of his administration had been indicted in a controversial patronage hiring scandal (The Associated Press, 2007). These events appear to have driven non-close election outcomes but were likely exogenous to the sorts of economic shocks and trends that one might worry have a direct effect on college graduates’ migration incentives.

3.4 Mechanisms for politically responsive migration

Above, we showed robust evidence that a Republican governor causally lowers immigration of college graduates, relative to having a Democratic governor. Is this because college graduates tend to dislike conservative policies, consistent with our descriptive evidence above and the literature on education polarization (Kuziemko et al., 2026)? Is it because of broader opposition to the Republican Party and perceptions of what it symbolizes (Nelson and Witko, 2022)? Is it because election outcomes signal things about the voters, and left-leaning college graduates would prefer to live near like-minded people (Koşar et al., 2022)? Or is it due to governors’ effects on the overall economy, and thereby reflects labor demand instead of labor supply? Here, we attempt to separate these explanations.

3.4.1 Policy considerations vs. party labels

We begin by attempting to separate specific policy preferences from broader views of the Republican Party. Above, we showed descriptive evidence that college graduates, relative to non-college respondents, prefer left-leaning social and economic policies. At the same time, however, college graduates’ broad, overall, feeling-based perception of the Republican Party has worsened substantially, particularly among the growing share of college graduates who identify as Democrats and Independents (see Appendix Figure B7). There remains debate in the literature about the extent to which these attitudinal shifts in affective polarization actually reflect true policy disagreements (Orr, Fowler, and Huber, 2023). Thus, it is possible that our results reflect general animosity towards the Republican Party as a whole, rather than aversion to the policies governors actually implement.

To separate these two explanations, we ask whether migration responses depend on the *size* of the policy shift.²⁷ If graduates are policy-motivated, then larger shifts in policy should see larger migration responses. If, on the other hand, voters are put off by “the Republican label” and a symbolic associations with the party and what it stands for, then migration responses should not depend on how much actual policy changes. To summarize

²⁷We thank an anonymous referee for this suggestion.

a large set of policies, we use the policy indices developed by Caughey and Warshaw (2016) and updated through 2024 by the Dynamic Democracy project. These indices summarize roughly 200 policies that vary across states and over time. Caughey and Warshaw (2016) aggregate policies into social and economic policy, and since we follow their classification, this perfectly matches the way we group issues in our descriptive analysis.²⁸

In Table 3, we test whether flips have different migration effects depending on the size of the resulting policy shift. Separately for D-to-R and R-to-D flips, and separately for social and economic policies, we group flips based on whether there was an above-median or a below-median change in policies during the four years following the election. This allows us to study whether migration responses are larger or smaller when the actual policy shifts that followed the election were larger or smaller.²⁹

Panel A splits Democrat-to-Republican flips based on whether the subsequent shift in economic policy was larger or smaller than the median shift. For events with below-median shifts, we estimate a zero effect on economic policy over the subsequent four years.³⁰ In these cases, we see a large 16% decline in the college in-migration rate (18 log points; $p < .05$). In other cases, however, we estimate a large $.36\sigma$ rightward shift in economic policies. In these cases, we estimate a smaller 8 log point decline in the college in-migration rate. While this point estimate is still negative, it is not significantly different from zero ($p = .119$), though it is also not significantly different from the point estimate we obtained for below-median shifts.

[Table 3 about here.]

Notwithstanding the wide confidence intervals from splitting the sample in this way, we see this as evidence that strong rightward economic policy shifts are able to offset about half of the aversion to Republican governors that we see in the in-migration responses. This is consistent with existing evidence on migration responses to tax cuts (Moretti and Wilson, 2017), but somewhat inconsistent with the stated preference descriptive evidence we presented above.

²⁸The economic policies that contribute to the index include union, labor rights, and collective bargaining policies; minimum and prevailing wage policies; rent control policies; generosity and eligibility for welfare and other public assistance programs; tax exemptions and tax rates; and many more. The social policies that contribute to the index include abortion laws and regulations; laws covering LGBTQ marriage, adoption, discrimination, and participation in sports; gun control policies; immigration rules like language policies and access to driver's licenses; and many more.

²⁹Splitting the sample based on the estimated change in policies invalidates our confidence intervals for the *policy* treatment effects. However, our main interest is in the confidence intervals for the *migration* treatment effects, and our sample split does not mechanically depend on the estimated migration response.

³⁰Coefficient plots corresponding to all of these specifications are included in Appendix Figures C4-C7.

We find a similar result for Republican-to-Democrat transitions in Panel C. When we split based on the size of the economic policy shift, we find that large leftward swings in economic policy do not bring in new college graduates. Democrats who do not meaningfully shift economic policy, on the other hand, tend to see substantially larger increases in in-migration, although neither of these migration responses is significantly different from the other or from zero. Thus, we see this as suggestive evidence that college graduates do tend to respond to economic policy shifts, but unlike their stated preferences, their revealed preference migration behavior suggests that they prefer the bundle of tax cuts and pro-market reforms generally enacted by Republican governors.

When we split flips based on the size of the *social* policy shift, on the other hand, we see a very different pattern. Looking at Panels B and D, for both D-to-R and R-to-D flips, we find very similar in-migration responses regardless of how much social policy change there is. Whether Republicans preside over large or small rightward shifts in social policy, we estimate a 13-14 log point decline in the college in-migration rate.

One issue with the above analysis is that we are splitting the sample based on post-treatment changes in policy. As an alternative, in Appendix Table C4 we estimate the effects of governors separately by whether their party controlled the state legislature at the time they took office.³¹ When the governor takes office with an aligned legislature, we find robust evidence that *actual policy changes* are larger, but we find no evidence that *migration responses* are larger.³²

These results suggest that party labels matter more than actual policy preferences. College graduates may be responding to the “Republican brand” and broader associations with the Republican Party. Particularly in light of the nationalization of US politics (Hopkins, 2018), it is plausible that many voters think of all Republicans as one, despite meaningful differences in their policy choices. This is consistent with the overall worsening of college-educated Democrats’ attitudes towards the Republican Party as a whole (see Appendix Figure B7).

Of course, it is also possible that graduates respond to things our policy indices do not capture and which do not require legislation. This could include things best thought of as political theatrics, such as flying asylum seekers from Texas to Martha’s Vineyard (Florida Governor Ron DeSantis) or declaring that “immigration without assimilation is invasion”

³¹The disadvantage of this approach, however, is that governors often change policy without legislation. Moreover, heterogeneity by control of the state legislature does not let us distinguish between social policy changes and economic policy changes.

³²Across Table C4, in almost all cases, we find a significant difference in the effects on our policy indices, but we never find a significant difference in the effects on migration. Moreover, the magnitudes of the migration response are very similar regardless of control of the state legislature.

(Louisiana Governor Bobby Jindal). However, it could also reflect things that would traditionally be thought of as a form of policy-making, such as using the National Guard for aggressive border enforcement (Texas Governor Greg Abbott) or using administrative rules to undermine the implementation of public programs (Herd and Moynihan, 2025).

3.4.2 Avoiding Republican politicians vs. avoiding Republican voters

Particularly if our results reflect party labels or symbolism rather than actual policy, then it is important to distinction between aversion to Republican *politicians* and to Republican *voters*. Voters value living near like-minded neighbors (Brown et al., 2022; Koşar et al., 2022; Mummolo and Nall, 2017), so they may interpret Republican election victories as a signal that the state’s voters are more Republican than previously realized.

To test for this, Figure 5 compares migration effects in states that flipped from voting for Barack Obama in 2012 to Donald Trump in 2016 with states that switched from Democratic to Republican governors during roughly the same period (flips from 2014-2017). While we find large and pronounced declines in in-migration after switching to Republican governors, we find no visual or statistical evidence that migration declined after states switched their presidential choice, even though such switches were very salient. This is consistent with our argument that college graduates care about the identity of the politician in charge of the state (since Donald Trump was in charge of *all* states), but it is difficult to explain as a result of them drawing inferences about the state’s voters.

[Figure 5 about here.]

3.4.3 Labor supply vs. labor demand

Thus far, we have considered various reasons why college-educated workers’ political attitudes might shape their decision to move into states governed by Republicans or Democrats. These are labor supply explanations, but it is also possible that Republican and Democratic governors have economic effects that change the set of labor market opportunities available, and that college graduates’ migration behavior is simply responding to these opportunities. This would be a labor demand explanation.

We find this *ex ante* unlikely for two reasons. First, Republicans generally enact pro-market policies that existing evidence suggests would benefit college graduates (who tend to be higher income), at least in the short run. Moreover, when we estimate separate effects by the size of the economic policy shift, we find the largest in-migration declines were concentrated among the Republican governors who changed economic policy the least. Second, we estimate instantaneous migration effects which are stable over time. This is

difficult to explain through labor demand effects (which would take time to materialize), but easy to explain through labor supply effects (especially if they reflect symbolic views on party labels instead of actual policy changes) since the Republican governor becomes observable to job-seekers as soon as he/she is elected.

Nonetheless, in columns (3) and (6) of Table 3 we estimate the effects of governors on the unemployment rate for college graduates, separately depending on the size of the shift in social and economic policies. For both D-to-R and R-to-D transitions, and regardless of the size of shifts in economic or social policies, we estimate no effects on unemployment. These are fairly precisely estimated null effects, with confidence intervals usually ruling out increases or decreases of 0.4 percentage points or larger (off of a base of around 3.1%). For instance, while Republican governors who do not shift economic policies tend to see larger in-migration declines than Republican governors who push economic policies to the right, we see no difference in how college unemployment changes under these different groups of Republicans.³³ In interpreting the magnitudes of the null effects, it is important to keep in mind that migration responses to shifts in economic conditions tend to be small (Jia et al., 2023; Ransom, 2026), and so large changes in economic activity would be needed to explain the migration responses we estimate. As noted above, using the magnitudes estimated by Monras (2020), a 0.8 percentage point change would be required to generate the migration responses we estimate, and our confidence intervals rule this out.

4 Structural analysis

4.1 Model overview

Our model builds on a recent wave of migration models in which workers choose where to live based on the general amenities of a given state (e.g., Albouy 2016) and their stochastic draw of employment opportunities in that state (e.g., Bryan and Morten 2019). Similar to Monras (2020), workers receive an exogenous shock that affects *whether* they move, but *where* to move is endogenously determined by general state-level amenities, their idiosyncratic productivity draw, and equilibrium wages in each state. Within each state, college and non-college labor are combined in a constant elasticity of substitution (CES) production function, which determines labor demand. Our elasticity of substitution (taken from the existing literature) implies that an exogenous decline in the number of college graduates in

³³An alternative approach would consider job openings and the job-filling rate. Interestingly, this shows an immediate decline in job filling without any short-term change in job openings, exactly as a labor supply response would predict, although these are simple state-level aggregates without any additional detail to attempt to separate between college-like and non-college jobs.

a state pushes up their relative wage and pushes down the real wage of non-college workers.

Our key innovation is the addition of a “political amenity” that depends on the partisanship of the governor and differs between college and non-college workers. We calibrate this political amenity to match the reduced form migration responses we estimate above. Additional parameters are mainly calibrated following the approaches laid out by Hsieh et al. (2019) and Bryan and Morten (2019). We discuss these in detail in Appendix D.

4.2 Counterfactuals

The main purpose of the model is to understand the general equilibrium effects of plausible changes in the control of state government. In the Appendix, we simulate the effects of a flip in each state separately, each time holding fixed the partisanship of governors in all other states. The *proportional* effect of a governor on the migration *rate* is roughly constant across these counterfactuals, and mechanically matches our empirical estimates because this is a targeted moment.³⁴ However, empirically, states differ in how important interstate migration is for generating their stock of college graduates. For instance, over 5% of Nevada’s college-educated workers have moved within the last year alone. This is three times the rate in Michigan, where far more college graduates come from the state itself.³⁵ Thus, constant proportional effects on migration *rates* will have much larger effects on the *stock* of college-educated workers in Nevada than it will in Michigan.

For this reason, it is important that our counterfactuals reflect realistic possibilities. For instance, because Wyoming relies so heavily on migration for its college-educated workforce, our model implies that a Democratic governor would substantially increase the college population there. However, the most recent Democratic candidate for governor received only 18% of the vote, so these large effects are likely irrelevant.

To choose counterfactuals, we note that voters tend to make state-level political choices based on national-level debates, issues, and preferences, which Hopkins (2018) calls the “nationalization” of state politics. Figure 6 shows a very high correlation between the Democratic Party vote share from the 2020 Presidential election and the most recent gubernatorial election. However, the arbitrariness inherent in winner-take-all elections means

³⁴Our model includes various non-linearities and feedback mechanisms, so it is possible for a fixed preference shock to have very different effects on equilibrium migration decisions. Quantitatively, these turn out not to generate much variation in the proportional effect of partisanship on migration rates.

³⁵In general, states with strong public university systems tend to rely less on migration to generate a college-educated workforce (e.g., California, Michigan, Minnesota, Ohio, Texas, Wisconsin). They therefore have smaller model-implied effects of partisanship. One interesting policy implication is that Republican governors could blunt the effects we document by investing in universities to produce “home grown” college-educated workers. In practice, however, Republican governors and legislatures tend to reduce university funding.

that outcomes differ, and since many elections were close, modest shifts in public sentiment could easily sway the outcomes in many of these states.

[Figure 6 about here.]

Figure 6 shows the four counterfactuals we consider. First, a “weak red wave” could plausibly flip the three states won by Donald Trump but governed by a Democrat as of January 2024 (top left quadrant). Second, a “strong red wave” could also flip an additional four states with Democratic governors that Biden won narrowly (top right). Similarly, we consider a “weak blue wave” that flips the five states won by Biden but represented by a Republican governor (bottom right) and a “strong blue wave” that also flips four battleground states narrowly lost by Biden and represented by a Republican governor (bottom left).

4.3 General equilibrium effects

Our first question is how these plausible “waves” of partisan shifts might affect equilibrium economic activity and inequality. Panel A of Table 4 summarizes our model’s prediction for how the college-educated workforce would change in the states experiencing a gubernatorial flip. For all four counterfactuals, the average state experiencing a flip is predicted to have a 1.3% to 2.2% change in its college population. These magnitudes are similar to our reduced form estimates from above because we used those estimates in our calibration. However, as noted above, primarily because states differ in the extent to which they rely on migrants to make up their college-educated workforce, there is considerable heterogeneity in these effects. In some states, the change in the stock of college-educate workforce can be as small as 0.5 or 0.6%, while other states can see a change as large as 3.8%.

[Table 4 about here.]

In Panel B, we then ask what these effects imply for GDP per worker.³⁶ Across estimates, the elasticity of GDP per worker with respect to the size of the college-educated workforce is about .9, meaning that the 1.8% average decline in the college workforce we simulate in our “small red wave” counterfactual implies a 1.6% average decline in GDP per worker. These effects are modest but meaningful. The median state saw average annual GDP growth of 1.25% during the 2005-2019 period, so our estimated effects roughly correspond to one year of lost growth. Again, however, the heterogeneity is important, with some states seeing effects as small as a 0.5% increase under Democratic governance, while others see effects as large as a 2.6% increase.

³⁶Our model does not include non-workers, so GDP per worker is equivalent to GDP per capita.

We next turn to inequality, both because it is important in its own right and because it affects political dynamics, where rising inequality often fuels support for controversial populist candidates (Guriev and Papaioannou, 2022). Intuitively, the inequality implications of our results are straightforward. If a Republican governor makes college graduates more scarce, then decreasing the relative supply of college graduates should increase their relative price, and college/non-college inequality should rise (with the converse effects for a Democratic governor).

In Panel C, we present these effects by summarizing the change in college/non-college gaps in the log “wages,” by which we mean earnings per unit of productivity. Consistent with the supply and demand logic above, the counterfactuals that increase the relative supply of college graduates reduce the relative price of their labor, and vice versa for the counterfactuals that decrease their relative supply.³⁷

However, this simple intuition could be misleading because changes in the migration incentives also change the composition of college graduates. Workers are not randomly sorted across states, but choose their state after realizing idiosyncratic productivity draws for all states. This selection means that migrants will tend to have higher productivity than non-migrants because their idiosyncratic realization must have been sufficient to draw them to this state over all other states. Thus, while increasing the relative supply of college graduates through more in-migration will reduce their wage *per unit of productivity* (a price effect), it will also increase the average productivity of the graduates in the state (a composition effect). Thus, college/non-college inequality in *earnings* (which depends on both productivity and wages per unit of productivity) could go up or down.

In Panel D, we find that across the counterfactuals we consider, the composition effect dominates the price effect, and switching to a Democratic governor simultaneously increases the number of college graduates *and* their relative earnings, while switching to a Republican reduces both.³⁸ These magnitudes are modest on average, ranging from 0.4 to 0.6 log points, but can be large. In the largest case, our “small blue wave” counterfactual implies that a Democratic governor would increase the college/non-college earnings gap by 1.5 log points in Vermont. How large is this? Acemoglu and Autor (2011) estimate that from 1980-

³⁷As is common in the literature, we assume that the elasticity of substitution in production is greater than one, which implies that decreasing the supply of college graduates also reduces the real earnings of non-college workers (as in Moretti (2004) and Giannone (2022)). In Appendix Table D2, we show that across the counterfactuals, about 40% of the increase in inequality arises from the change in non-college wages with the other 60% arising from the change in college wages.

³⁸This is not always true. Appendix Figure D2 shows the inequality implications for the 50 single-state counterfactuals we simulated above. In two of the counterfactuals that imply the largest increase in the number of college graduates, the price effect dominates the composition effect, and college/non-college inequality falls under a Democratic governor.

2008—widely seen as a period of dramatically increasing inequality and college premia—the college/non-college earnings gap increased by roughly 1 log point per year. Thus, an additional 1.5 log point increase over four years, on top of any secular trends in inequality, would likely be substantial enough to ignite some resentment among non-college voters, particularly when combined with a visible increase in the number of college graduates.

These results are interesting partly because including endogenous selection overturns the standard intuition about the effects of a supply shock. This is relevant because many of the most liberal and progressive areas in the country (e.g., California, Seattle, New York) have seen rising shares of college graduates at the same time as rising inequality. In a simple model of homogeneous college graduates, increasing the labor supply of one type of worker cannot increase the relative earnings of those same workers. Thus, these trends have largely been interpreted as evidence for demand-side forces (e.g., agglomeration effects) rather than labor supply forces. In our model, these trends can emerge from inequality driven by college graduates moving into the state with particularly high-paying job offers, even without any industry-specific or demand-side forces.³⁹

The composition effect that increases inequality under Democratic governors is ultimately determined by how positively selected in-migrating college graduates are relative to home-state college graduates. In raw data, it is generally true that the migrant college graduates in a state have higher average wages than home-state college graduates, which is evidence that these composition effects are real. However, it is unclear whether they would become more or less quantitatively important in a richer model that includes multiple sectors that differ in wages, productivity, and the share of college graduates who migrated from another state. We leave this exploration to future work.

4.4 Cross-state spillover effects

Our model also lets us study how changes in the governor in one state affect other states. All states are linked through the migration decision since workers are deciding between different states in choosing where to live. Thus, changes in the incentives to live in one state might push those workers towards another state or pull workers away from that state.

We map some of these spillovers in Figure 7. In panel (a), we present the direct effects

³⁹The best reduced form evidence on how the supply of college graduates affects inequality likely comes from Fortin (2006). Consistent with the standard intuition about the effects of a supply shock, she finds that an increase in college graduates reduces college/non-college earnings inequality. However, Fortin’s identification comes from state-level policy changes in university funding and access, which implies that the marginal college graduates driving identification in her analysis were primarily from the state itself and not cross-state migrants. This means that Fortin’s results would not capture the composition channel that we discuss.

on the college-educated workforce for the nine states we consider electing Democratic governors in a strong blue wave. These effects vary, but it is worth noting that we estimate reasonably strong 1–2% increases in the three Southeastern states that we consider plausible flips (Florida, Georgia, and Virginia) and that these are large states.⁴⁰

[Figure 7 about here.]

In panel (b), we plot the indirect spillover effects on the college-educated workforce in states that *do not* change their governor. Most states show small effects, but not always. For instance, in South Carolina (a relatively small state bordering Georgia and having high excess migration with Virginia and Florida), we estimate a .7% decline in the college graduate population. Thus, while spillovers are typically small, when several large states in the same area all flip at once, it can cause a substantial ripple effect through the region.⁴¹

Another implication of these cross-state spillovers is that flips among “nearby” states (in terms of migration costs) can cancel one another out. To illustrate this, consider the flips in Figure 8. In panel (a), we present a set of flips in which each state within a Census division has the same gubernatorial partisanship. Because nearby and similar states all share the same gubernatorial partisanship, the effects of a flip should be diminished; college graduates who wish to avoid a Republican governor cannot easily choose a different nearby state as a substitute, and so will be less responsive to governor partisanship. To quantify the importance of this, we can compare these region-wide flips to a set of flips in which each Census division becomes roughly 50/50 Democrat/Republican, shown in panel (c).

[Figure 8 about here.]

We find that the changes in GDP per worker, college in-migration rates, and total college population from *region-wide* flips are about 1/6 smaller than the flips that keep every Census division balanced. One practical implication of this is that increasing regional dominance of one specific party—a salient geographic pattern of recent political polarization—will blunt the type of politically responsive migration that we document here, but only modestly.

5 Conclusion

College education is increasingly the fulcrum of political disagreement in the United States. On both social and economic issues, college graduates report views that are well to

⁴⁰Our data on governor partisanship was updated as of 2024, but as of writing, Virginia has a Democratic governor; Abigail Spanberger was elected in 2025.

⁴¹In Appendix Figure D3, we perform the same exercise to calculate the spillovers induced by a strong red wave. Because these states are not concentrated in a single area, they are not amplified into a large regional effect.

the left of non-college voters. As a result, Republican governance deters college graduates from moving to a state, driving down total human capital in the state and reducing economic growth. To the extent that Republican policies themselves are pro-growth, these migration responses offset some of those gains.

What might conservative politicians do to reduce this growth penalty? One option is to increase support for universities in order to increase the stock of “home-grown” college graduates (as in Fortin (2006) or Kennan (2015)) who would presumably show greater sympathies for the policies preferred by local voters. Instead, however, Republican governors and legislatures tend to cut funding and support for universities, exacerbating the local supply shortages induced by lower in-migration.

Alternatively, conservative politicians might aim to win back the college-educated electorate. Thus far, this does not appear to be a priority. Influential party leaders tend to use “college-educated,” “liberal,” and “elites” as almost interchangeable terms to refer to their enemies.⁴² If anything, the regional economic inequality exacerbated by education polarization and migration appears to be a rhetorical victory for conservative politicians, who frequently emphasize these gaps to win votes. Evidence of rising inequality in Democratic-governed states, which our model can explain through labor supply composition effects, creates further opportunities to criticize Democratic Party redistributive policies as ineffective.

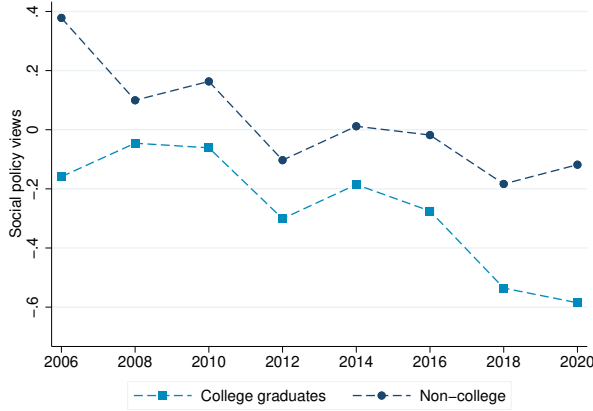
Finally, an open question is how the COVID-19 pandemic and its aftermath will interact with education polarization. We intentionally end our analysis in 2019 because we see migration data from 2020-2022 as unreliable. However, a salient feature of the post-pandemic United States is that working remotely will be common, particularly for better-educated workers (Barrero, Bloom, and Davis, 2026). This is likely to exacerbate our findings since it allows college-educated workers to live in the states they prefer without having to switch to a local employer, and thus without general equilibrium wage pressures to offset political migration incentives.

The implications for Republican-led states could be dramatic. Conservative states tend to raise more revenue through sales taxes and typically have low income and corporate tax rates, while liberal states rely more on income taxation for revenue. Thus, even if the firm locates in a conservative state, the state will receive little revenue from taxing the company and little revenue from sales taxes since many of the state’s workers do not actually live and consume within the state. Liberal states reliant on income taxes, on the other hand, will

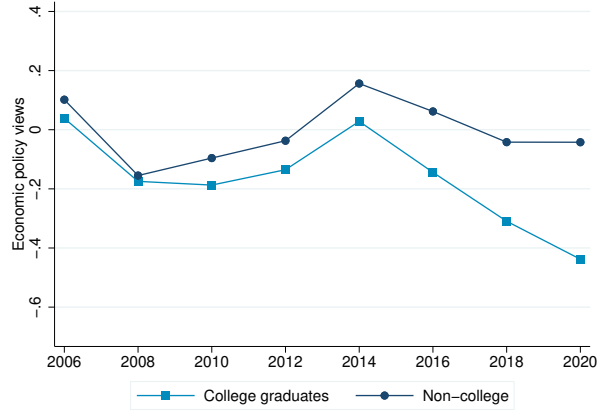
⁴²For instance, Vice President JD Vance has said, “If any of us wants to do the things that we want to do for our country, and for the people who live in it, we have to honestly and aggressively attack the universities in this country,” which he describes as “very hostile” to conservative aims and dedicated to “deceit and lies, not the truth” (Knott, 2024).

essentially benefit from a revenue windfall as more high-earning workers relocate there than otherwise would, given the relatively high corporate tax rates disincentivizing firms from locating there. With this in mind, future research should account for education polarization (and the possibility that it continues to grow) when assessing the implications of post-pandemic work arrangements for heterogeneous growth in different regions of the country.

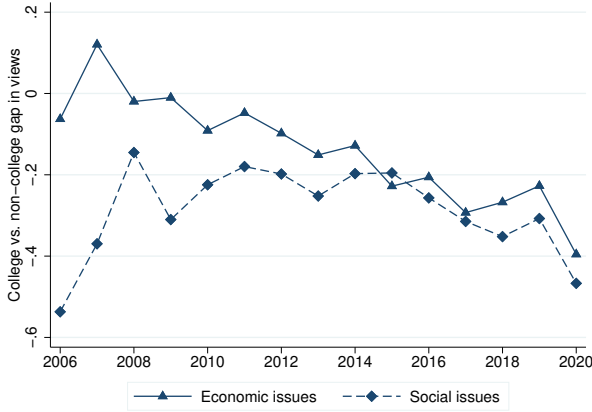
Figure 1: Differences in policy views by education



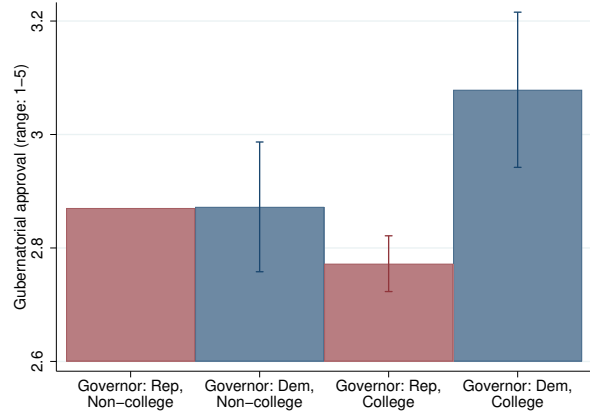
(a) Social policy views



(b) Economic policy views



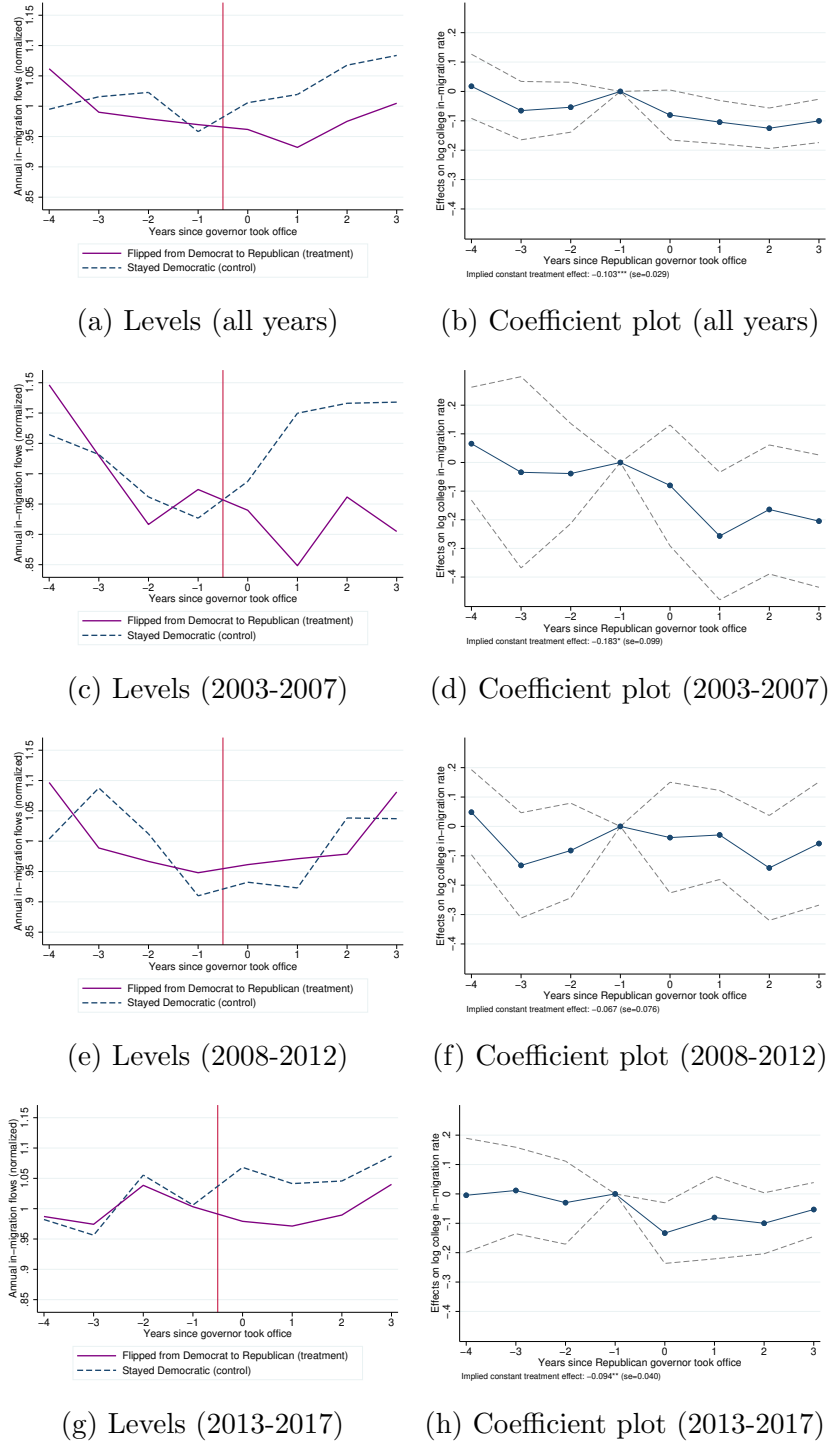
(c) College vs. Non-college gaps



(d) Gubernatorial approval

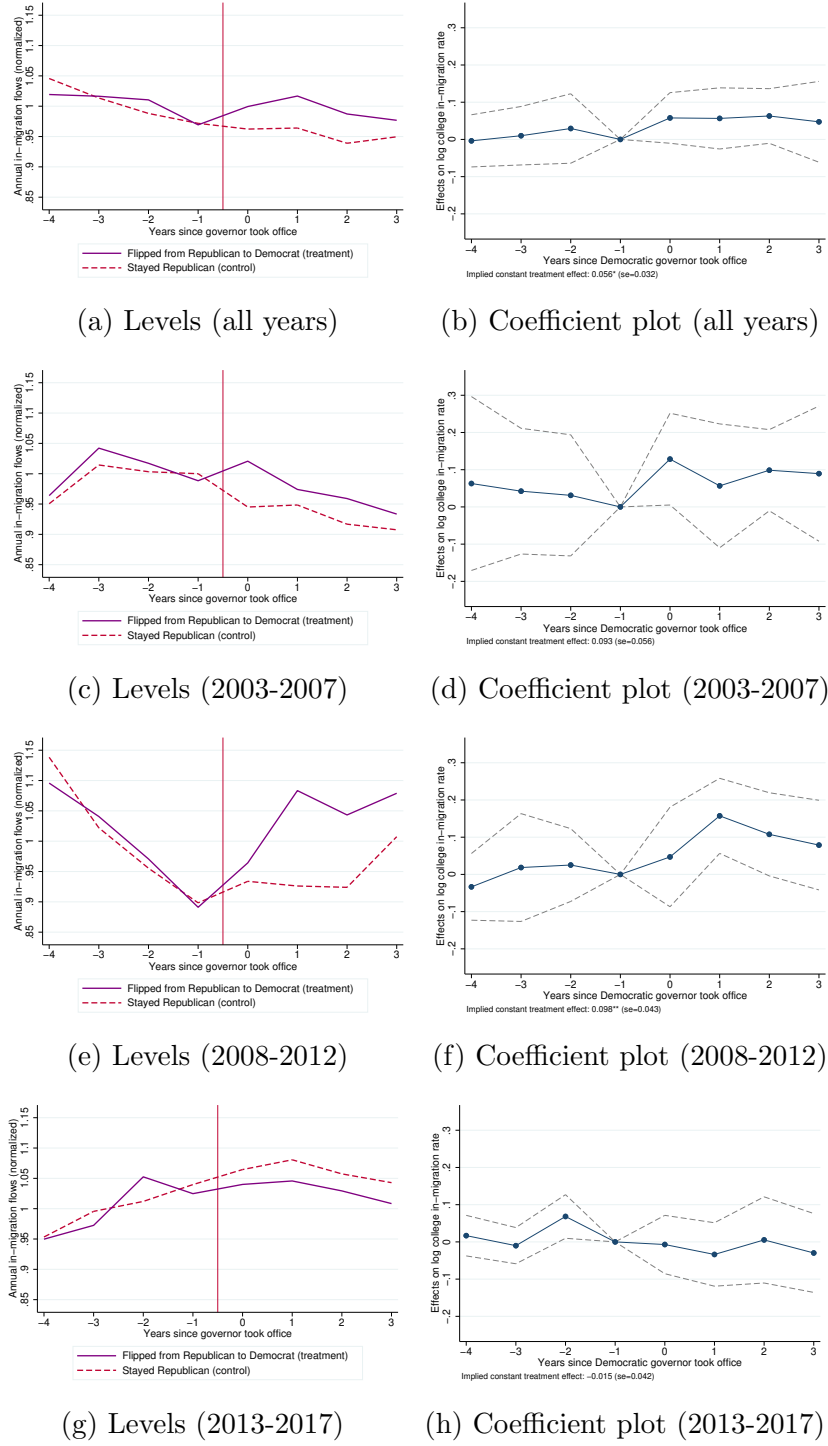
Notes: All calculations based on the Cooperative Election Study (CES). Policy view indices are based on Item Response Theory estimates using the approach developed by Imai et al. (2016). Policy questions are divided into social and economic policy issues using the scheme developed by Caughey and Warshaw (2018). Indices are normalized to have mean of zero and a unit standard deviation across all respondents across all years, with more positive values indicating more right-leaning views. Trends in panels (a)-(c) are based on respondents aged 26-45 where interstate migration is concentrated (see Appendix Figures B1 and B4). All calculations use sample weights. Panels (a) and (b) keep only even numbered years (when the sample size is larger) to smooth out idiosyncratic year-to-year fluctuations in levels. Panel (c) uses all years because college/non-college gaps do not exhibit idiosyncratic year-to-year fluctuations. Panel (d) is based on coefficients from a regression that includes state and year fixed effects, and presents point estimates and 95% confidence intervals (where standard errors are clustered at the state-level).

Figure 2: College graduates' in-migration responses Democrat-to-Republican transitions



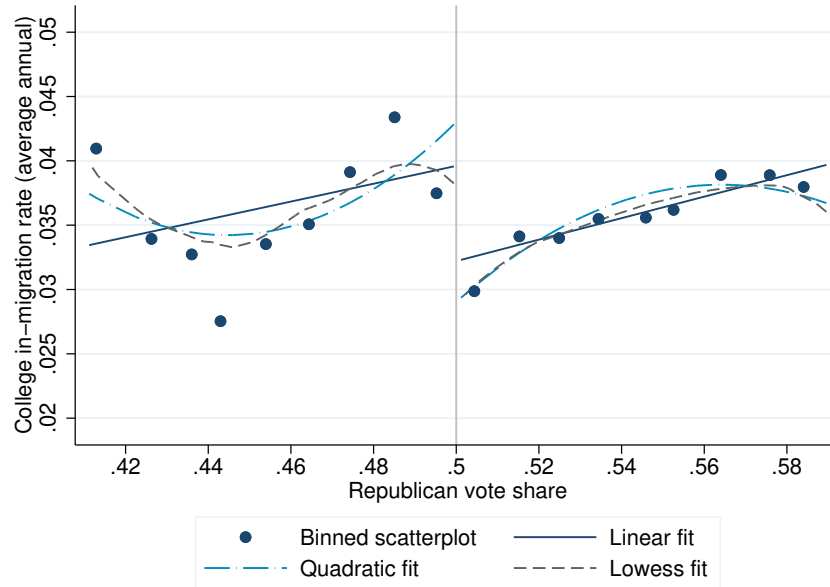
Notes: Panels (a), (c), (e), and (g) present levels of college graduate in-migration, normalized by pre-treatment levels, for treatment and control group states. Panels (b), (d), (f), and (h) present coefficient estimates and 95% confidence intervals corresponding to estimating equation (1). Panels (c)-(h) divide the sample based on the year in which the new governor took office; the pre- and post-treatment windows in these different subsamples can overlap one another. In each panel, the sample is based on college graduates aged 26+ employed in the private sector (see column (3) of Table 1).

Figure 3: College graduates' in-migration responses Republican-to-Democrat transitions



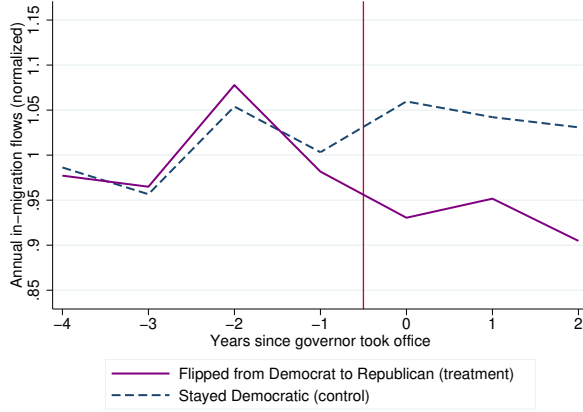
Notes: Panels (a), (c), (e), and (g) present levels of college graduate in-migration, normalized by pre-treatment levels, for treatment and control group states. Panels (b), (d), (f), and (h) present coefficient estimates and 95% confidence intervals corresponding to estimating equation (1). Panels (c)-(h) divide the sample based on the year in which the new governor took office; the pre- and post-treatment windows in these different subsamples can overlap one another. In each panel, the sample is based on college graduates aged 26+ employed in the private sector (see column (3) of Table 1).

Figure 4: Regression discontinuity estimates of partisanship and college in-migration

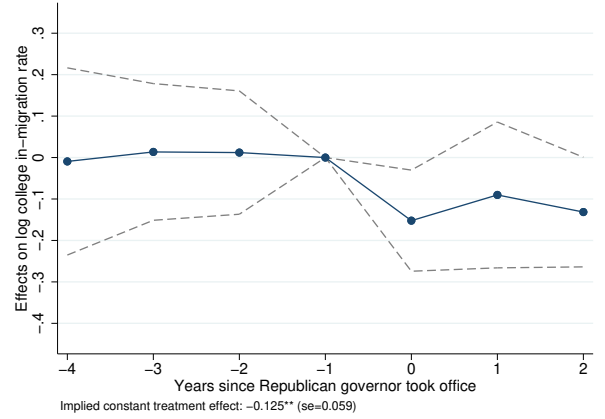


Notes: Figure presents a binned scatterplot showing college graduates' average in-migration rates (the fraction of college graduates who lived in a different state the prior year) during the four post-election years as a function of the Republican vote share in the election. The sample consists of college graduates aged 26+ employed in the private sector and elections 2000-2016.

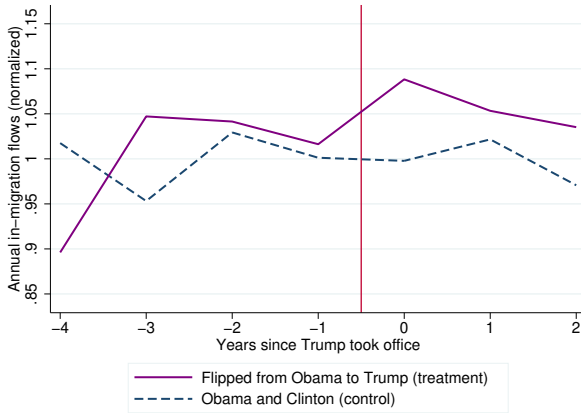
Figure 5: Migration responses to Gubernatorial vs. Presidential flips



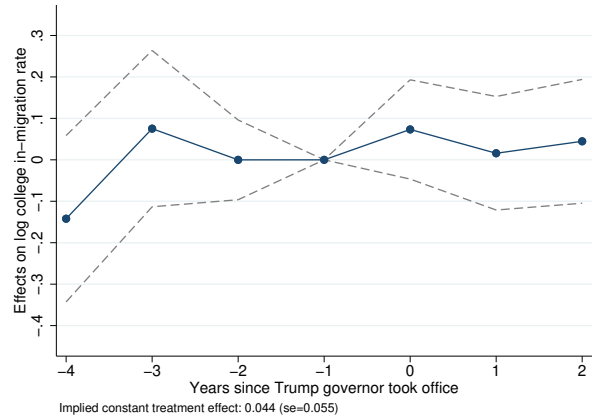
(a) D-to-R Governor flips 2014-2017 (DiD)



(b) D-to-R Governor flips 2014-2017 (coefficients)



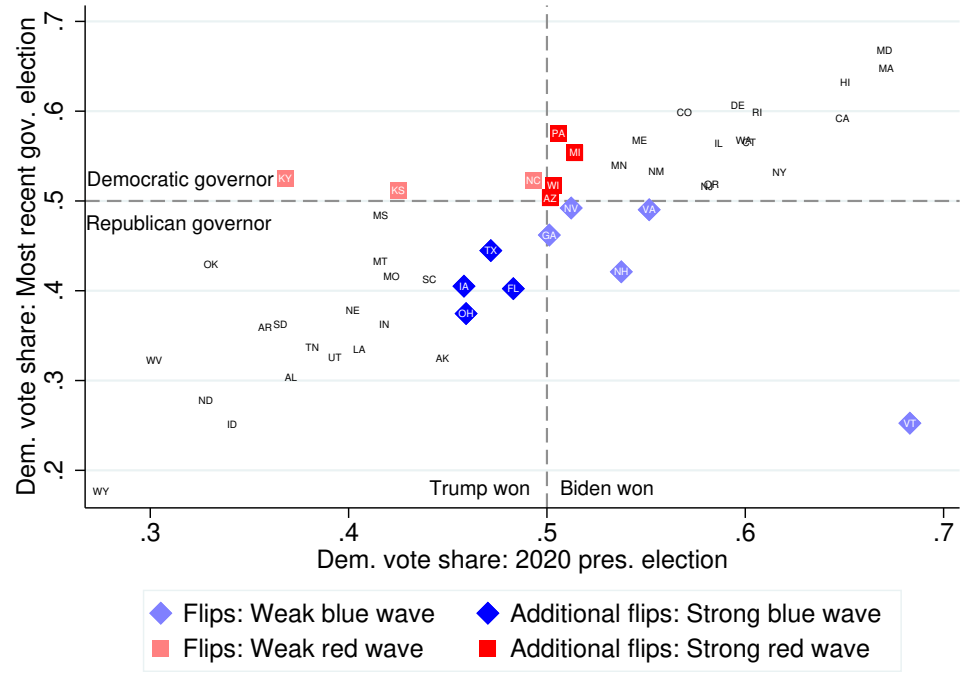
(c) Obama-Trump voting flips (DiD)



(d) Obama-Trump voting flips (coefficients)

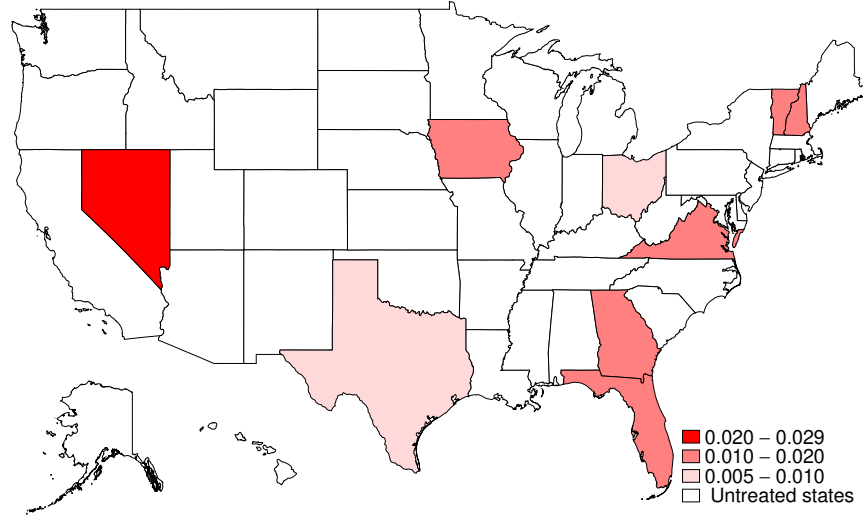
Notes: Panels (a) and (c) present levels of college graduate in-migration, normalized by pre-treatment levels, for treatment and control group states. Panels (b) and (d) present coefficient estimates and 95% confidence intervals corresponding to estimating equation (1). In panels (a) and (b), treatment is switching from a Democratic to a Republican governor, and we restrict to flips that happened 2014-2017. In panels (c) and (d), treatment is switching from voting for Barack Obama in 2012 to Donald Trump in 2016, and 2017 is the first post-treatment year for all events. Migration from 2020 onward is never used. In each panel, the sample is based on college graduates aged 26+ employed in the private sector.

Figure 6: Counterfactual switching states relative to 2024 gubernatorial partisanship

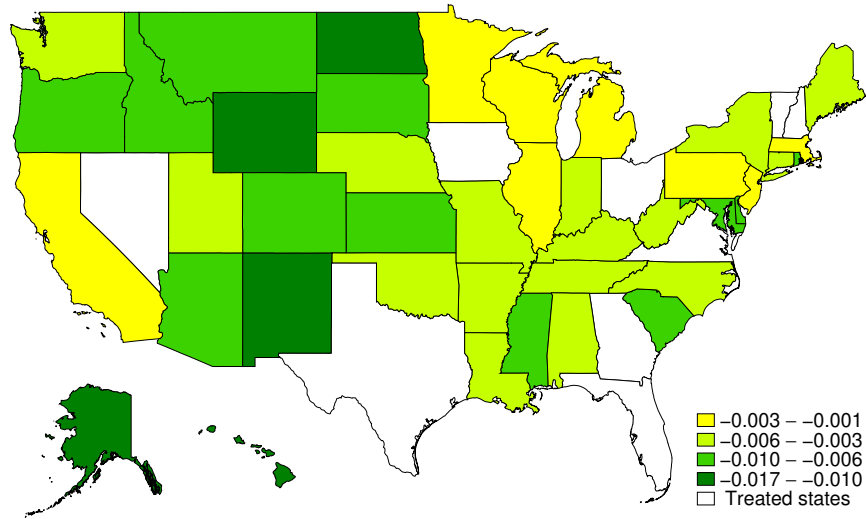


Notes: Figure plots the relationship between the 2020 Presidential vote share and the most recent (as of January 2024) gubernatorial election vote share. All vote shares are based on two-party vote share. Figure identifies the 16 states we consider plausible flips in our four counterfactuals.

Figure 7: Direct and spillover effects of big blue wave



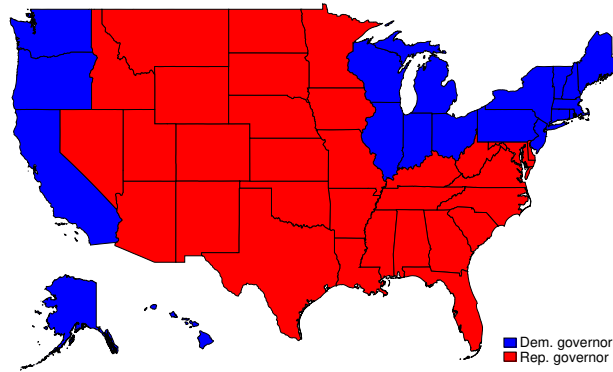
(a) Direct effects: Treated states



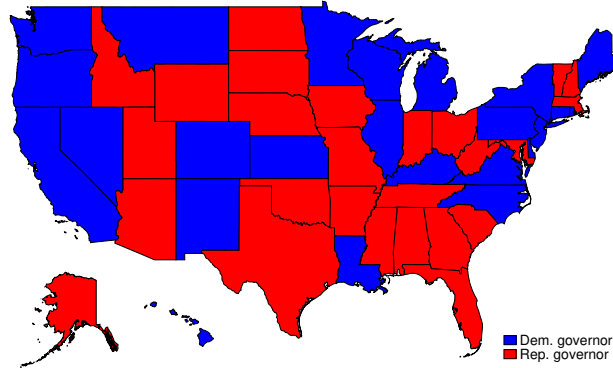
(b) Indirect spillover effects

Notes: Map based on the strong blue wave (see Figure 6) in which nine states governed by Republicans in 2024 are simulated as flipping to Democratic governors. Percentage changes in the college-educated workforce for the treated states are mapped in Panel (a), and percentage changes for un-treated states are mapped in Panel (b).

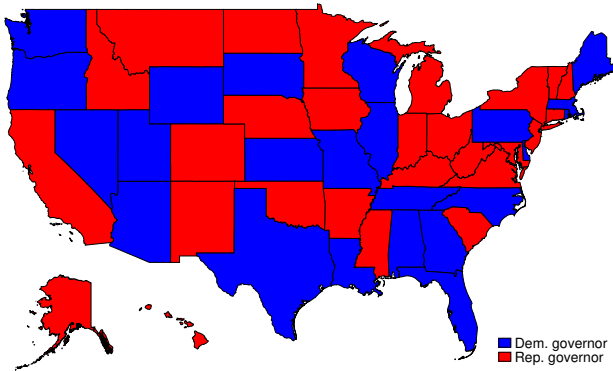
Figure 8: Regionally Concentrated vs. Diffuse Flips



(a) Counterfactual: Region-wide flips



(b) Actual governors in 2021



(c) Counterfactual: Region-balanced flips

Notes: Panel (b) is based on actual governors in office in January 2021. Panel (a) is based on switching governors' partisanship based on whichever party holds the most governorships within the Census division (9 divisions that group similar states together). Panel (c) is based on switching governors' partisanship randomly to obtain a 50-50 Democrat-Republican split within each Census division.

Table 1: Difference-in-difference estimates of governor partisanship and migration

	(1)	(2)	(3)	(4)	(5)
DV:	Net log migration	Log out-migration	Log in-migration	Log in-migration	Col - NonColl in-migration
College:	Yes	Yes	Yes	Yes	
Employed:	Priv. sector	Priv. sector	Priv. sector	Priv. sector	Priv. sector
Ages:	26+	26+	26+	26-45	26-45
US Citizen:				Yes	Yes
Panel A: All flips					
Republican Governor	-0.088** (0.040) [$p : .052$]	0.009 (0.029) [$p : .761$]	-0.079*** (0.024) [$p : .002$]	-0.095*** (0.029) [$p : .002$]	-0.101*** (0.030) [$p : .002$]
N	2172	2172	2174	2174	2174
Panel B: Democrat-to-Republican flips					
Republican Governor	-0.113** (0.055) [$p : .052$]	0.009 (0.041) [$p : .827$]	-0.103*** (0.029) [$p : .004$]	-0.129*** (0.040) [$p : .004$]	-0.162*** (0.054) [$p : .003$]
N	851	851	853	853	853
Panel C: Republican-to-Democrat flips					
Democratic Governor	0.065 (0.054) [$p : .356$]	-0.008 (0.035) [$p : .857$]	0.056* (0.032) [$p : .119$]	0.062 (0.037) [$p : .144$]	0.045 (0.040) [$p : .305$]
N	1321	1321	1321	1321	1321

* $p < .10$, ** $p < .05$, *** $p < .01$. Estimates are based on a staggered rollout different-in-difference estimator; see equations (2) and (3). We consider elections from 2002-2016. Standard errors in parentheses are clustered at the state-level and p -values in brackets are from a cluster wild bootstrapped test of the null of zero treatment effect (Roodman et al., 2019), since we sometimes have relatively few states in the samples.

Table 2: College in-migration responses to governor partisanship over time

	(1)	(2)	(3)	(4)	(5)
Elections:	2003-2017	2003-2007	2008-2012	2013-2017	2013-2017
Panel A: Democrat-to-Republican flips					
Republican Governor	-0.103*** (0.029) [$p : .003$]	-0.183* (0.099) [$p : .092$]	-0.067 (0.076) [$p : .417$]	-0.094** (0.040) [$p : .062$]	
N	853	211	312	330	
N. states (clusters)	42	18	23	20	
Panel B: Republican-to-Democrat flips					
Democratic Governor	0.056* (0.032) [$p : .116$]	0.093 (0.056) [$p : .177$]	0.098** (0.043) [$p : .063$]	-0.119* (0.069) [$p : .304$]	-0.015 (0.042) [$p : .757$]
N	1321	436	368	517	509
N. states (clusters)	44	29	22	23	22
Exclude Louisiana					Yes

* $p < .10$, ** $p < .05$, *** $p < .01$. Estimates are based on a staggered rollout different-in-difference estimator; see equations (2) and (3). We consider elections from 2002-2016. Standard errors in parentheses are clustered at the state-level and p -values in brackets are from a cluster wild bootstrapped test of the null of zero treatment effect (Roodman et al., 2019), since we sometimes have relatively few states in the samples. Columns (3)-(5) divide the sample based on the year in which the new governor took office; the pre- and post-treatment windows in these different subsamples can overlap one another. In all columns, the sample consists of college graduates aged 26+ employed in the private sector (see column (3) of Table 1). Column (5) excludes Louisiana; see Appendix Figure C3.

Table 3: College in-migration responses by magnitude of policy shift

	(1)	(2)	(3)	(4)	(5)	(6)
DV:	Policy index	Log of coll. in-migration	College unemp.	Policy index	Log of coll. in-migration	College unemp.
Policy shift:	Small (below median)			Large (above median)		
	Panel A: Democrat-to-Republican flips, by economic policy shift					
Republican Governor	0.032 (0.054)	-0.175** (0.065)	0.051 (0.171)	0.360*** (0.055)	-0.079 (0.049)	-0.005 (0.146)
N	713	713	713	708	708	708
$\bar{y}$	-0.659	-3.066	3.108	-0.619	-3.093	3.138
σ_y	0.93	0.40	1.09	0.90	0.38	1.09
	Panel B: Democrat-to-Republican flips, by social policy shift					
Republican Governor	-0.021 (0.039)	-0.133* (0.066)	-0.083 (0.154)	0.294*** (0.042)	-0.142*** (0.043)	0.124 (0.141)
N	706	706	706	715	715	715
$\bar{y}$	-0.672	-3.098	3.149	-0.512	-3.061	3.097
σ_y	0.84	0.40	1.12	0.90	0.38	1.05
	Panel C: Republican-to-Democrat flips, by economic policy shift					
Democratic Governor	-0.039 (0.049)	0.099 (0.070)	0.074 (0.193)	-0.283*** (0.026)	0.030 (0.050)	-0.156 (0.107)
N	1199	1199	1199	1206	1206	1206
$\bar{y}$	0.851	-3.143	2.741	0.825	-3.133	2.758
σ_y	0.84	0.41	1.10	0.86	0.41	1.13
	Panel D: Republican-to-Democrat flips, by social policy shift					
Democratic Governor	-0.046 (0.036)	0.076 (0.074)	0.079 (0.170)	-0.546*** (0.056)	0.065 (0.039)	-0.131 (0.104)
N	1207	1207	1207	1198	1198	1198
$\bar{y}$	0.701	-3.141	2.732	0.702	-3.135	2.767
σ_y	0.85	0.41	1.11	0.84	0.41	1.12

* $p < .10$, ** $p < .05$, *** $p < .01$. Estimates are based on a staggered rollout different-in-difference estimator; see equation (2). We consider elections from 2002-2016. Standard errors in parentheses are clustered at the state-level. In all columns, the sample consists of college graduates aged 26+ employed in the private sector (see column (3) of Table 1). Standard errors in parentheses are clustered at the state-level. Policy indices are drawn from Caughey and Warshaw (2016); more positive numbers reflect more conservative policies in place.

Table 4: General equilibrium effects of counterfactual gubernatorial switches

	(1)	(2)	(3)	(4)
Counterfactual	Small blue	Big blue	Small red	Big red
Number of states	5	9	3	7
Panel A: Total college-educated labor force				
Average effects	2.2%	1.4%	-1.8%	-1.3%
(min, max)	(1.6, 3.8)	(0.5, 2.9)	(-2.1, -1.6)	(-2.3, -0.6)
Panel B: GDP per worker				
Average effects	2.0%	1.2%	-1.6%	-1.2%
(min, max)	(1.3, 2.6)	(0.5, 2.1)	(-1.9, -1.4)	(-1.8, -0.6)
Panel C: Inequality in log wages per unit of productivity				
Average effects	-1.8 log points	-1.1 pts.	1.5 pts.	1.1 pts.
(min, max)	(-2.6, -1.2)	(-2.0, -0.5)	(1.3, 1.7)	(0.6, 1.6)
Panel D: Inequality in log earnings per worker				
Average effects	0.6 log points	0.4 pts.	-0.6 pts.	-0.5 pts.
(min, max)	(0.3, 1.5)	(0.1, 1.2)	(-0.8, -0.4)	(-0.7, -0.3)

Table presents changes in equilibrium outcomes for “treated” states (i.e., those changing their governors) in our four counterfactuals (Figure 6 presents the states we simulate as flipping in each counterfactual). Panels A and B present percent changes in the college-educated workforce and GDP per worker, respectively. Panels C and D present changes in the college/non-college difference in log human capital price and log average earnings per worker, respectively.

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A Data

Migration data: All reduced form analyses use migration data from the American Community Survey, accessed via IPUMS (Ruggles et al., 2022).

Political data: Data on gubernatorial vote shares is from Caughey, Xu, and Warshaw (2017), updated and extended beyond 2012 by our own data collection, which was mostly based on Wikipedia. Data on control of the state legislature is from Klarner (2018), updated and extended by our own data collection based on the National Conference of State Legislatures.⁴³ Data on policy indices is from Caughey and Warshaw (2018), updated and extended by the authors and available on Christopher Warshaw’s website.

Migration data:

[Table A1 about here.]

[Table A2 about here.]

⁴³<https://www.ncsl.org/about-state-legislatures/state-partisan-composition>

Table A1: Social Policy Questions

Category	N. questions ever	Most common question each year
Immigration	23	4.7
Abortion and birth control	5	2.7
Gun control	6	2.1
Gender and sexuality	7	1.5
Crime and policing	12	1.1
Affirmative action	1	0.7
Supreme Court	5	0.4
Education	3	0.2
Impeachment	2	0.1

Table displays characteristics of social policy questions. Topics are classified as social policy based on a scheme developed by Caughey and Warshaw (2018). Many of the questions appear in grid format, where the literal question itself is simply the last part of a longer sentence (the first part being shown above). We have shortened and reworded the questions we present here into standard language (also sometimes eliminating preambles before the questions). It is sometimes difficult to cleanly separate questions across these categories (e.g., a question related to taxes *and* government spending, or to immigration *and* crime), however, our analysis does not use these category labels *within* social/economic policy groups. It is typically simple to determine whether a question belongs to the social group or the economic group. “Number of questions ever” refers to the total number of unique questions appearing about the topic between 2006 and 2020. “Number of questions each year” refers to the average number of questions asked about the topic during any given year, 2006-2020.

Table A2: Economic Policy Questions

Category	N. questions ever	each year	Most common question
Government spending	23	4.7	If your state were to have a budget deficit this year, what would you prefer more, raising taxes or cutting spending?
Environment	10	2.9	Do you support giving the Environmental Protection Agency the power to regulate Carbon Dioxide?
Health care	14	2.3	Do you support repealing the Affordable Care Act?
Taxes	5	2.1	If the state had to raise taxes, what share of the tax increase should come from increased income taxes and what share from increased sales taxes?
Trade	8	1.2	Should the United States withdraw from the Trans-Pacific Partnership
Minimum wage	2	0.4	Do you support raising the minimum wage to \$15 an hour?
Deregulation	2	0.3	Would you support an executive action requiring that with each new regulation enacted, two must be cut?

Table displays characteristics of economic policy questions. Topics are classified as economic policy based on a scheme developed by Caughey and Warshaw (2018). Many of the questions appear in grid format, where the literal question itself is simply the last part of a longer sentence (the first part being shown above). We have shortened and reworded the questions we present here into standard language (also sometimes eliminating preambles before the questions). It is sometimes difficult to cleanly separate questions across these categories (e.g., a question related to taxes *and* government spending, or to immigration *and* crime), however, our analysis does not use these category labels *within* social/economic policy groups. It is typically simple to determine whether a question belongs to the social group or the economic group. “Number of questions ever” refers to the total number of questions about the topic ever asked from 2006-2020. “Number of questions each year” refers to the average number of questions asked about the topic during any given year, 2006-2020.

B Descriptive results

B.1 Migration and population change

[Figure B1 about here.]

[Figure B2 about here.]

[Figure B3 about here.]

B.2 Political attitudes

B.2.1 Basic facts

[Figure B4 about here.]

[Figure B5 about here.]

[Figure B6 about here.]

[Figure B7 about here.]

B.2.2 Differences in policy preferences by earnings

Most state policymakers care about attracting a skilled, productive workforce, which may not be the same as attracting college graduates. If the college/non-college gaps we see are driven by a subset of left-leaning graduates with low earnings, then a conservative policymaker might be unconcerned, and reasonable minds may disagree about the costs to economic growth. If, on the other hand, the gap in policy views holds broadly when comparing high- and low-earners, then virtually all policymakers would be concerned about the consequences of pushing away these workers.

One challenge in assessing this is that the CES does not collect information on individual-level earnings or income.⁴⁴ For most recent years, however, it does collect respondents' sector of employment (2-digit NAICS), in addition to education and age.⁴⁵ This information is sufficient to impute earnings with high accuracy. To do so, we rely on an approach that adjusts earnings for geography. Coastal areas and large cities are more liberal and have

⁴⁴The CES *does* collect information on household income, but *i*) it is collected only for relatively broad income ranges, and *ii*) household income includes income from both partners in married or cohabiting families, and patterns of assortative matching and marriage rates differ by education and have changed over time.

⁴⁵Industry is collected in 2011-2014 and 2016-2020.

higher nominal wages, and we would not want to conflate political differences by earnings with a spurious correlation between regional earnings differences and local political dynamics. Using the American Community Survey (ACS), we regress log annual earnings on geography-by-time fixed effects and age-by-education-by-industry fixed effects:

$$\ln(y_i) = \alpha_{m,s,t} + \theta_{a,n,e} + \varepsilon_i \quad (4)$$

where i denotes individual, m denotes MSA, s denotes state, t denotes year, a denotes age range (25-32, 33-40, 41-55), n denotes industry (2-digit NAICS), and e denotes education (college graduate vs. non-college). The MSA-by-state-by-year fixed effects ($\alpha_{m,s,t}$) account for heterogeneous levels and trends in nominal earnings in a flexible way, and the age-by-industry-by-education fixed effects ($\theta_{a,n,e}$) therefore represent persistent earnings differences between different types of workers, regardless of where they live. We see these estimated $\hat{\theta}_{a,n,e}$ fixed effects as CES respondents' *earnings capacity*, which is relevant for a state attempting to attract high-productivity workers.⁴⁶ In this way, we determine the earnings capacity for all employed CES respondents (ignoring non-employed respondents) and normalize our measure to have unit standard deviation across all employed respondents across all years.

Figure B8 shows the basic patterns of earnings polarization. In Panel (a), we plot the coefficients on our earnings measure from year-by-year regressions of social and economic policy views on earnings capacity (again for workers aged 26-45). Earnings had little correlation with policy views through 2014, though in 2011, higher earners were slightly more conservative on economic policy issues. However, since then, higher earners have become substantially more left-leaning on both social and economic issues. By 2020, a 1σ increase in earnings capacity is associated with a $.12\sigma$ shift left in policy views on both sets of issues.

This relationship is entirely driven by education. Panel (b) presents coefficients that come from regressing views on both earnings and education. Once we control for education, high earners are more conservative on both economic and social policy issues, and this relationship has not changed at all over time. The trends in “earnings polarization” arise entirely from the college/non-college gap, which, once we control for earnings, has increased from roughly $.25\sigma$ in 2011 to $.55\sigma$ -. $.65\sigma$ in 2020.

[Figure B8 about here.]

Finally, panels (c) and (d) show the distribution of earnings capacity and recent policy views across all age-education-industry cells. There is a striking separation between college

⁴⁶One advantage of this approach relative to simple average earnings is that changes over time in the relationship between policy views and earnings capacity reflect changes in different groups' views, holding fixed the earnings position of any given “group” (i.e., age-industry-education cell).

and non-college workers, with almost zero overlap. At virtually any point in the earnings distribution, all types of college graduates are to the left of all groups of non-college workers on both economic and social policy issues. The figure shows that differences between education groups are dramatically larger than any differences within the college and non-college population, at least among groups that we can measure in the data.

Earnings and political views differ within age-industry-education cell, and we can say nothing about the within-cell correlation in views. Existing causal evidence from the University of California’s admissions policies, however, shows that attending a more prestigious university increases earnings (Bleemer, 2021) *and* pushes students further to the left (Firoozi, 2026). These effects are consistent with the correlation across all universities, which shows that students at more prestigious universities are generally further to the left (Firoozi, 2026). Moreover, college graduates with post-graduate degrees are further left than those with a Bachelor’s only (see Appendix Figure B5). In other data, we can test differences across college majors, and there is very little correlation across majors between earnings and political views. This is shown in Figure B9. Together, all of this implies that the correlation between ideology and earnings *within* age-industry-education cell likely yields a correlation between earnings and political attitudes that is even stronger than what we can document *between* cells.

[Figure B9 about here.]

B.2.3 Index-free evidence of educational polarization

Our primary measures of political opinion are based on Item Response Theory (IRT) indices. These aggregate over an imbalanced panel of questions. This creates indices that are comparable over time, but potentially obscures the effects of the changing set of questions. In this section, we present measures of polarization between college and non-college CES respondents for every divisive question in a fully disaggregated and transparent way.

Consider K binary policy questions numbered $k = 1, \dots, K$. Let $\tilde{x}_{i,k,t}$ denote respondent i ’s answer to question k during year t . Note that some k ’s are not observed during some t ’s.

Without loss of generality, define $x_{i,k,t}$ to be 1 if respondent i chose the answer to question k that is more among Democrats than Republicans and 0 if respondent i chose the answer that is more common among Republicans than Democrats. Thus, $x_{i,k,t}$ summarizes whether the respondent agrees with the position that is more popular among Democrats than Republicans.

We then calculate the difference between college and non-college respondents’ views, relative to the difference between Democrats’ and Republicans’ views. We refer to this as

the “polarization coefficient,” and calculate it as follows:

$$PolarCoe f_{k,t} = \frac{\bar{x}_{col,k,t} - \bar{x}_{nc,k,t}}{\bar{x}_{DEM,k} - \bar{x}_{REP,k}} \quad (5)$$

where *i*) the denominator is fixed across t so that we can ensure trends are driven by changes in college/non-college gaps and not trends in Democratic/Republican gaps, and *ii*) we exclude questions where partisan gaps are 10 percentage points or less, to avoid very small denominators. Note that this amounts to excluding questions that are not very divisive along partisan lines, and there are very few of these.

If the polarization coefficient is, say, 0.4 then that means that the college voters are more likely than non-college ones to hold the position more popular among Democrats (since the gap is positive), and the magnitude of the difference between the average college and non-college respondent is 40% as large as the difference between the average Democrat and the average Republican. Figure B10 presents the question-by-question polarization coefficient, separately for social and economic issues.

Regarding social issues, three things are worth noting. First, for most questions, college graduates are more likely than non-college respondents to hold the Democratic Party’s views (i.e., the polarization coefficient is mostly positive). Second, most questions show a decline from 2006-2012 and an increase from 2014-2020, both of which are consistent with the non-monotonic trends we reported in the text for our aggregate IRT index. Third, newly appearing lines (which reflect newly added questions) tend to coincide fairly well with the continuing lines. This suggests that the additions of new questions are not distorting the trends in our IRT index.

[Figure B10 about here.]

Regarding economic issues, four things are worth noting. First, most questions show an upward trend, consistent with the trends in our aggregate IRT index showing college graduates moving left (relative to non-college) monotonically over time. Second, early in our sample, most questions had negative polarization coefficients (college graduates tended to agree with Republicans), while by the end, most had positive polarization coefficients (college graduates tended to agree with Democrats). This is consistent with the levels of our aggregate IRT index. Third, unlike with social issues, the newly added questions in 2014 are somewhat dissimilar to previous questions. However, the simultaneous addition of new environmental policy questions (where college graduates tended to agree with Democrats) and new spending questions (where college graduates tended to agree with Republicans) mutes the extent to which these additions affect the aggregate trend. Moreover, it is clear

that all of these new questions, once they are included, show the continuous movement over time of college graduates towards the Democratic Party’s position (consistent with trends in our aggregate IRT index). Fourth, environmental policy questions are not representative of other economic policy issues. Nonetheless, they are important and divisive issues that have played a key role in recent political debates, and so we consider them an important input to our index. We have re-estimated our IRT index excluding these questions, and the overall trends change very little, though the levels of education polarization on economic issues become smaller.

It is worth noting that many of the polarization coefficients are quite large. For both social and economic policy issues, it is common for the college/non-college gap to be as much as one-third of the Democratic/Republican gap. Put differently, at a time when American politics is almost universally characterized as highly polarized between parties, the gaps between a representative college graduate and a representative non-college respondent (irrespective of their partisanship) are one-third as large as the gaps between parties. While these question-by-question patterns are generally indicative of the overall signs and trends in education polarization, it is difficult to interpret magnitudes from them. Our IRT index solves this problem in a data-driven way that accounts for changes in questions over time and corresponds to mainstream practice in political science.

B.2.4 Explaining education polarization

Although it is not our goal to explain education polarization, our data allows us to speak to several prominent explanations, and we find some evidence in favor of most. It is worth noting that these explanations aim to explain slightly different facts, such as the rise in affective polarization between Democrats and Republicans (Bishop, 2009; Klein, 2020), college vs. non-college support for the Democratic Party (Gethin et al., 2022; Kuziemko et al., 2026), or aggregate changes in views (Schumpeter, 1942). Nonetheless, all have some bearing on our results about changing views between college and non-college workers.

First, conservatives often allege that universities have become increasingly left-wing, increasing both the treatment effects (the college experience pushing students left) and selection effects (more left-wing students choosing to go to college) of higher education. In panel (a) of Figure B11, we separate education polarization into within-cohort and between-cohort changes. More recent birth cohorts indeed show larger college/non-college gaps at age 26. However, these trends are almost perfectly matched by widening gaps within previously educated cohorts. This is inconsistent with changes in the university experience and points instead towards changes in our political climate that college and non-college voters respond to differently.

[Figure B11 about here.]

One possible such change, suggested by Gethin, Martínez-Toledano, and Piketty (2022), is that the increasing importance of social issues rather than economic ones has led college graduates towards the Democratic Party since (as we showed above) college graduates have long been to the left of non-college voters on social issues. In panel (b), we predict party choice using social and economic policy views, separately for college and non-college voters, and plot the estimated coefficients over time. Consistent with Gethin et al. (2022), since 2010, the importance of social issues for explaining party choice has increased by one-third, while the importance of economic issues has fallen by half. All else equal, this shift will increase college graduates’ support for the Democratic Party and non-graduates’ support for the Republican Party. However, as we showed above, there are also dramatic shifts in views on these issues themselves, which this story cannot explain.

A very different explanation – advanced by Mason (2015) and popularized by Klein (2020) – is that the distinction between economic and social issues is increasingly unimportant as voters’ political identities have collapsed into a single dimension. This can exacerbate affective polarization and hostility by eroding the common ground that voters of different groups might otherwise have shared on different sets of issues. Panel (c) is consistent with this, as both college and non-college voters show a rise in the correlation between social and economic policy views since 2009. However, without some additional theory for reinforcing pressures in opinion formation, this cannot explain why views themselves have diverged between college and non-college voters.

One possible type of reinforcing pressure is associated with Bishop (2009). He shows Democrats and Republicans increasingly live in different places and argues that if exposure to neighbors affects political views, then the clustering of like-minded voters can exacerbate polarization via an echo chamber-like effect.⁴⁷ Looking at social issues (where educational gaps are longstanding), panel (d) does show that states with more college graduates prior to 2010 saw greater divergence between college and non-college voters after 2010.

Another explanation aims to explain changes in views over time, specifically for economic policy. Kuziemko, Longuet-Marx, and Naidu (2026) show that college graduates tend to support redistribution (taxes and transfers) while non-college voters support “predistribution” (market intervention and regulation). They argue that as the Democratic Party has increasingly prioritized redistribution over predistribution, this has pulled in college graduates and

⁴⁷While the book was controversial at the time (Abrams and Fiorina, 2012), more recent evidence has largely supported both the stylized fact of increased partisan sorting (Kaplan, Spinkuch, and Sullivan, 2022), as well as his key mechanism that exposure to neighbors affects political attitudes (Perez-Truglia and Cruces, 2017; Cantoni and Pons, 2022).

pushed away non-graduates. If political debates about economic issues have also shifted from predistribution to redistribution, then our indices of economic policy views (which capture the main content of political debates at the time) would show a divergence between college and non-college voters, even if there were no changes in underlying views on fixed issues. In panel (e), we present college and non-college support for increasing taxes (redistribution) and the minimum wage (predistribution). Consistent with Kuziemko et al. (2026), non-college voters are more likely to support increasing the minimum wage, while college voters are more likely to support tax increases. Thus, holding views fixed, these shifts in the Democratic Party’s priorities will attract college graduates and repel non-graduates. However, it is also worth noting that the college/non-college gap in support for predistribution has collapsed, suggesting that changes in underlying opinions are also important.

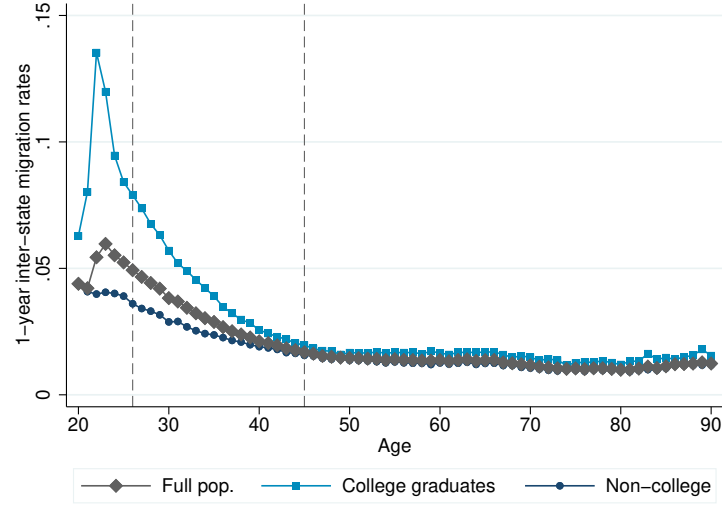
Finally, over 80 years ago, Austrian economist Joseph Schumpeter (1942) predicted that economic growth would lead to a class of intellectuals, journalists, and bureaucrats who are sufficiently far removed from production, innovation, and market competition to abandon capitalism for socialism.⁴⁸ Panel (f) presents trends in the economic policy attitudes of college graduates working in select industries, measured relative to the nationwide average non-college voter during the same year.

Perhaps consistent with Schumpeter’s prediction, college graduates working in education and arts/entertainment have long been well to the left of non-college voters and have trended substantially further. However, in professional and business services (which includes many fast-growing and innovative industries), we see the same trends despite starting to the right of non-college voters. Even in finance – where workers surely continue to think about capitalism, economic growth, and market competition – college graduates have drifted from being $.25\sigma$ to the right of non-college voters in 2012 to being $.25\sigma$ to the left by 2020. Only in manufacturing (the quintessential production sector) are college graduates roughly in the same place as the average non-college respondent.

Indeed, then, we find support for five common claims, although none offer a complete explanation for the patterns we document. Indeed, we do not believe there *is* a singular explanation for the rise in education polarization. For this reason, below, we do not model endogenous polarization.

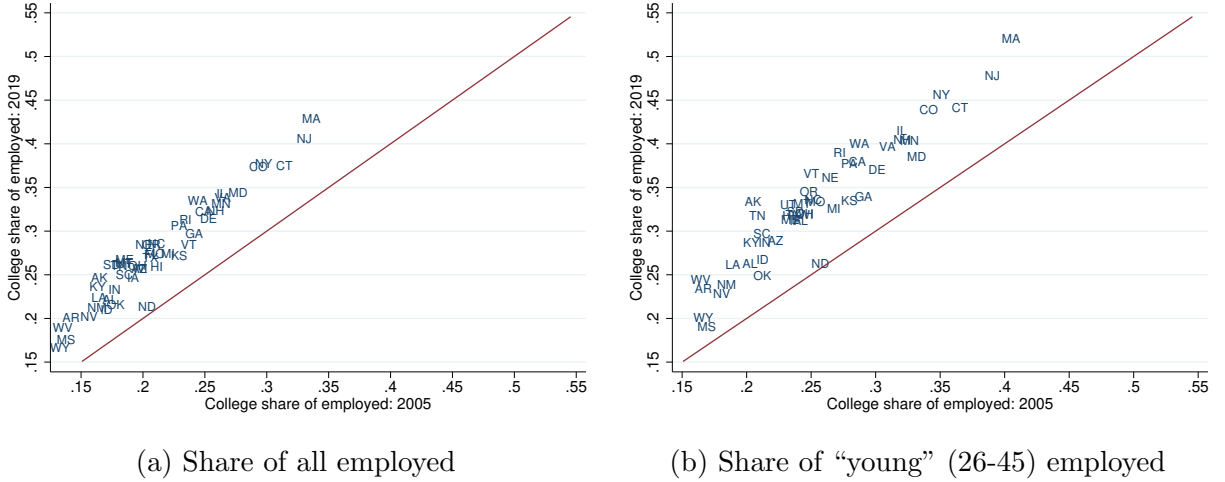
⁴⁸See Sobel (2021) for some helpful discussion. At the time (1942), Schumpeter argued that this trend had already begun (without quantitative evidence). More recently, using data back to the late 1940s, Gethin et al. (2022) show that the shift of better-educated voters towards the Democratic Party had indeed begun as early as their data can measure.

Figure B1: interstate migration rates by age and education (2005-2019)



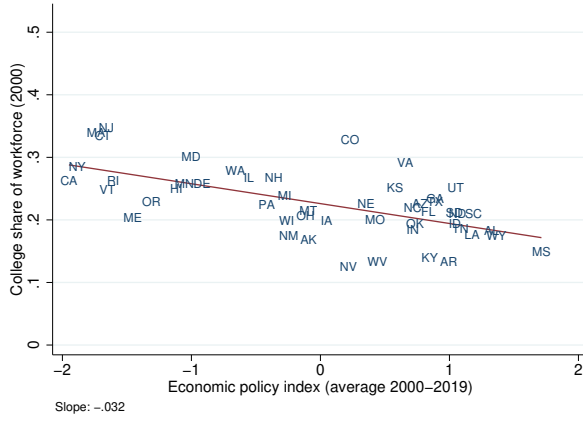
Notes: Calculations based on one-year interstate migration rates from the American Community Survey (ACS). All calculations use sample weights. Vertical lines indicate the 26-45 sample that we use for some analyses.

Figure B2: Changes in college-shares by state over time

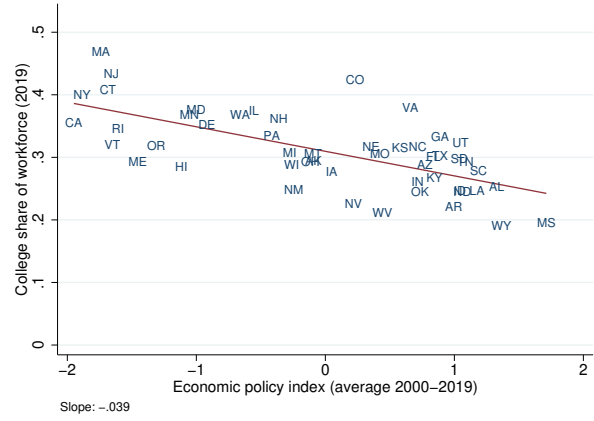


Notes: Calculations based on American Community Survey (ACS) respondents employed in the private sector. All calculations use sample weights. Red line is a 45-degree line. The slope relating 2019 college share to 2005 college share is significantly greater than one ($p < .01$).

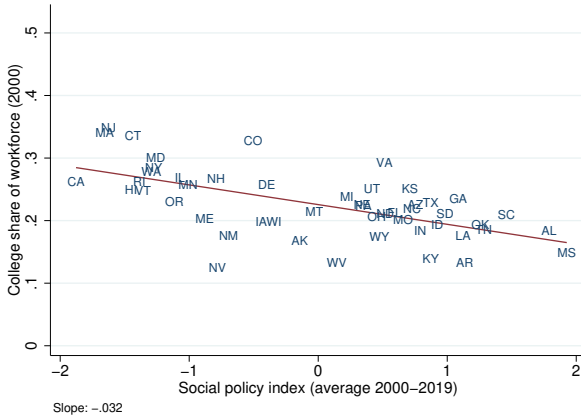
Figure B3: College share by state policy over time



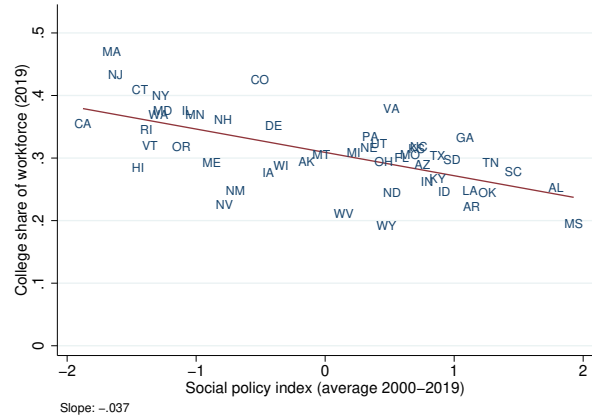
(a) By economic policy, 2000



(b) By economic policy, 2019



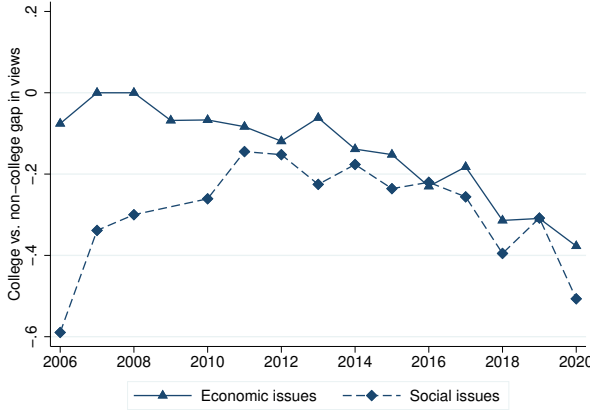
(c) By social policy, 2000



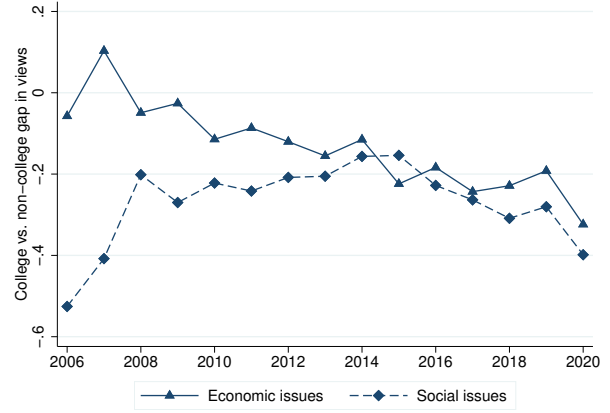
(d) By social policy, 2019

Notes: College shares are calculated among private sector, employed ACS respondents age 26 and older. Policy indices describe the overall set of policies in place in the state, are drawn from Caughey and Warshaw (2016), are averaged over the 2000-2019 interval, and are normalized to have standard deviation of one across states.

Figure B4: Robustness: College vs. Non-college gaps in views



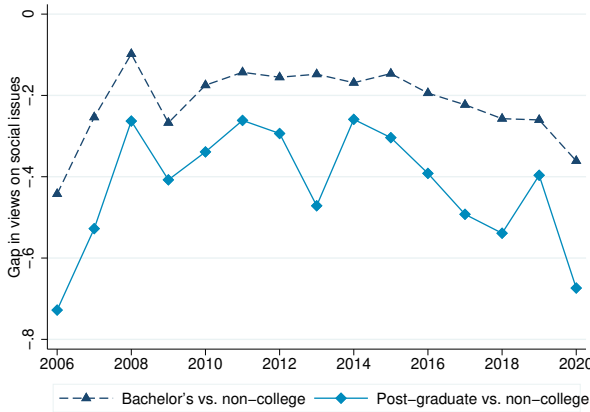
(a) Gap in median views, 26-45



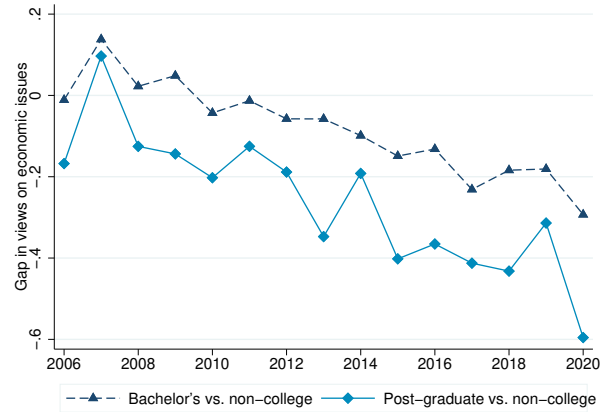
(b) Gap in average views, all ages

Notes: All calculations are based on Cooperative Election Study (CES) respondents. Policy view indices are based on Item Response Theory estimates using the approach developed by Imai et al. (2016). Policy questions are divided into social and economic policy issues using the scheme developed by Caughey and Warshaw (2018). Indices are normalized to have mean zero and unit standard deviation across all respondents across all years, with more positive values indicating more conservative views. All calculations use sample weights.

Figure B5: Education polarization for Bachelor's and Post-graduate respondents



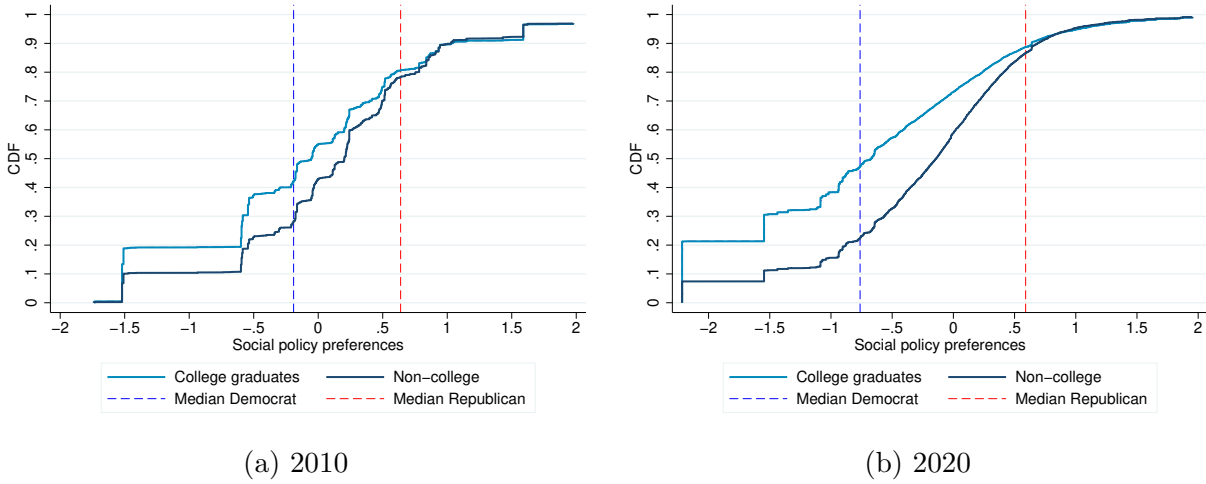
(a) Social issues



(b) Economic issues

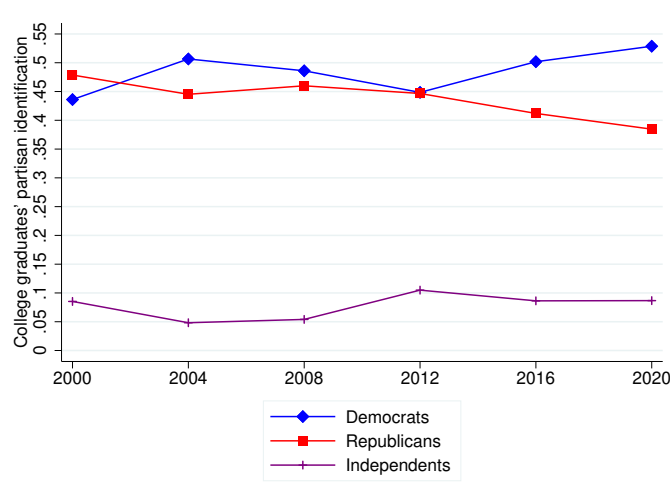
Notes: All calculations are based on Cooperative Election Study (CES) respondents aged 26-45. Policy view indices are based on Item Response Theory estimates using the approach developed by Imai et al. (2016). Policy questions are divided into social and economic policy issues using the scheme developed by Caughey and Warshaw (2018). Indices are normalized to have mean zero and unit standard deviation across all respondents across all years, with more positive values indicating more conservative views. All calculations use sample weights.

Figure B6: Distribution of social policy views: 2010-2020

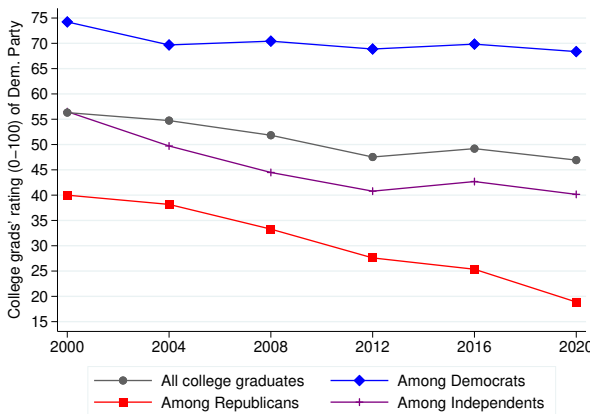


Notes: All calculations are based on Cooperative Election Study (CES) respondents aged 26-45. Policy view indices are based on Item Response Theory estimates using the approach developed by Imai et al. (2016). Policy questions are divided into social and economic policy issues using the scheme developed by Caughey and Warshaw (2018). Indices are normalized to have mean zero and unit standard deviation across all respondents across all years, with more positive values indicating more conservative views. All calculations use sample weights.

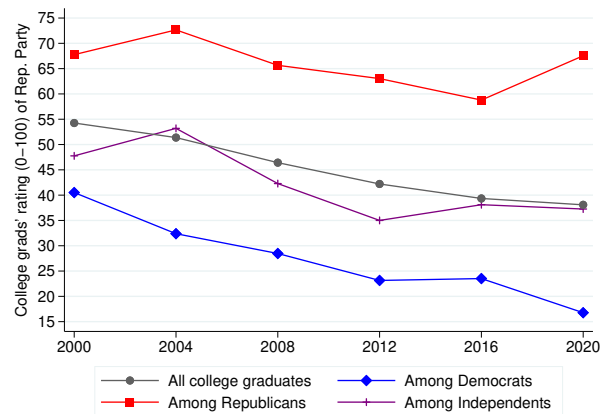
Figure B7: College graduates' attitudes towards parties, by partisan affiliation



(a) Partisan identification among college graduates



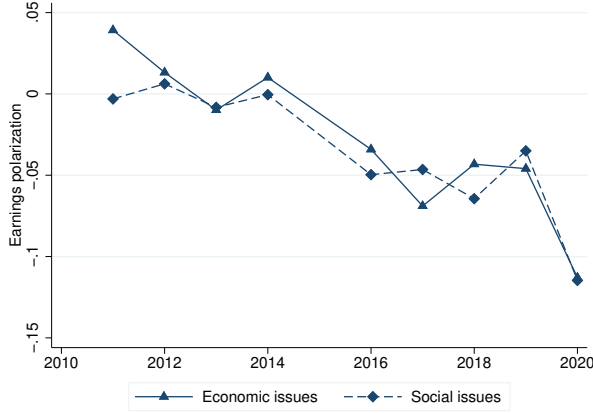
(b) Feeling thermometer: Democratic Party



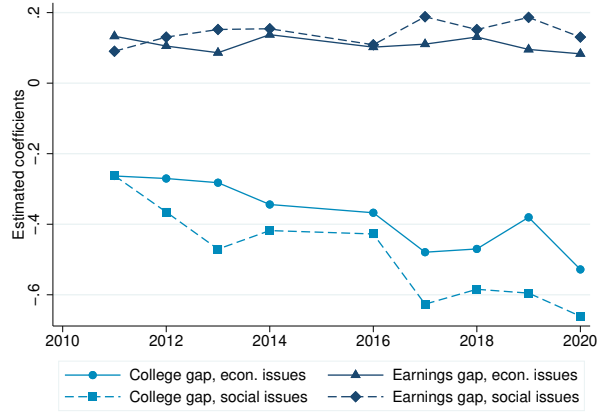
(c) Feeling thermometer: Republican Party

Notes: All calculations are based on the American National Election Study (ANES). Panel (a) is based on partisan identification, while panels (b) and (c) present average "feeling thermometer" scores of college graduates. Feeling thermometer scores measure how respondents feel, overall, towards the party and are a common measure of affective polarization (Boxell, Gentzkow, and Shapiro, 2024).

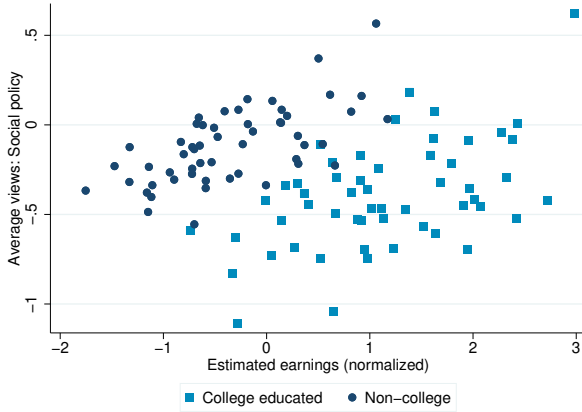
Figure B8: Differences in policy views by earnings



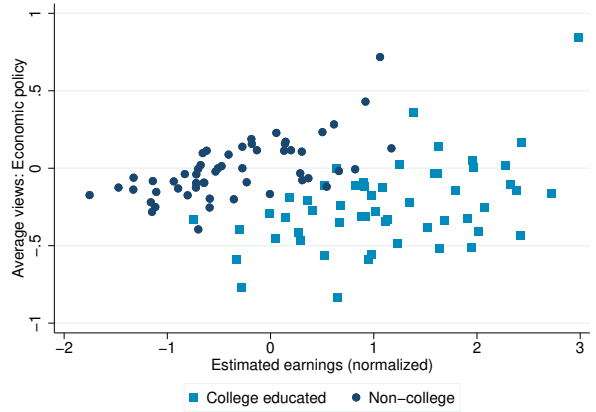
(a) Earnings polarization over time



(b) Coefficients on earnings and education



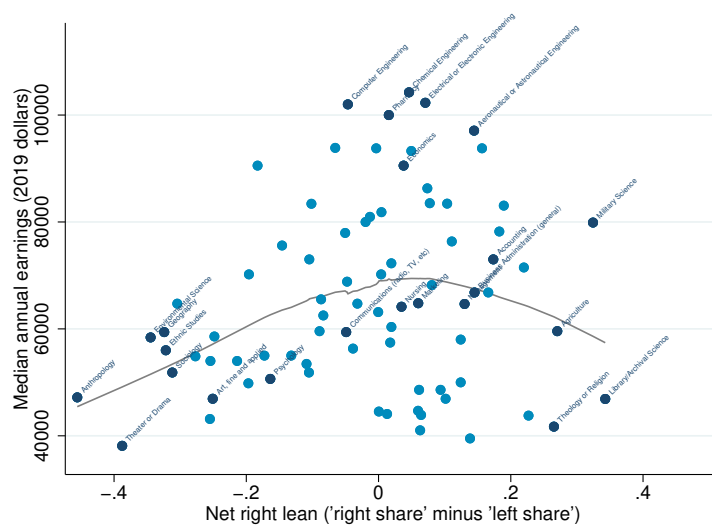
(c) Social policy views
by age-industry-college



(d) Economic policy views
by age-industry-college

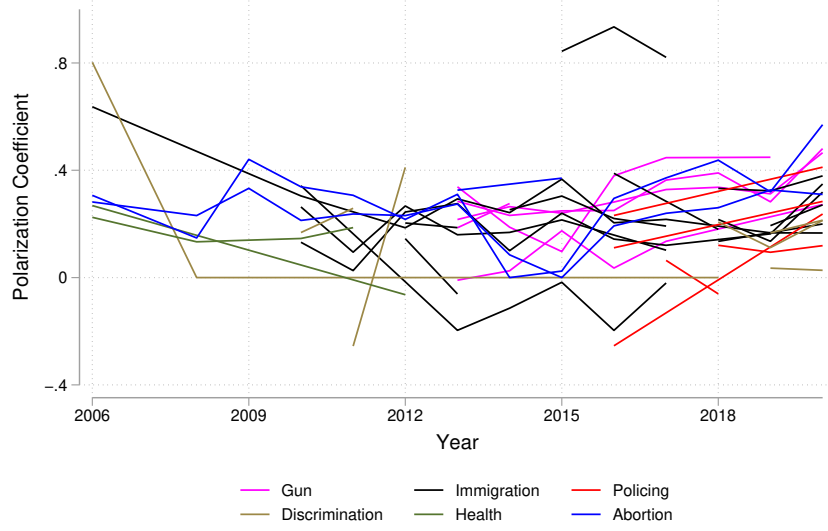
Notes: All calculations based on Cooperative Election Study (CES) respondents aged 26-45 and employed in the private sector. Earnings are imputed from the American Community Survey (ACS) based on age, education, and industry, using the approach proposed in the text that accounts for geographic pay differences (see equation (4)). Policy view indices are based on Item Response Theory estimates using the approach developed by Imai et al. (2016). Policy questions are divided into social and economic policy issues using the scheme developed by Caughey and Warshaw (2018). Indices are normalized to have mean zero and unit standard deviation across all respondents across all years, with more positive values indicating more conservative views. All calculations use sample weights. Panels (c) and (d) based on respondents from 2017 onward.

Figure B9: Self-described political leanings and earnings by college major

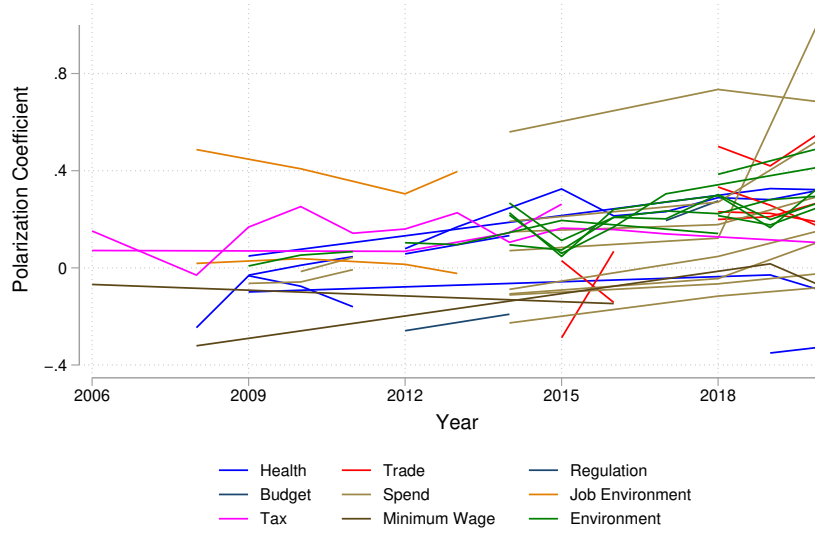


Notes: Median earnings (y -axis) are based on American Community Survey (ACS) respondents from 2009-2019. Net right lean (x -axis) is the difference between the share of survey respondents (college seniors) who identify with “far right” or “conservative” minus the share who identify with “far left” or “liberal” political views (1994-2008). Line represents a Lowess non-parametric fit.

Figure B10: Question-specific education polarization over time



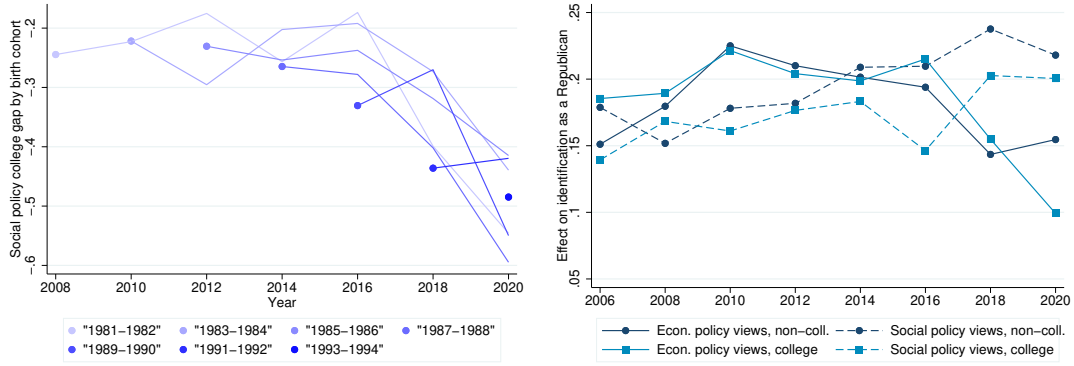
(a) Social issues



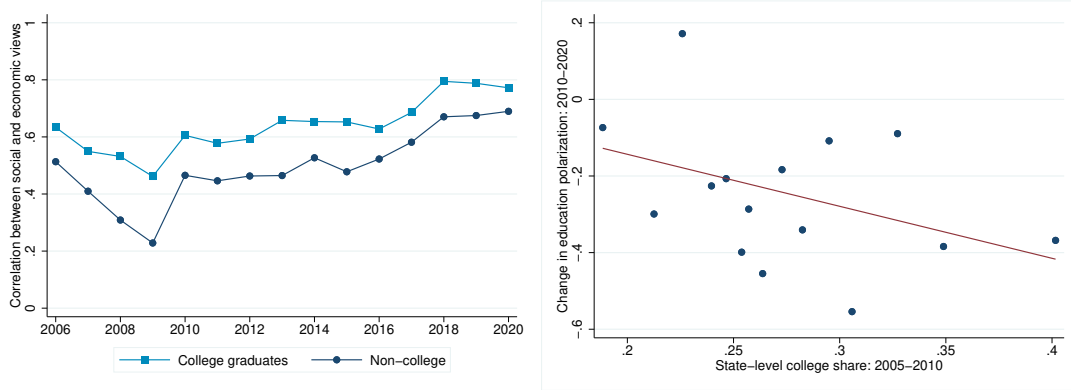
(b) Economic issues

Notes: The figure reports question-specific education polarization coefficients for questions from the Cooperative Election Study. The coefficient is defined as the difference between the average college graduate and the average non-college respondent, relative to the difference between the average Democrat and the average Republican. Positive values indicate that college graduates are more aligned with Democratic positions relative to non-college respondents. See equation (5).

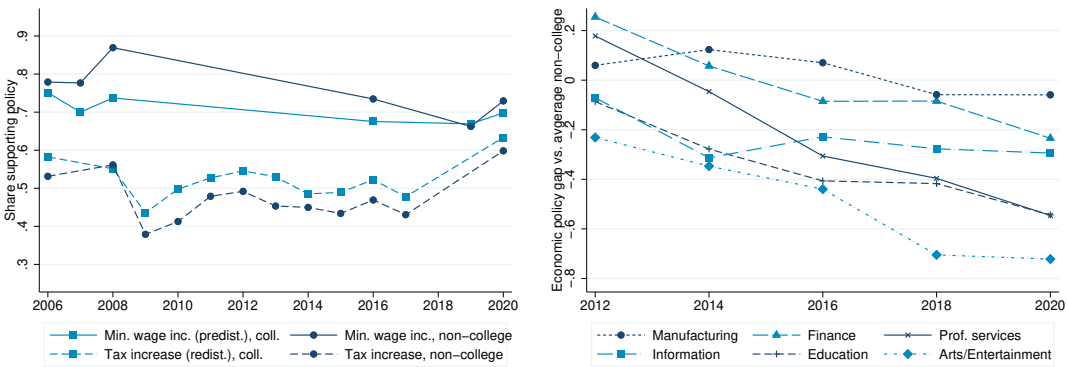
Figure B11: Assessing explanations for education polarization



(a) Within- vs. between-cohort changes (b) Social vs. economic policy importance



(c) Social and economic policy correlation (d) Polarization and echo chambers



(e) Redistribution vs. Predistribution (f) Trends by industry

Notes: All calculations are based on Cooperative Election Study (CES) respondents aged 26-45. Policy view indices are based on Item Response Theory estimates using the approach developed by Imai et al. (2016). Policy questions are divided into social and economic policy issues using the scheme developed by Caughey and Warshaw (2018). Indices are normalized to have mean zero and unit standard deviation across all respondents across all years, with more positive values indicating more conservative views. Because of some small industry-level samples, panel (f) pools consecutive years together. All calculations use sample weights.

C Additional reduced form results

C.1 Identifying variation

[Figure C1 about here.]

C.2 DiD results

C.2.1 Non-college workers

[Table C1 about here.]

[Figure C2 about here.]

C.2.2 College graduates

[Table C2 about here.]

[Table C3 about here.]

[Figure C3 about here.]

C.2.3 Effects by size of policy shift

[Figure C4 about here.]

[Figure C5 about here.]

[Figure C6 about here.]

[Figure C7 about here.]

[Table C4 about here.]

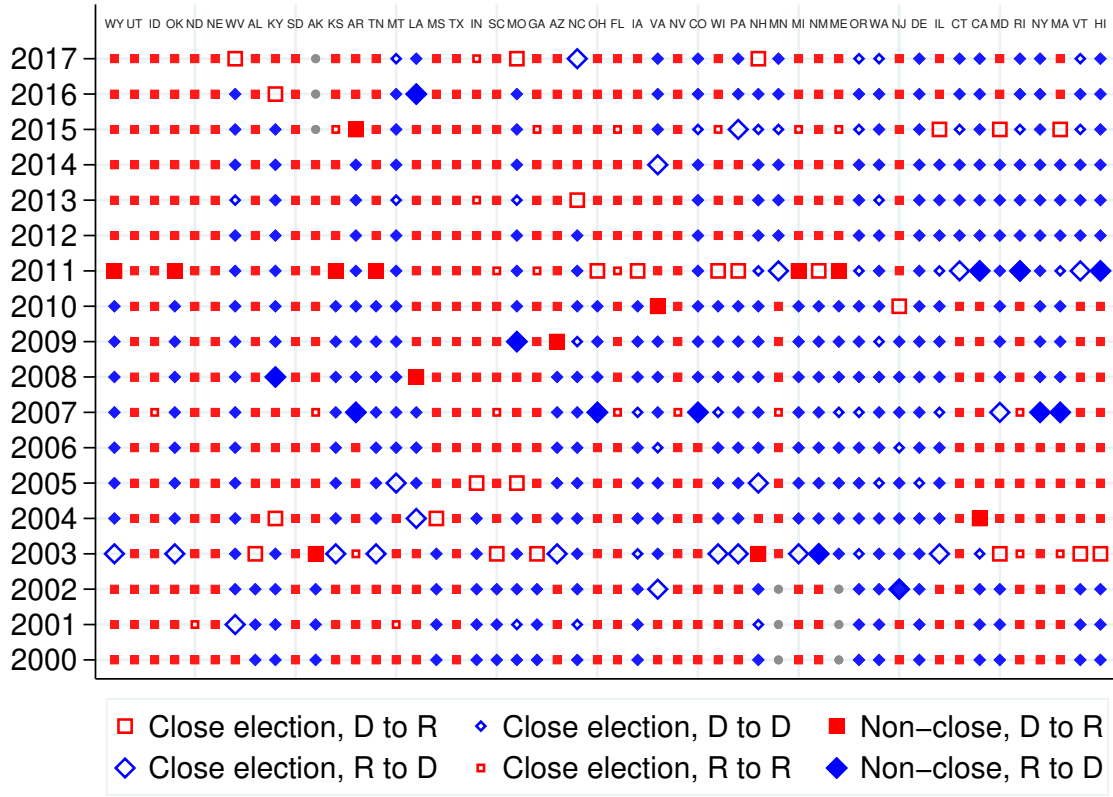
[Figure C8 about here.]

C.3 Regression discontinuity results

[Table C5 about here.]

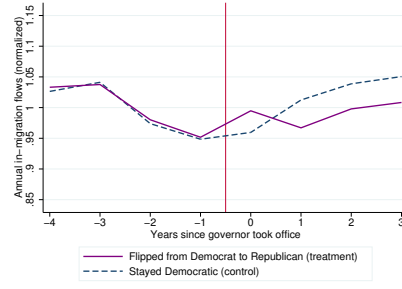
[Figure C9 about here.]

Figure C1: Identifying variation: Close elections and partisan flips

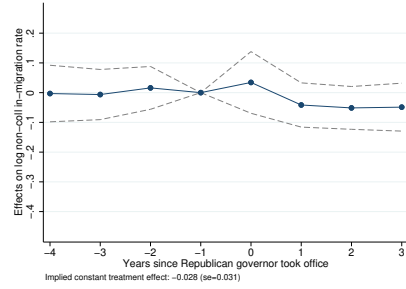


Notes: Figure shows the partisanship of the governor of each state over time, with states ordered by increasing Democratic vote share in presidential elections 2000-2020. Blue diamonds indicate Democratic governors, red squares indicate Republicans, and gray circles indicate independents. Large shapes indicate the first year of a flip from one party to the other (which are the source of identifying variation in our DiD specifications) and hollow shapes indicate that the last year saw a close election (which are the source of identifying variation in our RDD specifications). For this figure, a “close election” is defined as a margin of victory of 6 percentage points or less, which is close to the bandwidths selected by the Calonico et al. (2014) estimator in various specifications (see Appendix Table C5).

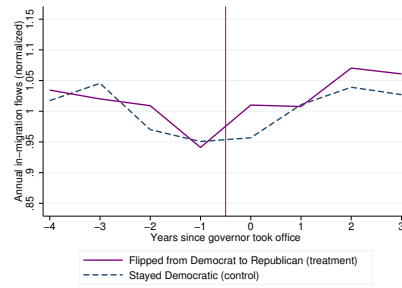
Figure C2: Non-college in-migration responses gubernatorial transitions



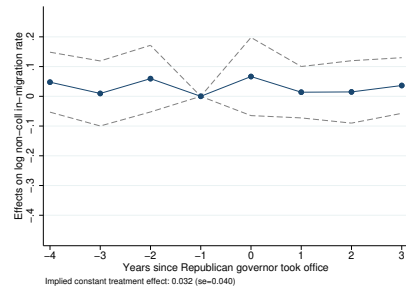
(a) Democrat-to-Republican,
private sector emp., 26+



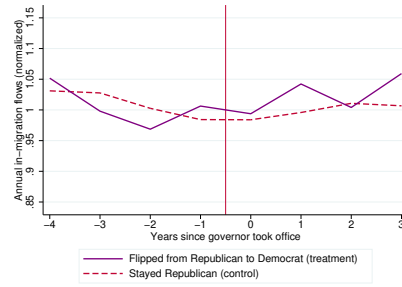
(b) Democrat-to-Republican,
private sector emp., 26+



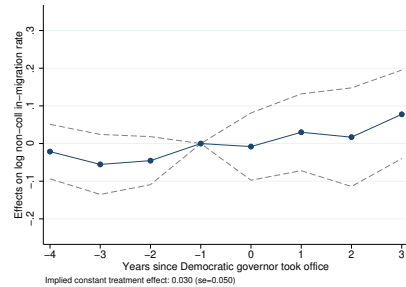
(c) Democrat-to-Republican,
private sector citizens, 26-45



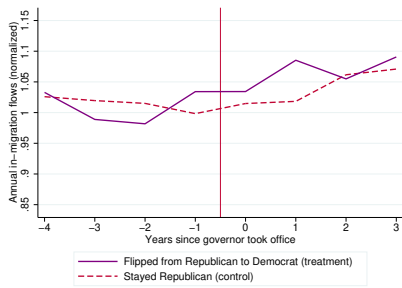
(d) Democrat-to-Republican,
private sector citizens, 26-45



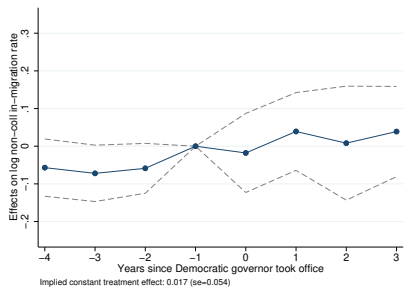
(e) Republican-to-Democrat,
private sector emp., 26+



(f) Republican-to-Democrat,
private sector emp., 26+



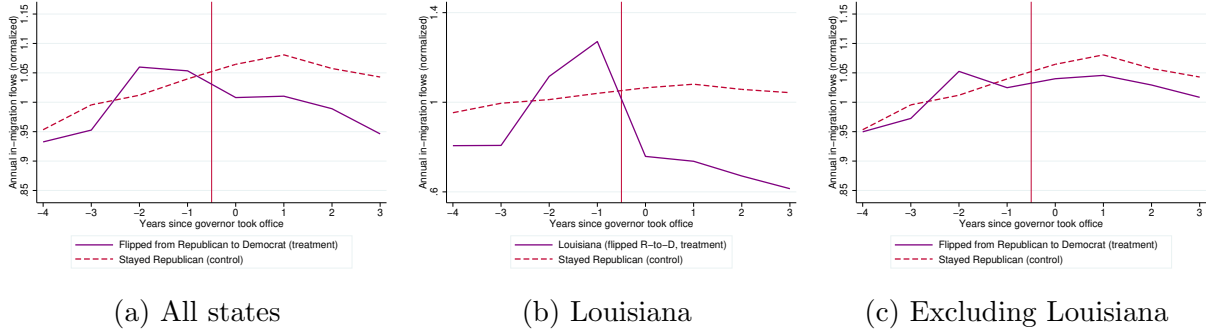
(g) Republican-to-Democrat,
private sector citizens, 26-45



(h) Republican-to-Democrat,
private sector citizens, 26-45

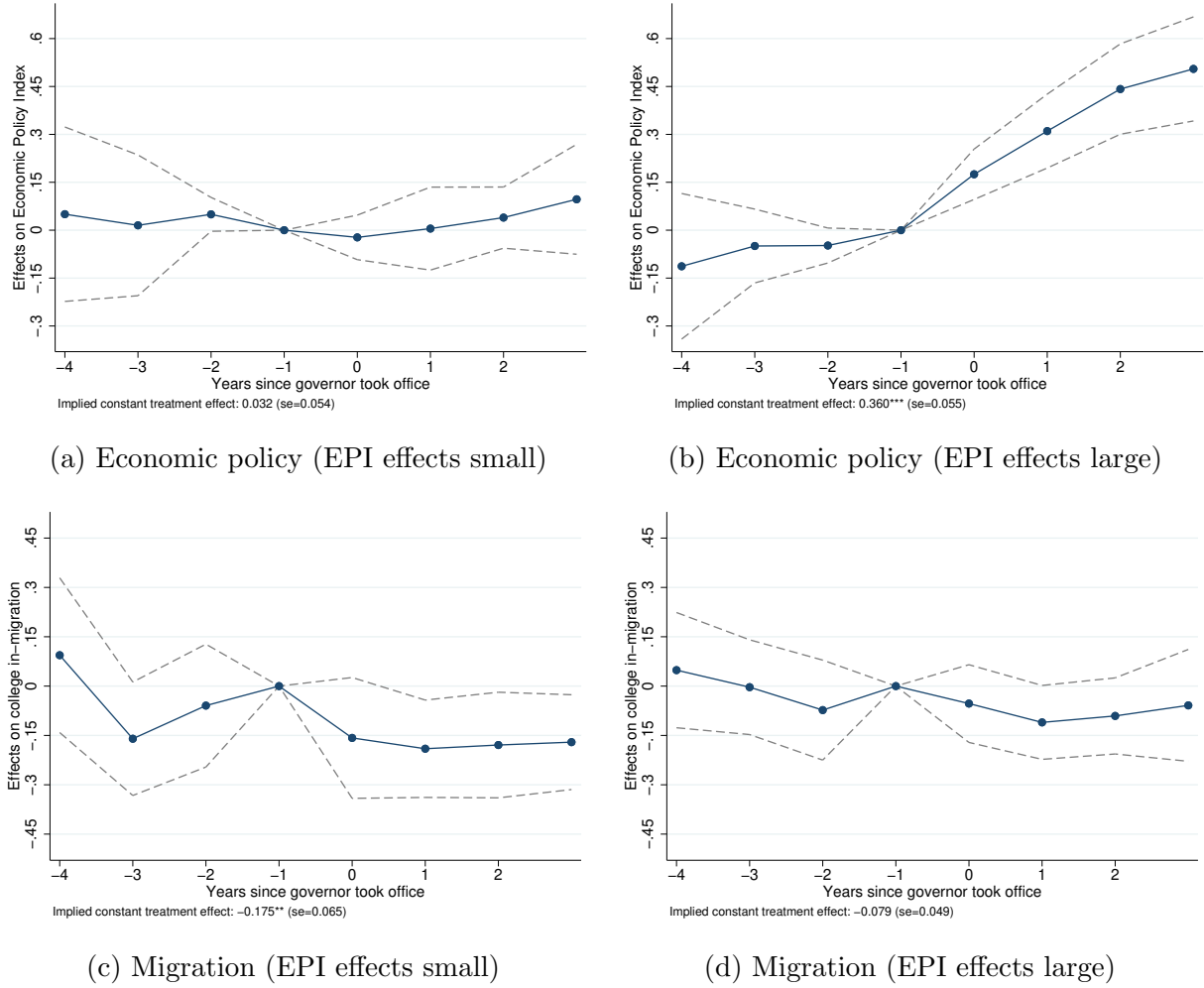
Notes: Panels (a), (c), (e), and (g) present levels of non-college in-migration, normalized by pre-treatment levels, for treatment and control group states. Panels (b), (d), (f), and (h) present coefficient estimates and 95% confidence intervals corresponding to estimating equation (1).

Figure C3: Louisiana and other R-to-D transitions from 2013-2017



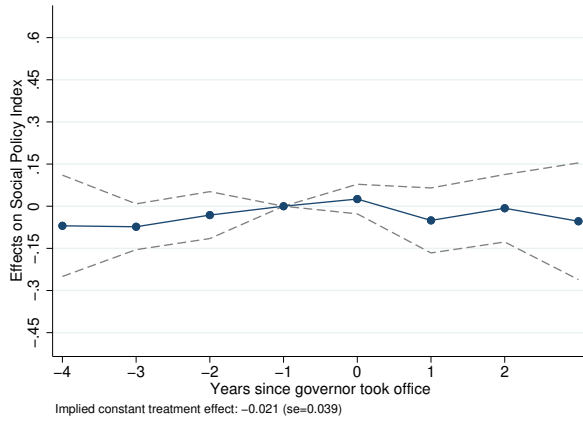
Notes: Panel (a) presents estimated treatment effects for all states switching from Republican to Democratic governors from 2013 to 2017. Panel (b) presents the comparable treatment effect for Louisiana, where John Bel Edwards was elected as a Democratic governor and took office in 2016. Panel (c) presents the results for the sample from Panel (a), excluding Louisiana. This is the figure that is included in the text in Figure 3. Panel (b) illustrates a clear and anomalous spike in in-migration leading up to Edwards' election, which makes it difficult to draw any causal conclusions in the full sample in panel (a). As shown in Table 2, ignoring these divergent pre-trends, estimates based on the sample used in panel (a), which includes Louisiana, imply a decrease in in-migration under Democratic governors. Excluding Louisiana, which restores the reasonable pre-trends shown in panel (c), on the other hand, imply little change in in-migration after a Democratic governor takes office during this period.

Figure C4: Effects of Democratic-to-Republican Transitions, by Economic Policy Effects

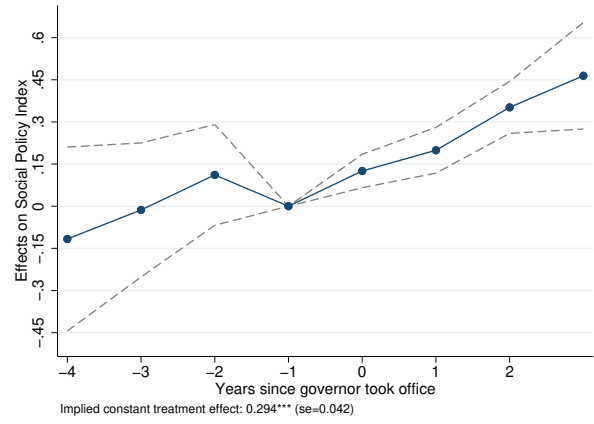


Notes: Figure presents coefficient estimates and 95% confidence intervals corresponding to estimating equation (1). Treatment events are switches from Democratic to Republican governors occurring 2003-2017, and states are divided according to the size of the estimated effects on economic policy. Migration calculations are based on college graduates aged 26+ employed in the private sector. Policy indices are drawn from Caughey and Warshaw (2016); more positive numbers reflect more conservative policies in place.

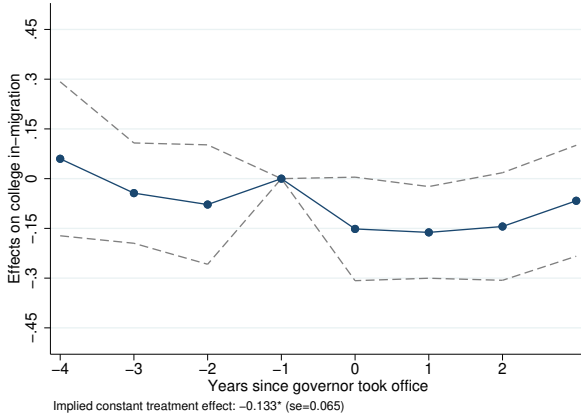
Figure C5: Effects of Democratic-to-Republican Transitions, by Social Policy Effects



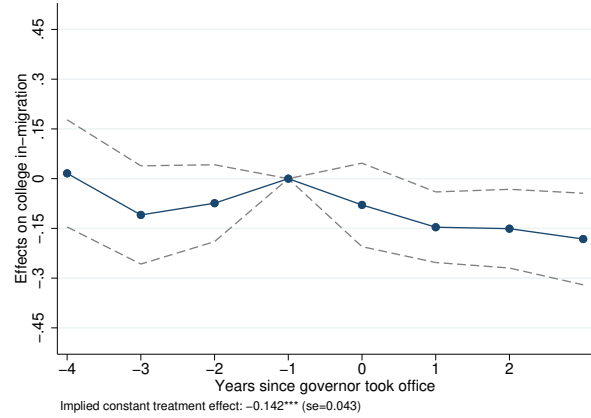
(a) Social policy (SPI effects small)



(b) Social policy (SPI effects large)



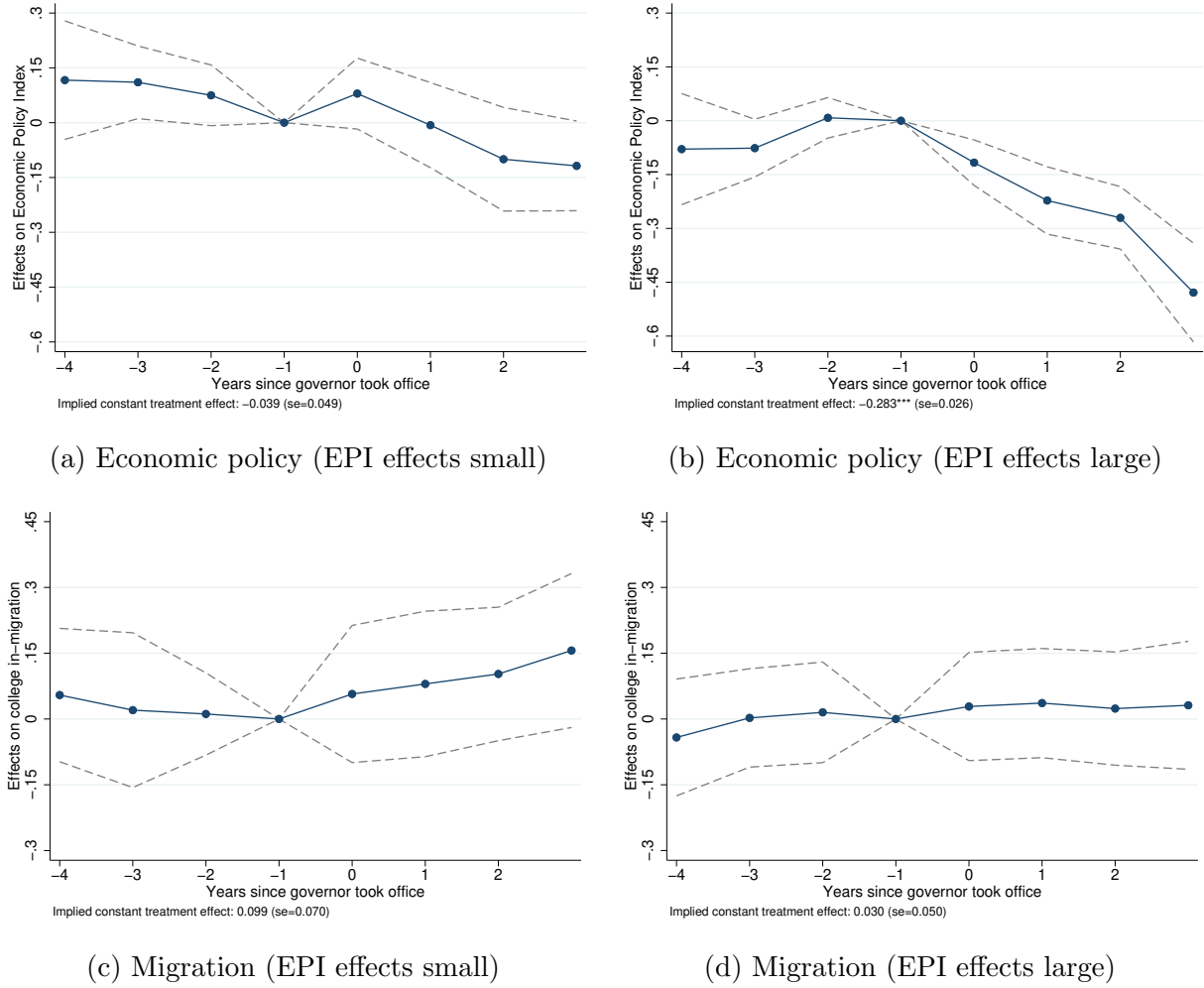
(c) Migration (SPI effects small)



(d) Migration (SPI effects large)

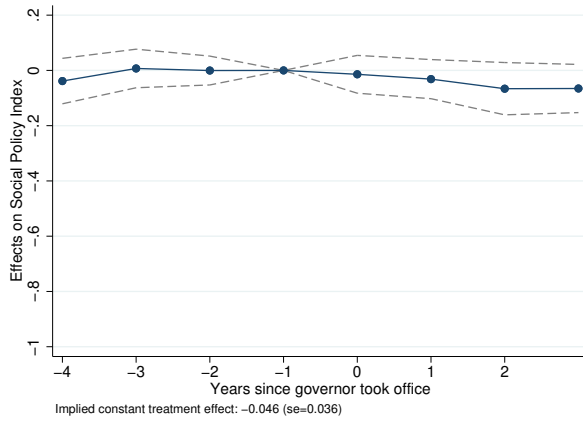
Notes: Figure presents coefficient estimates and 95% confidence intervals corresponding to estimating equation (1). Treatment events are switches from Democratic to Republican governors occurring 2003-2017, and states are divided according to the size of the estimated effects on social policy. Migration calculations are based on college graduates aged 26+ employed in the private sector. Policy indices are drawn from Caughey and Warshaw (2016); more positive numbers reflect more conservative policies in place.

Figure C6: Effects of Republican-to-Democratic Transitions, by Economic Policy Effects

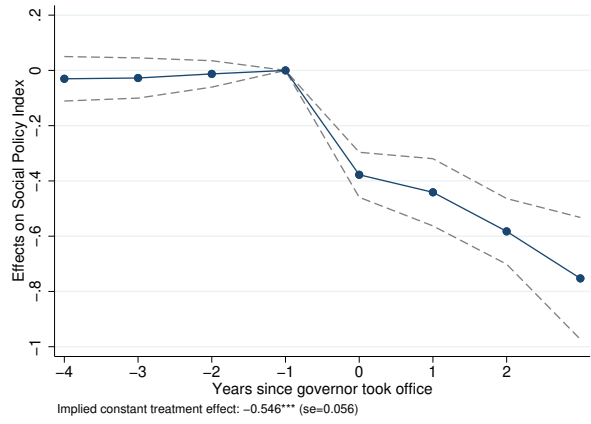


Notes: Figure presents coefficient estimates and 95% confidence intervals corresponding to estimating equation (1). Treatment events are switches from Republican to Democratic governors occurring 2003-2017, and states are divided according to the size of the estimated effects on economic policy. Migration calculations are based on college graduates aged 26+ employed in the private sector. Policy indices are drawn from Caughey and Warshaw (2016); more positive numbers reflect more conservative policies in place.

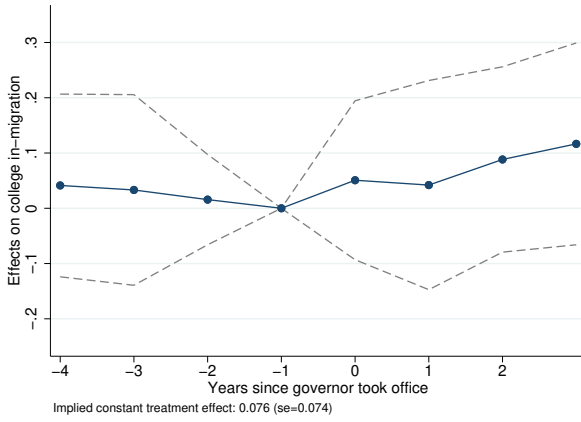
Figure C7: Effects of Republican-to-Democratic Transitions, by Social Policy Effects



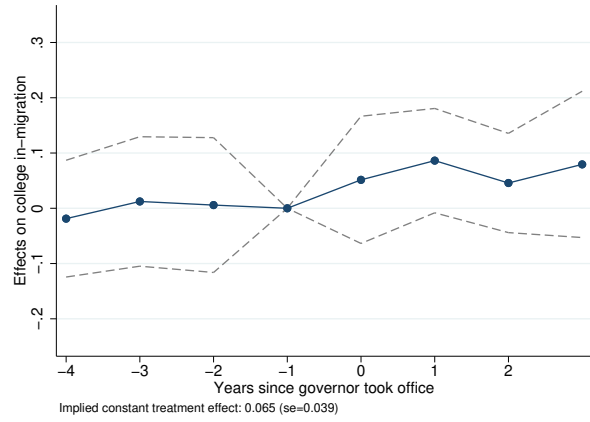
(a) Social policy (SPI effects small)



(b) Social policy (SPI effects large)



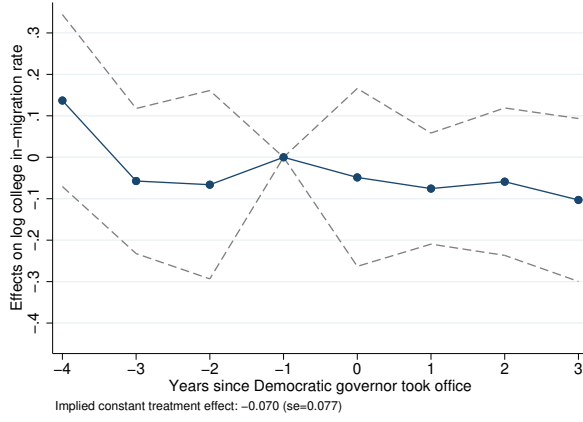
(c) Migration (SPI effects small)



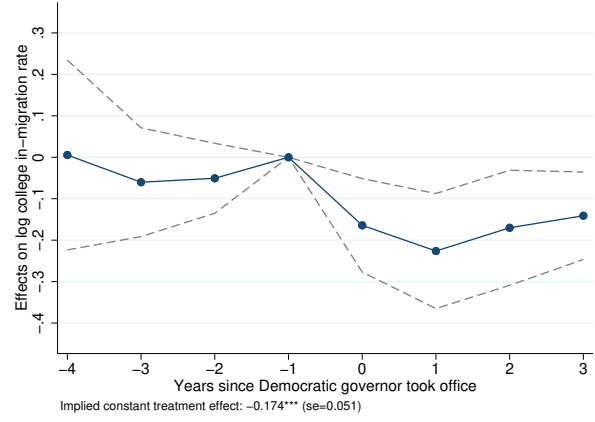
(d) Migration (SPI effects large)

Notes: Figure presents coefficient estimates and 95% confidence intervals corresponding to estimating equation (1). Treatment events are switches from Republican to Democratic governors occurring 2003-2017, and states are divided according to the size of the estimated effects on social policy. Migration calculations are based on college graduates aged 26+ employed in the private sector. Policy indices are drawn from Caughey and Warshaw (2016); more positive numbers reflect more conservative policies in place.

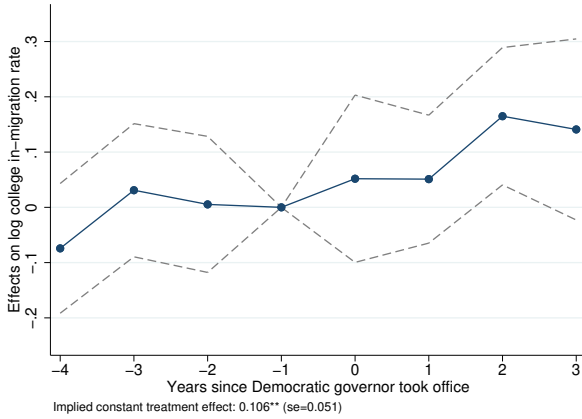
Figure C8: College in-migration responses by control of state legislature



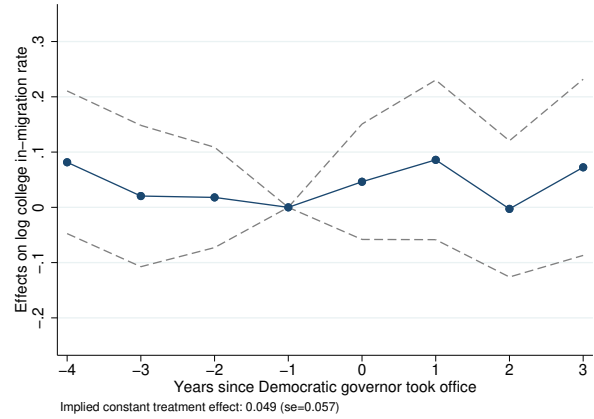
(a) D-to-R, aligned legislature



(b) D-to-R, non-aligned legislature



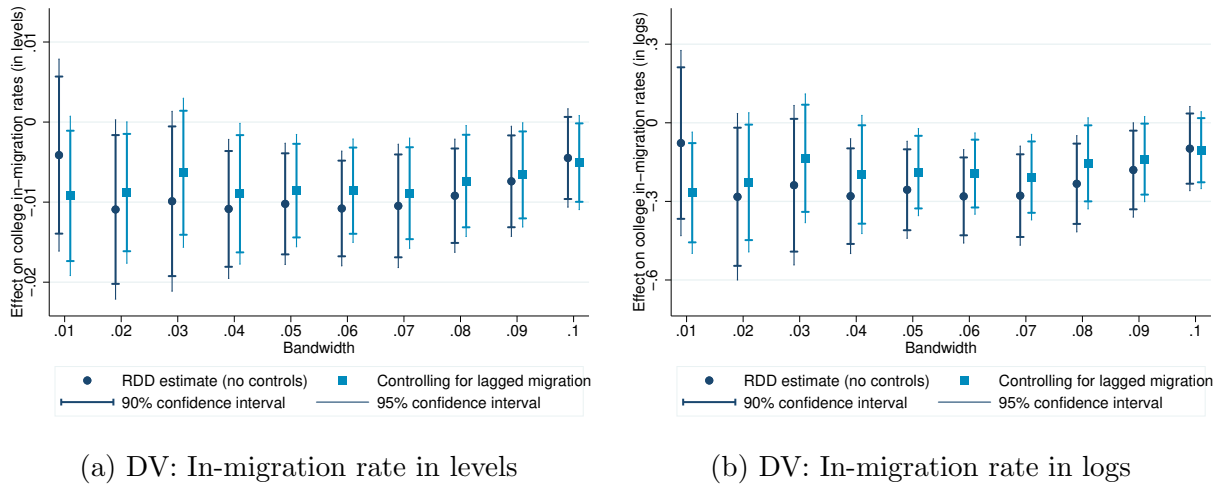
(c) R-to-D, aligned legislature



(d) R-to-D, non-aligned legislature

Notes: Figure presents coefficient estimates and 95% confidence intervals corresponding to estimating equation (1). Treatment events include any switch occurring 2003-2017. States are divided according to whether both chambers of the state legislature share the party of the new governor during the first post-treatment year. Migration calculations are based on college graduates aged 26+ employed in the private sector. Policy indices are drawn from Caughey and Warshaw (2016); more positive numbers reflect more conservative policies in place.

Figure C9: Robustness of regression discontinuity estimates to controls and bandwidth choices



Notes: Figure displays point estimates (with 90% and 95% confidence intervals) for regression discontinuity estimates of the effect of a Republican win on college graduates' average in-migration rates (the fraction of college graduates who lived in a different state the prior year) during the four post-election years. The x -axis depicts integer bandwidths ranging from .01 (i.e., including only elections where the Republican got between 49% and 51% of the two-party vote share) to .10. In panel (a), the dependent variable is the in-migration rate; in panel (b) the dependent variable is the log in-migration rate. Dark blue circles do not include controls; light blue squares control for the in-migration rate, panel (a), or log in-migration rate, panel (b), during the election year (i.e., the last pre-election year). All specifications use linear polynomials and uniform kernels.

Table C1: Governor partisanship and in-migration for non-college workers

	(1)	(2)	(3)	(4)	(5)	(6)
DV:	Log in-migration rate (non-coll., emp. in priv. sector)					
Ages:	26+	26+	26+	26-45	26-45	26-45
US Citizen:				Yes	Yes	Yes
Flips:	All	D-to-R	R-to-D	All	D-to-R	R-to-D
Republican Governor	-0.029 (0.029)			0.006 (0.031)		
Republican Governor		-0.028 (0.031)			0.032 (0.040)	
Democratic Governor			0.030 (0.050)			0.017 (0.054)
R^2	0.936	0.942	0.930	0.894	0.898	0.890
N	2174	853	1321	2174	853	1321
N. states	50	42	44	50	42	44

* $p < .10$, ** $p < .05$, *** $p < .01$. Estimates are based on a staggered rollout different-in-difference estimator; see equations (2) and (3). We consider elections from 2002-2016. Standard errors in parentheses are clustered at the state-level.

Table C2: Difference-in-difference estimates of governor partisanship and migration (robustness to a simpler specification)

	(1)	(2)	(3)	(4)	(5)
DV:	Net log migration	Log out-migration	Log in-migration	Log in-migration	C-NC in-mig.
College:	Yes	Yes	Yes	Yes	
Employed:	Priv. sector	Priv. sector	Priv. sector	Priv. sector	Priv. sector
Ages:	26+	26+	26+	26-45	26-45
US Citizen:				Yes	Yes
	All flips				
Republican Governor	-0.065* (0.037) [$p : .121$]	0.004 (0.019) [$p : .835$]	-0.061** (0.023) [$p : .021$]	-0.080*** (0.026) [$p : .006$]	-0.051 (0.031) [$p : .112$]
N	2172	2172	2174	2174	2174

* $p < .10$, ** $p < .05$, *** $p < .01$. Estimates are based on a staggered rollout different-in-difference estimator. Sample and measurement choices are identical to Panel A of Table 1, but the estimating equation is simpler: $Y_{s,t,\tau,F} = \alpha_{s,\tau,F} + \delta_{t,\tau,F} + \beta_R \text{Republican}_{s,t,\tau} + \varepsilon_{s,t,\tau,F}$. Unlike equation (3), which is used in Table 1, this equation suppresses the pre-trend coefficients.

Table C3: Heterogeneous migration responses to governor partisanship

	(1)	(2)	(3)	(4)	(5)	(6)
DV:	Log in-migration rate					
Employed:	Priv.	Any	Priv.	Priv.	Priv.	Priv.
Has children:			No	Yes		
Gender:					Men	Women
Panel A: All flips						
Republican Governor	-0.079*** (0.024)	-0.058*** (0.018)	-0.090*** (0.026)	-0.053 (0.039)	-0.072** (0.033)	-0.074* (0.040)
N	2174	2174	2174	2173	2174	2171
$\bar{y}$	0.032	0.029	0.042	0.021	0.034	0.030
Panel B: Democrat-to-Republican flips						
Republican Governor	-0.103*** (0.029)	-0.089*** (0.025)	-0.142*** (0.033)	-0.019 (0.048)	-0.062 (0.043)	-0.153** (0.059)
N	853	853	853	852	853	853
$\bar{y}$	0.033	0.029	0.043	0.021	0.035	0.030
Panel C: Republican-to-Democrat flips						
Democratic Governor	0.056* (0.032)	0.029 (0.024)	0.042 (0.031)	0.086 (0.054)	0.083** (0.041)	0.000 (0.042)
N	1321	1321	1321	1321	1321	1318
$\bar{y}$	0.031	0.029	0.041	0.021	0.033	0.029

* $p < .10$, ** $p < .05$, *** $p < .01$. Estimates are based on a staggered rollout different-in-difference estimator; see equations (2) and (3). We consider elections from 2002-2016. Standard errors in parentheses are clustered at the state-level. Except for column (2), the sample is based on college graduates age 26+ and employed in the private sector. $\bar{y}$ refers to the average annual migration rate of this sample (not the logarithm of the migration rate).

Table C4: College in-migration responses by control of state legislature

	(1)	(2)	(3)	(4)	(5)	(6)
DV:	Economic policy	Social policy	Log of coll. in-migration	Economic policy	Social policy	Log of coll. in-migration
Legislature:	Aligned			Not aligned		
Panel A: All flips						
Republican Governor	0.236*** (0.032) [<i>p</i> : .000]	0.344*** (0.060) [<i>p</i> : .000]	-0.089* (0.045) [<i>p</i> : .070]	0.133*** (0.040) [<i>p</i> : .010]	0.134*** (0.045) [<i>p</i> : .004]	-0.102** (0.043) [<i>p</i> : .038]
N	996	996	996	1627	1627	1627
N. states	46	46	46	48	48	48
Panel B: Democrat-to-Republican flips						
Republican Governor	0.260*** (0.052) [<i>p</i> : .000]	0.196*** (0.046) [<i>p</i> : .000]	-0.070 (0.077) [<i>p</i> : .402]	0.157** (0.075) [<i>p</i> : .122]	0.075 (0.064) [<i>p</i> : .290]	-0.174*** (0.051) [<i>p</i> : .017]
N	490	490	490	494	494	494
N. states	31	31	31	27	27	27
Panel C: Republican-to-Democrat flips						
Democratic Governor	-0.214*** (0.042) [<i>p</i> : .000]	-0.485*** (0.093) [<i>p</i> : .001]	0.106** (0.051) [<i>p</i> : .080]	-0.115** (0.042) [<i>p</i> : .053]	-0.177** (0.068) [<i>p</i> : .032]	0.049 (0.057) [<i>p</i> : .482]
N	506	506	506	1133	1133	1133
N. states	27	27	27	35	35	35

* $p < .10$, ** $p < .05$, *** $p < .01$. Estimates are based on a staggered rollout different-in-difference estimator; see equations (2) and (3). We consider elections from 2002-2016. States are divided according to whether both chambers of the state legislature share the party of the new governor during the first post-treatment year. Standard errors in parentheses are clustered at the state-level and p -values in brackets are from a cluster wild bootstrapped test of the null of zero treatment effect (Roodman et al., 2019), since we sometimes have relatively few states in the samples. Migration calculations are based on college graduates aged 26+ employed in the private sector. Policy indices are drawn from Caughey and Warshaw (2016); more positive numbers reflect more conservative policies in place.

Table C5: Regression discontinuity estimates of governor partisanship and migration

	(1)	(2)	(3)	(4)	(5)	(6)
DV:	In-migration, college graduates, private sector employed					
Estimator:	OLS	OLS	OLS	OLS	CCT	CCT
Kernel:	Uniform	Uniform	Uniform	Uniform	Uniform	Triangle
Bandwidth:	.02	.04	.06	.08	.043	.071
Panel A: In-migration rate in levels						
Republican win	-0.0088* (0.0043) [$p : .042$]	-0.0090** (0.0043) [$p : .048$]	-0.0086** (0.0032) [$p : .012$]	-0.0074** (0.0034) [$p : .044$]	-0.0092** (0.0042) [$p : .024$]	-0.0084** (0.0035) [$p : .023$]
N	39	83	106	136	85	119
Panel B: In-migration rate in logs						
Bandwidth:	.02	.04	.06	.08	.054	.059
Republican win	-0.228* (0.129) [$p : .100$]	-0.196* (0.111) [$p : .083$]	-0.192** (0.076) [$p : .012$]	-0.153* (0.086) [$p : .090$]	-0.166* (0.086) [$p : .032$]	-0.174* (0.090) [$p : .053$]
N	39	83	106	136	102	105

* $p < .10$, ** $p < .05$, *** $p < .01$. CCT estimates refer to Calonico et al. (2014), and use the optimally selected bandwidth. All columns control for lagged migration rates (see Figure C9). Standard errors in parentheses are clustered at the state-level and p -values in brackets are from a cluster wild bootstrapped test of the null of zero treatment effect (Roodman et al., 2019), since we sometimes have relatively few states in the samples.

D Structural analysis

D.1 Model overview

Our model of migration is based on Bryan and Morten (2019). There are other models of spatial sorting (e.g., Diamond (2016)). An advantage of the Bryan-Morton model is that identification and estimation are based on observed migration flows, whereas some other models infer migration behavior from relative changes in city size over long periods. Given that our reduced form estimates focus specifically on migration flows, this model provides a more natural framework to incorporate those results.

There are two types of agents: College-educated workers and non-college workers. These two types have different wage rates, productivity, appreciation of amenities, migration costs, and political preferences. However, within the education type, all workers are *ex ante* identical, although they will realize different productivity draws and migration choices.

Workers are infinitely-lived agents who, during each period, statically maximize their utility each period. Each period (one year) consists of two sub-periods. First, the worker draws a stochastic, idiosyncratic shock which determines whether she will migrate to a new state. This shock reflects idiosyncratic events like job dissatisfaction that leads a worker to look for new employment opportunities; a spouse’s beginning or finishing graduate school; children facing key transitions into, out of, or between schools, etc. We allow the probability of realizing this “migration shock” to differ across states of residence to match the empirical fact that different states see very different out-migration rates. This shock is helpful because it allows our model to generate asymmetric responses in which in-migration responds, but out-migration does not, which standard models do not generate (Monras, 2020), but which is a salient feature of our empirical results above.

Second, workers who receive the “migration shock” (and will move to a new state) draw a vector of idiosyncratic human capital opportunities available in these new states, excluding the one where they currently live. We see this as reflecting potential employment opportunities across those states. Like much of the migration literature, we gain substantial tractability by assuming that these idiosyncratic draws are independent across locations and follow a Fréchet distribution (discussed below).

A human capital draw reflects the idiosyncratic productivity of a worker in a particular state. Each draw is valued at the local wage rate in that state, which depends on productivity and relative skills supplies (discussed below). In addition to wages, a worker considering moving from state o (origin) to d (destination) considers three additional factors. First, she considers one factor unique to the o - d pair. This reflects migration costs (as is standard in the literature), but also idiosyncratic features of d making it attractive to those moving from

o (such as cultural similarities). Thus, our model allows residents from liberal states, for instance, to prefer moving to other liberal states for general cultural reasons – rather than solely due to the partisanship of the governor – in order to avoid misattributing broader cultural similarities to partisan politics.

Second, she considers the general amenities in the state, which could reflect how attractive the state is in terms of weather, local restaurants, crime rates, school quality, or natural attractions. Amenities like these are standard in the literature and have been shown to be important determinants of migration decisions (Diamond and Gaubert, 2022). We allow college and non-college workers to have different valuations of the amenities in the state but hold these amenities fixed across all workers within each education-type, regardless of their origin state.

Finally, she considers a political amenity, which depends on the partisanship of the governor. This parameter is our key conceptual departure from the previous literature.

D.2 Setup

The **timing** of the model is as follows: there are K states. In period $t = 0$, each worker (indexed by i) is born in her origin state of o . Each worker receives a destination-specific skill draw $s_{id,t=0}$ for each destination d and migrates to a state $d_{i,t=0}^*$ maximizing her utility based on wage rates, productivity, appreciation of amenities, migration costs, and political preference. From $t \geq 1$, origin-specific migration shock μ_o^g activates. Workers who do not receive the shock will stay in the previous location $d_{i,t-1}^*$; workers who receive the shock will get a new skill draw $s_{id,t}$, for $d \neq o$, and migrate based only on the utility in t .

D.2.1 Utility

Let the utility of worker i belonging to education group g and moving from origin o to destination d in period $t = 0$ be given by:

$$U_{ido,t}^g = c_{id,t}^g \alpha_d^g \eta_{do}^g (1 - \gamma_{p(d,t)}^g)$$

where $c_{id,t}^g$ is her consumption and α_d^g is the general amenity in d for education group $g \in \{C, N\}$. We consider these amenities fixed across all individuals with the same education over time, but we allow all parameters to vary across education groups, denoted by $g = C$ for college graduates and $g = N$ for non-college. Define η_{do}^g to be both the migration costs of moving from o to d , as well as idiosyncratic features of d that make it particularly attractive to those moving from o .

The term $\gamma_{p(d,t)}^g$ captures a group-specific preference wedge emerging from group g 's disutility of living under a governor of partisanship $p(d,t) \in \{\text{Republican, Democrat}\}$. When this wedge is positive, workers must be compensated with $1/(1 - \gamma_{p(d,t)}^g)$ times greater consumption in order to be indifferent. Since college-educated workers are to the left of non-college ones, we normalize $\gamma_{Dem}^C = 0$ and $\gamma_{Rep}^N = 0$ so that γ_{Rep}^C captures college graduates' disutility of Republican governance, relative to Democratic governance, and conversely for non-college workers' disutility of Democratic governance.⁴⁹

Workers who get the migration shock and will move across states receive one idiosyncratic skill draw for each potential destination state. Our representation of exogenous migration shocks is similar to Monras (2020), while our representation of idiosyncratic human capital draws is similar to Bryan and Morten (2019) and Hsieh et al. (2019). We think of this as a match-specific human capital draw reflecting the best employment opportunity available in that state. We assume these idiosyncratic opportunities, denoted $s_{id,t}^g$, are drawn from a multivariate Fréchet distribution:

$$F(s_{1,t}^g, s_{2,t}^g, \dots, s_{K,t}^g) = \exp \left(- \left[\sum_{d=1}^K s_{d,t}^{-\theta^g} \right] \right)$$

The distribution of $s_{id,t}^g$ is governed by the scale parameter θ^g . A higher value of θ^g implies less skill dispersion across locations, such as would be the case if all states afforded the worker an equally good employment match. As θ^g decreases, there is a greater difference between the match-specific human capital realizations a worker sees in different locations. This parameter is key because it determines how close to indifferent workers are between the employment opportunities available in different states. Note that this formulation assumes each worker's draws are independent across states, although state-specific factor supplies and wage rates will lead to a correlation in *earnings* across different workers within the same state.

We derive the indirect utility function for worker i in group g who moved from o to d in period $t = 0$ as:

$$V_{ido,t}^g = \alpha_d^g \eta_{do}^g (1 - \gamma_{p(d,t)}^g) w_{d,t}^g s_{id,t}^g \quad (6)$$

⁴⁹This parameter varies across education groups but not across individuals within the group. It is stark and unrealistic to assume that all individuals of the same education have the same political preferences. More realistically, we could assume that only some share $\chi < 1$ of college graduates are liberal. But with this formulation, the decline in college in-migration that we estimated above would have been generated entirely by this χ share of liberal college graduates. This would require a larger value of the γ disutility parameter, since this subset's behavioral responses would have to be larger, but would not change the aggregate size of the migration response (since this is a targeted moment). This would change the welfare implications of our model (since it would increase the intensity and heterogeneity in preferences), but likely would not affect other implications, which is where our analysis focuses.

where $w_{d,t}^g$ is the human capital price in d for group g in period t . That is, if $w_{d,t}^g > w_{d',t}^g$, then d has higher “wages” (human capital prices) than d' : a worker with the same quality of employment opportunity in both d and d' (i.e., the same human capital draw: $s_{id,t}^g = s_{id',t}^g$) will have higher earnings and consumption in d since it is a higher wage state.

It is convenient to define the overall utility returns of destination d for a worker of group g and origin o as $\tilde{w}_{do,t}^g = \alpha_d^g \eta_{do}^g (1 - \gamma_{p(d,t)}^g) w_{d,t}^g$ so that $V_{ido,t}^g = \tilde{w}_{do,t}^g s_{id,t}^g$.

Our key assumption is that workers choose the location that yields the highest indirect utility. Note that the only idiosyncratic component that varies across individuals is their vector of skill draws. Since these are Fréchet distributed, a property of the Fréchet distribution is that the share of individuals from origin o who choose to work in destination d in period $t = 0$ can be written as:

$$\hat{\pi}_{do,t}^g \equiv \frac{L_{do,t}^g}{L_{o,t}^g} = \frac{\tilde{w}_{do,t}^{g\theta g}}{\sum_{j=1}^N \tilde{w}_{jo,t}^{g\theta g}} \quad (7)$$

where $L_{o,t}^g$ denotes the number of group- g workers from origin o in period t , and $L_{do,t}^g$ denotes the number who move to d in the same period. Relative returns, including non-wage returns in terms of higher amenities or lower migration costs drive these migration flows.

A property of the Fréchet distribution is that conditional on the workers’ optimal choice of where to live, the average skill of workers choosing to move from o to d can be written as:

$$\mathbb{E}(s_{d,t}^g \mid \text{choose } d \text{ from } o) = \bar{\Gamma}^g \left(\frac{1}{\hat{\pi}_{do,t}^g} \right)^{\frac{1}{\theta g}} \quad (8)$$

where $\bar{\Gamma}^g = \Gamma(\frac{\theta g - 1}{\theta g})$ and $\Gamma(\cdot)$ is the gamma function. This property is important, as it allows us to infer unobserved human capital from observed migration rates. Human capital supplied is the product of the skill conditional on migrating from and the number of workers migrating from o to d summed over all origins

$$H_{d,t}^g(\text{supply}) = \sum_{o=1}^N L_{o,t}^g \hat{\pi}_{do,t}^g \bar{\Gamma}^g \left(\frac{1}{\hat{\pi}_{do,t}^g} \right)^{\frac{1}{\theta g}} \quad (9)$$

Starting from $t = 1$, workers who receive a migration shock, which occurs with probability μ_o^g , will migrate to other states, and the rest will stay in the previous location. The indirect utility function in the period of $t \geq 1$ is :

$$V_{ido,t}^g = (1 - \mu_o^g) V_{ido,t-1}^g + \max_{d, d \neq o} \{ \mu_o^g (\tilde{w}_{do,t}^g s_{id,t}^g) \}$$

The share of individuals from origin o who choose to work in destination d in $t \geq 1$ is:

$$\pi_{do,t}^g \equiv \frac{L_{do,t}^g}{\mu_o^g L_{o,t-1}^g} = \frac{(\tilde{w}_{do,t}^g)^{\theta^g}}{\sum_{j \neq o} (\tilde{w}_{jo,t}^g)^{\theta^g}} \quad (10)$$

The average skill of workers choosing to move from o to d in t is

$$\mathbb{E}(s_{d,t}^g \mid \text{choose } d \text{ from } o) = \bar{\Gamma}^g \left(\frac{1}{\pi_{do,t}^g} \right)^{\frac{1}{\theta^g}}$$

Total efficiency units of labor supply in $t \geq 1$ is

$$H_{d,t}^g(\text{supply}) = \underbrace{(1 - \mu_d^g) H_{d,t-1}^g(\text{supply})}_{\text{stayers}} + \underbrace{\sum_{o=1}^N L_{o,t-1}^g \mu_o^g \pi_{do,t}^g \bar{\Gamma}^g \left(\frac{1}{\pi_{do,t}^g} \right)^{\frac{1}{\theta^g}}}_{\text{movers}}$$

D.2.2 Production

Firms produce a single final good, which is costlessly traded and is chosen as the numeraire ($p = 1$). Output is produced by perfectly competitive firms. They combine the effective labor (i.e., total human capital) of the two education groups using a Constant Elasticity of Substitution (CES) production function.

Total output in state d in period t is given by:

$$Y_{d,t} = A_{d,t} \left[(H_{d,t}^C)^{\frac{\sigma-1}{\sigma}} + (H_{d,t}^N)^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}$$

where $A_{d,t}$ is the exogenous location-specific total factor productivity (TFP) and $\sigma \geq 1$ is the elasticity of substitution between college and non-college workers. $H_{jd,t}^g$ is the total efficiency units of labor employed by firm j in time t , which is simply the sum of skill draws for workers working at firm j :

$$H_{jd,t}^g = \sum_{i \in P_{j,t}} s_{id,t}^g$$

where $P_{j,t}$ denotes the set of workers employed at firm j .

The profits for a representative firm in location d are given by:

$$\Pi_{jd,t} = Y_{jd,t} - w_{jd,t}^C H_{jd,t}^C - w_{jd,t}^N H_{jd,t}^N$$

where $w_{jd,t}$ is the wage paid per effective unit of labor by firm j . In this economy, firms compete for each type of worker by setting wages. Since the labor market is perfectly

competitive, in equilibrium, $\Pi_{jd,t} = 0$ and $w_{jd,t}^g = w_{d,t}^g \quad \forall j$. This prevailing wage (or the price of human capital) is the same $w_{d,t}^g$ discussed in the worker's problem above.

D.2.3 General equilibrium

A competitive equilibrium in this economy consists of destination choices, total efficiency of labor in each destination $H_{d,t}^g$, and a wage $w_{d,t}^g$ such that in each period t :

- Workers choose the workplace that maximizes their utility
- Firms choose efficient labor $H_{d,t}^g$ to maximize profit
- $w_{d,t}^g$ clears labor market for each destination

D.3 Additional derivations

D.3.1 The probability that an individual chooses one state

$$V_{ido} = \alpha_d^g(1 - \gamma)w_{d,t}\eta_{do}^g\epsilon_{do}s_{id} = \tilde{w}_{do}s_{id}$$

By extreme value theory, if $U(\cdot)$ is Frechet then so is $\max u(\cdot)$, which is $V(\cdot)$. Without loss of generality, consider the probability that worker chooses destination 1 and denote this by π_{1o} :

$$\begin{aligned}\pi_{1o} &= Pr[w_{1o}s_{i1} \geq \tilde{w}_{so}s_{is}] \quad \forall s \neq 1 \\ &= Pr\left[s_{is} \leq \frac{w_{1o}s_{i1}}{\tilde{w}_{so}}\right] \quad \forall s \neq 1 \\ &= \int F_1(s_i, T_2s_i, \dots, T_Ns_i)ds_i\end{aligned}$$

where $F_1(\cdot)$ is the derivative of cdf with respect to its first argument and $T_l \equiv \frac{w_{1o}}{\tilde{w}_{lo}}$. Recall that

$$F(s_1, s_2, \dots, s_N) = \exp\left(-\left[\sum_{d=1}^N s_d^{-\theta}\right]\right)$$

Taking the derivative with respect to s_1 gives

$$F_1(s_i, T_2s_i, \dots, T_Ns_i) = \theta s_i^{-\theta-1} \exp(-\bar{T}_r s_i^{-\theta})$$

where $\bar{T}_r \equiv \sum_{s=1}^N (\frac{w_{ro}}{\tilde{w}_{so}})^{-\theta}$.

$$\begin{aligned}
\pi_{1o} &= \int F_1(s_i, T_2 s_i, \dots, T_N s_i) ds_i \\
&= \frac{1}{\bar{T}_r} \int \bar{T}_r \theta s_i^{-\theta-1} \exp(-\bar{T}_r s_i^{-\theta}) dF s_i \\
&= \frac{1}{\bar{T}_r} \int dF s_i \\
&= \frac{1}{\bar{T}_r} \\
&= \frac{\tilde{w}_{ro}^\theta}{\sum_{s=1}^N \tilde{w}_{so}^\theta}
\end{aligned}$$

D.3.2 Average skill of workers

To calculate this conditional expectation, we use the extreme value properties of the Frechet distribution. Let $y_d = \tilde{w}_{d,t} s_d$ denote the key destination choice term.

$$y^* \equiv \max_s \{y_s\} = \max_s \{s_s / \zeta_s\} = s^* / \zeta^*.$$

$$\begin{aligned}
Pr[y^* < z] &= \prod_{s=1}^N Pr[y_s < z] \\
&= \prod_{s=1}^N Pr[\tilde{w}_s s_s < z] \\
&= \prod_{s=1}^N Pr[s_s < z / \tilde{w}_s] \\
&= \prod_{s=1}^N \exp[-(z / \tilde{w}_s)^{-\theta}] \\
&= \exp\left\{-\sum_{s=1}^N (z / \tilde{w}_s)^{-\theta}\right\} \\
&= \exp\{-\bar{T} z^{-\theta}\}
\end{aligned}$$

where $\bar{T} \equiv \sum_{s=1}^N \tilde{w}_s^\theta$

The ability of people in their chosen place is also Frechet distributed:

$$F(x) \equiv Pr[s^* < x] = \exp\{-T^* x^{-\theta}\} \quad (11)$$

where $T^* \equiv \sum_{s=1}^N (\frac{\tilde{w}_s}{w^*})^\theta = \frac{1}{\pi}$.

The expected skill of the chosen d is

$$\begin{aligned} E(s_d) &= \int_0^\infty s dF(s) \\ &= \int_0^\infty \theta T^* s^{-\theta} e^{-T^* s^{-\theta}} ds \end{aligned} \tag{12}$$

Recall the Gamma function is $\Gamma(\alpha) \equiv \int_0^\infty x^{\alpha-1} e^{-x} dx$. We replace $x = T^* s^{-\theta}$, $dx = \theta T^* s^{(-\theta-1)} ds$, $s = (\frac{T^*}{x})^{\frac{1}{\theta}}$. We can show that

$$\begin{aligned} E(s_d) &= (T^*)^{\frac{1}{\theta}} \int_0^\infty x^{-\frac{1}{\theta}} e^{-x} dx \\ &= (T^*)^{\frac{1}{\theta}} \Gamma\left(\frac{\theta-1}{\theta}\right) \end{aligned} \tag{13}$$

Applying this result to our equation, we have

$$\begin{aligned} E(s_d^\theta \mid \text{choose } d \text{ from } o) &= (T^*)^{\frac{1}{\theta}} \Gamma\left(\frac{\theta-1}{\theta}\right) \\ &= \pi_{do}^{-\frac{1}{\theta}} \Gamma\left(\frac{\theta-1}{\theta}\right) \end{aligned} \tag{14}$$

D.4 Identification

D.4.1 Elasticity of skills substitution: σ

A large literature estimates the elasticity of substitution between skills. We use $\sigma = 2.6$, which is the estimate from Jerzmanowski and Tamura (2020) for college and non-college workers. Acemoglu and Autor (2011) for some discussion, who calibrate a value of $\sigma = 2.9$ for the 1964-2008 period. It is worth noting that we use a value such that $\sigma > 1$. This implies that a decrease in the supply of college-educated workers will increase inequality through two forces. First, it will raise the relative wages of college graduates (i.e., college and non-college workers are gross substitutes). Second, it will reduce the real wage of non-college workers (i.e., they are q -complements).

D.4.2 Fréchet parameter: θ^g

Workers' earnings are determined by state-specific human capital prices (the wage rate paid per effective unit of labor) and the individual human capital draw with which workers move to the state. Looking at equation (8), once one accounts for the cross-state variation in the returns to human capital and variation across od in the degree of worker self-selection, the remaining variation in earnings is informative about the underlying dispersion of human

capital draws. This insight was pointed out by Hsieh et al. (2019), and our approach to estimation follows theirs.

Specifically, we regress log annual earnings in the ACS on destination-by-origin fixed effects, while also controlling for age, gender, and fixed effects for year, occupation, and industry. We do this separately for college and non-college workers and calculate residual log earnings.

Since we control for destination-by-origin fixed effects, these residualized earnings have been purged of differential cross-state selection and heterogeneous returns to human capital. Since we control for individual-level covariates, the remaining dispersion reflects the variation in place-adjusted earnings that one worker could plausibly see across different potential states of residence. In this way, we can write a non-linear function of θ as a non-linear function of the distribution of residual wages:

$$\frac{\text{Variance}(\tilde{s}_i^g)}{(\text{Mean}(\tilde{s}_i^g))^2} = \frac{\Gamma\left(1 - \frac{2}{\theta^g}\right)}{\left(\Gamma\left(1 - \frac{1}{\theta^g}\right)\right)^2} - 1 \quad (15)$$

where $\tilde{s}_i^g$ is individual i 's residual log earnings, exponentiated.⁵⁰

D.4.3 Migration shock: μ_o^g

Since “migration shock” starts from $t \geq 1$, we precisely match the group-origin migration outflows from the data to calculate μ_o^g , the average of $\mu_{o,t}^g$ where

$$\mu_{o,t}^g = \frac{\sum_{j \neq o} L_{jo,t}^g}{L_{o,t-1}^g} \quad (16)$$

D.4.4 Moving cost: η_{do}^g

Taking the log of equation (10), we have

$$\ln(\pi_{do,t}^g) = \underbrace{\theta^g \ln(\alpha_d^g)}_{\text{destination fixed effect}} + \underbrace{\theta^g \ln(w_{d,t}^g)}_{\text{destination-time fixed effect}} + \underbrace{\theta^g \ln(1 - \gamma_{p(d,t)}^g)}_{\text{party-time fixed effect}} - \underbrace{\ln\left(\sum_{j=1, j \neq o}^N \left(w_{jo,t}^g\right)^{\theta^g}\right)}_{\text{origin-time fixed effect}} + \theta^g \ln \eta_{do}^g \quad (17)$$

After controlling for origin, destination, time, and party fixed effects, migration costs act as residuals that match the migration inflows.

⁵⁰Residual log earnings are mechanically mean zero across od , but exponentiated residual log earnings are not mean one because the exponential function is non-linear.

D.4.5 Productivity and wages: $A_{d,t}, w_{d,t}^g$

We jointly identify productivity and wages using two conditions. First, the assumption of perfect competition in the product market and the law of one price implies that price equals marginal cost, which yields:

$$\left(\frac{w_{d,t}^C}{A_{d,t}}\right)^{1-\sigma} + \left(\frac{w_{d,t}^N}{A_{d,t}}\right)^{1-\sigma} = 1 \quad (18)$$

Second, labor market clearing implies that total human capital demanded equals total human capital supplied. Human capital demanded can be written from the CES production function as a function of wages and total output.

$$H_{d,t}^g(demand) = \left(\frac{A_{d,t}^{\frac{\sigma-1}{\sigma}}}{w_{d,t}^g}\right)^{\sigma} Y_{d,t} = H_{d,t}^g(supply) \quad (19)$$

Given the above estimate of θ^g , and observable output and migration, this delivers three equations (since equation (19) must hold for $g \in \{C, N\}$) and three unknowns ($A_{d,t}, w_{d,t}^C, w_{d,t}^N$) for each state in each period.

Fundamentally, TFP and wages are pinned down by variation in GDP. Given an estimate of skill dispersion, our model tells us how human capital stocks can be determined by population size and migration (since migration entails selection: higher rates of in-migration crowd-in marginally lower human capital draws). Thus, surprisingly high levels of GDP (given the human capital stock) imply a high total factor productivity, and because all firms are assumed to compete in the same product market and face the same production function, there is a stable relationship between productivity and wages across states. Thus, variation in productivity combined with the available supply of human capital determines the wages.

D.4.6 General amenities: α_d^g

When governors in d and d' belong to the same party, politics will not affect workers' decisions about whether to migrate from o to d as opposed to o to d' . Considering equation (10) and taking the difference in migration between two same-governor destination states, we have

$$[\ln(\pi_{do,t}^g) - \ln(\pi_{d'o,t}^g)] = \theta^g \ln(\alpha_d^g/\alpha_{d'}^g) + \theta^g \ln(w_{d,t}^g/w_{d'}^g) + \theta^g \ln(\eta_{do}^g/(\eta_{d'o}^g)) \quad (20)$$

With θ^g , $w_{d,t}^g$, η_{do}^g at hand, we can identify relative amenities, relative to some reference

state d' (discussed below). Essentially, state d 's amenities are identified by higher migration into d and lower out-migration out of d than one would expect given wages and estimated migration costs.

D.4.7 Political amenities: $\gamma_{\text{Rep}}^C, \gamma_{\text{Dem}}^N$

To calculate our values of political amenities, we perform a grid search over different values for the pair γ_{Rep}^C and γ_{Dem}^N such that our model-implied effects of switching governors match our reduced form estimates. More specifically, our primary reduced form estimates are identified from eight states switching from a Democratic to a Republican governor between 2015 and 2017. For each value pair of values for γ_{Rep}^C and γ_{Dem}^N , we simulate two counterfactuals from our model. First, we simulate outcomes where all governors are held fixed as they were in 2014. Second, we simulate outcomes where all governors are held fixed as they were in 2017.

Let S denote the set of states that switched from a Democratic to a Republican governor between 2015 and 2017. The purpose of difference-in-difference is to identify the expected difference in potential outcomes between treated states under treatment and treated states in the absence of treatment: $E[Y_d(\text{Rep}) - Y_d(\text{Dem})|d \in S]$, which can be rewritten as $E[Y_d(\text{Rep})|d \in S] - E[Y_d(\text{Dem})|d \in S]$. Given some choice of the parameters γ_{Rep}^C and γ_{Dem}^N , our two simulations allow us to recover these two quantities by averaging (over the eight treated states that identify our reduced form estimates) the change in log in-migration rates between the counterfactual where governors are as they were in 2014 – an estimate of $E[Y_d(\text{Dem})|d \in S]$ – and where governors were as they were in 2017 – an estimate of $E[Y_d(\text{Rep})|d \in S]$.

For each education group g and each choice of γ_{Rep}^C and γ_{Dem}^N , these two counterfactuals are used to generate one moment:

$$\varphi^g = \frac{1}{|S|} \sum_{d \in S} \left[\underbrace{\ln \left(\sum_{o=1, o \neq d}^N L_{o,t-1}^g \pi_{do,t}^g(p(d, t)') \right)}_{Y_d(\text{Rep})} - \underbrace{\ln \left(\sum_{o=1, o \neq d}^N L_{o,t-1}^g \pi_{do,t}^g(p(d, t)) \right)}_{Y_d(\text{Dem})} \right] \quad (21)$$

According to our difference-in-difference analysis, when a Democratic governor is replaced by a Republican governor, the migration flow of college-educated workers decreases by 12.5%, while the migration flow of non-college-educated workers does not show a significant change. Therefore, we choose γ_{Rep}^C and γ_{Dem}^N to match the target of $\hat{\varphi}^C = -12.5\%$ and $\hat{\varphi}^N = 0$.

D.5 Estimation

D.5.1 Adjusting migration flows

The matrix of migration flows is central to our estimation procedure. Using the American Community Survey (ACS), we calculate the matrix of cross-state flows, year-by-year, separately for college and non-college workers. However, the ACS is only a 1% sample and so flows are measured with error. This causes a problem when flows are low, in which case there is a chance that the 1% sample misses the people who moved from o to d , and we observe zero flows. This is an issue because many key moments derive from the *log* of migration flows. As is well-known, ignoring these missing flows can bias our estimates, and so we build on the approach suggested by Silva and Tenreyro (2006) and assume that the realized number of college graduates observed in the ACS migrating from o to d in year t is drawn from a Poisson process. We can then estimate the expected number of true flows by imposing a functional form assumption on how these flows relate to state characteristics.

Key to this approach is that we have administrative data made publicly available by the Internal Revenue Service (IRS) on the total number of tax filers who moved from origin o to destination d in year t . This data is not suitable for our main reduced form regressions because it includes no demographic information (such as education), but it can be used to infer flows for our structural estimates. To do so for each origin-destination pair in each year, we assume that the *ratio* of college graduate movers to IRS movers is a function of an origin-fixed effect, a destination-fixed effect, a time-fixed effect, the log distance between o and d , and the ratios of per capita GDP, group-specific earnings from the ACS, and college populations between the two states. This is a very flexible approach that assumes that we can describe very well the *relationship* between college migration and total tax-filer migration and then imposes no assumptions or structure on the realizations of tax-filer migration across pairs or over time. Under these assumptions, we can write the expected number of college migrants per tax-filing migrant as:

$$\frac{E[L_{dot}^g]}{IRS_{dot}} = \exp\left(\theta_o + \eta_d + t + \beta_1 \log(distance_{do}) + \beta_2 \frac{GDPpc_d}{GDPpc_o} + \beta_3 \frac{Wage_{dt}^g}{Wage_{ot}^g} + \beta_4 \frac{L_{dt}^g}{L_{ot}^g} + \epsilon_{dot}\right) \quad (22)$$

where IRS_{dot} refers to observed migration in the IRS administrative data. Equation (22) can be estimated using a Poisson pseudo-maximum likelihood procedure (Correia, Guimarães, and Zylkin, 2021, 2020), and we can calculate the expected number of college graduate migrants $\tilde{L}_{dot}^g$ from the observed IRS migrants (where we do not know education).

It is important to note that our reduced form estimates are not based on the transformed migration data, but use only ACS estimates.

D.5.2 Estimation procedure

In this section, we describe the procedure for estimating the parameters $\{\theta^g, \mu_o^g, w_{d,t}^g, \eta_{do}^g, \alpha_d^g, \gamma_{\text{Rep}}^C, \gamma_{\text{Dem}}^N, A_{d,t}\}$. Unless otherwise specified, all calculations are based on the sample of ACS respondents from 2011-2019, age 26 or older, and employed in the private sector. We consider 2011 as the initial period $t = 0$. In all cases, o refers to state of residence last year, and d refers to current state of residence. In all calculations based on worker earnings, we restrict to full-time, full-year employees since ACS earnings are reported annually.

We provide a detailed explanation of estimating all the parameters in six steps. In all of these steps, we use the estimated migration flows based on the two adjustments to the raw ACS flows described in the previous section.

The first step involves estimating θ^g using equation (15) based on the dispersion of residual wages. We use workers' real earnings, which refers to the nominal wage obtained from ACS divided by the price deflators estimated by the Bureau of Economic Analysis at year and state levels. We calculate the wage residuals from a cross-sectional regression of log annual wage on destination-by-origin fixed effects, age squared, gender, marital status, education, race and fixed effects for year, occupation, and industry.

Next, we estimate migration shock μ_o^g by taking the average of the migration outflows for each group-origin from 2012 to 2019, using equation (16).

The third step is to estimate the matrix of migration costs, η_{do}^g , using equation (17). This process relies on the previously estimated value of θ^g and the matrix of migration flows.

After estimating θ^g, μ_o^g and η_{do}^g , the following step is to recover A_d and w_d^g which are determined using the system of equations (18) and (19). For output $Y_{d,t}$, we adjust nominal state-level GDP using the year-level and state-level price deflators we used to adjust workers' nominal earnings.

To determine α_d , it is necessary to select a reference state that experienced a change in the governor's party between 2012 and 2019, as α_d can only be estimated if the governors of d and the reference state belong to the same party. We compile a list of states that have experienced such a switch and select as a reference state the state with the highest migration flows for college-educated and non-college workers: North Carolina. Given this, we can recover α_d using equation (20). For any state d , it is often the case that multiple years exist in which the same governor serves with the reference state. In such situations, we compute the average of $\hat{\alpha}_{dt}$ values to recover α_d .

The final step is to recover the last two parameters, γ_{Rep}^C and γ_{Dem}^N , which are chosen to match the reduced form results. We use a grid search method that allows each γ to take on values ranging from -0.5 to 0.5, with a grid size 0.001. We estimate γ_{Rep}^C and γ_{Dem}^N that

produce the minimum sum of absolute errors when we solve for equation (21).

Estimation results and parameter values are presented in Table D1.

[Table D1 about here.]

D.6 State-specific counterfactuals

We begin with a set of 50 counterfactuals, each of which flips only a single state. For each flip, we calculate the equilibrium number of college graduates four years later,⁵¹ and Figure D1 maps the percent change in the number of college graduates. Blue states have Democratic governors in 2024, and so we are simulating the effects of a Republican governor who reduces the number of college graduates. Red states currently have a Republican governor, so we are simulating the effects of a Democratic governor. In both cases, darker colors reflect larger predicted effects.

[Figure D1 about here.]

How do these flips affect inequality? In Figure D2, we answer this, considering both “wage” inequality (earnings per unit of human capital, which cannot be observed in data) and “earnings” inequality (average earnings of college and non-college workers, which can be observed in data).

[Figure D2 about here.]

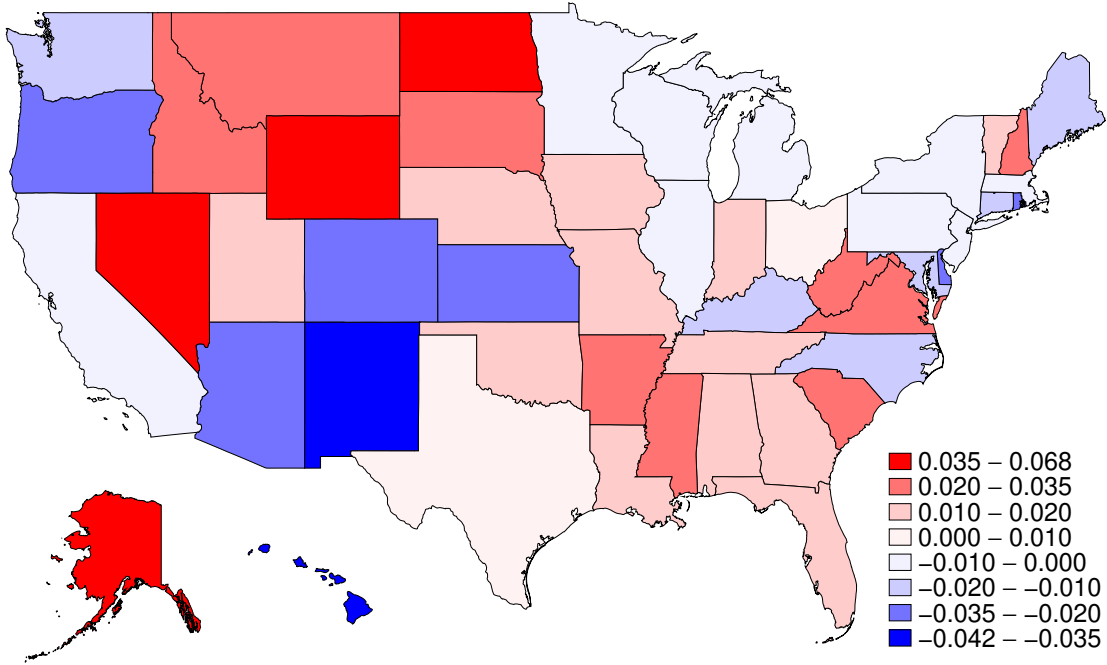
D.7 Further results

[Table D2 about here.]

[Figure D3 about here.]

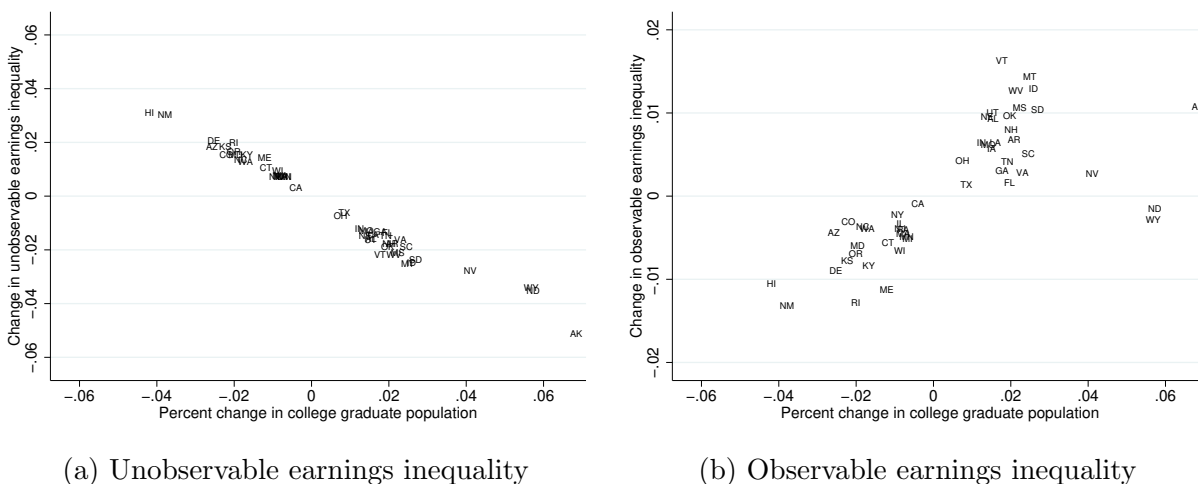
⁵¹We see these four-year effects as short-run effects and make no attempt to model changes in industry structure or firm location, endogenous changes in common-value amenities, or changes in future political conditions or attitudes, even though all of these may be affected by a change in the educational composition of the population. We see all of these as plausible long-run outcomes, but where further empirical work would be needed to discipline a quantitative model.

Figure D1: Heterogeneous effects of single states switching governors



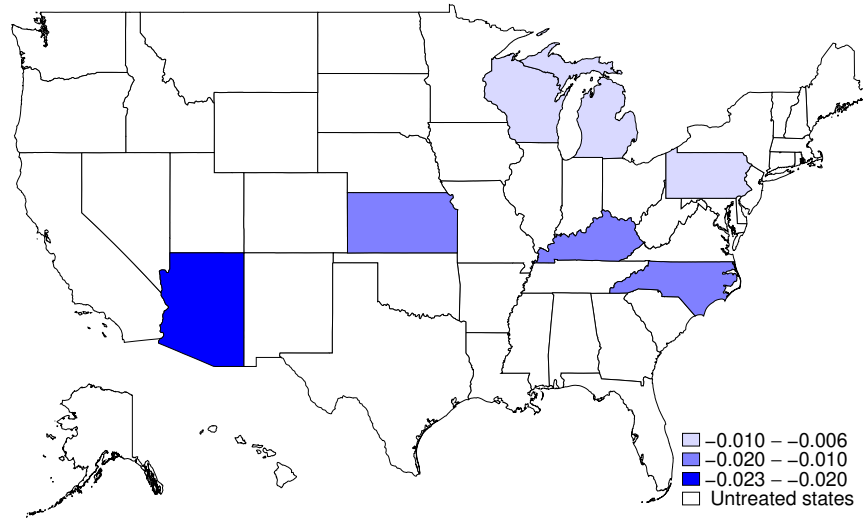
Notes: Map based on 50 separate counterfactuals, each of which flips only one state independently, relative to the set of governors in office in 2024. Red states have Republican governors in 2024, so we simulate the effects of a Democrat taking office. Blue states have Democratic governors in 2024, so we simulate the effects of a Republican taking office. Darker colors indicate a larger percentage change in the college graduate workforce predicted by our model.

Figure D2: Inequality effects of single states switching governors

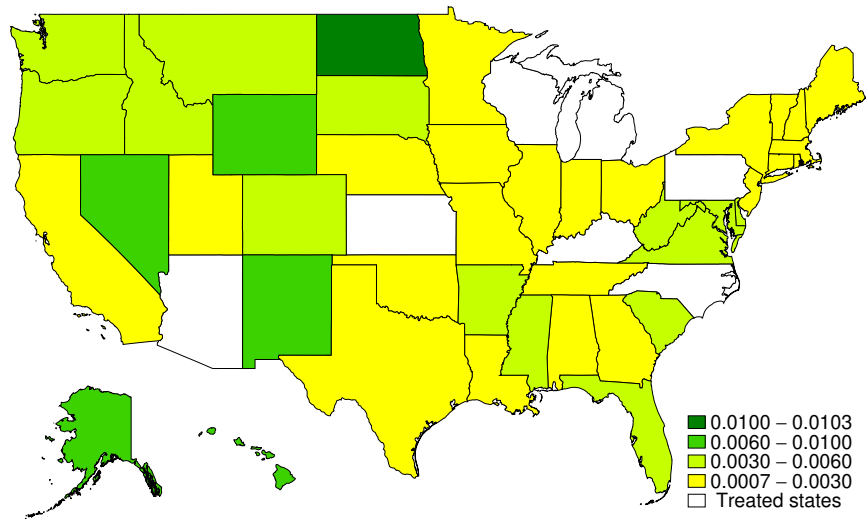


Notes: Figure based on 50 separate counterfactuals, each of which flips only one state independently, relative to the set of governors in office in 2024. In both panels, the x -axis refers to the model-implied percentage change in the size of the college graduate workforce. In panel (a), the y -axis refers to the change in the college/non-college gap in the log human capital price; in panel (b), the y -axis refers to the change in the college/non-college gap in log earnings per worker (see Table 4).

Figure D3: Direct and spillover effects of big red wave



(a) Direct effects: Treated states



(b) Indirect spillover effects

Notes: Map based on the strong red wave (see Figure 6) in which seven states governed by Democrats in 2024 are simulated as flipping to Republican governors. Percentage changes in the college-educated workforce for the treated states are mapped in Panel (a), and percentage changes for un-treated states are mapped in Panel (b).

Table D1: Parameter values

Description	Determination	Parameter	Value	
EoS across groups	Jerzmanowski-Tamura ‘20	σ	2.6	
Skill dispersion	Wage dispersion	θ^C	2.85	
		θ^N	2.96	
Political wedge	Reduced-form moments	$\gamma_{p(d)}^C$	0.053	
		$\gamma_{p(d)}^N$	0.001	
			Mean	Std. Dev.
Migration shock	Migration flow	μ_{do}^C	0.04	0.02
		μ_{do}^N	0.02	0.01
Migration cost	Migration flow difference	η_{do}^C	1.05	0.39
		η_{do}^N	1.06	0.42
Amenities	Same party governors	α_d^C	0.66	0.24
		α_d^N	0.69	0.24
Human capital prices	Market clear + common price	w_d^C	650.93	142.08
		w_d^N	525.18	174.51
TFP	Market clear + common price	A_d	373.67	99.60

This table reports the parameter values used for the model. The elasticity parameter σ is calibrated from the existing literature. The remaining parameters are estimated from target moments. Section D.5.2 provides the details of the estimation procedure. Human capital prices and total factor productivity (TFP) are reported in thousands of dollars.

Table D2: Decomposing effects on human capital price inequality

	(1)	(2)	(3)	(4)
Counterfactual	Small blue	Big blue	Small red	Big red
Number of states	5	9	3	7
Panel A: Change in log human capital price for college-educated workers: $\ln(w_d^C)$				
Average effects	-1.0 log points	-0.6 pts.	0.9 pts.	0.6 pts.
(min, max)	(-1.6, -0.7)	(-1.2, -0.3)	(0.8, 1.0)	(0.4, 0.9)
Panel B: Change in log human capital price for non-college workers: $\ln(w_d^N)$				
Average effects	0.7 log points	0.5 pts.	-0.6 pts.	-0.4 pts.
(min, max)	(0.5, 0.9)	(0.2, 0.7)	(-0.7, -0.5)	(-0.7, -0.2)
Panel C: Share attributable to non-college wages				
Share	.421	.412	.406	.405

Table presents changes in log human capital prices for “treated” states (i.e., those changing their governors) in our four counterfactuals (Figure 6 presents the states we simulate as flipping in each counterfactual). At the bottom, “share attributable to non-college wages” presents the share of the growth in college/non-college inequality that comes from falling real non-college wages, for the average treated state in the counterfactual.